MILITARY ENGINEER SERVICES

NAME OF WORK:DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN

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Signature of Contractor
Date:

AAD (Contracts)
for Accepting Officer

Telephone: 0832-2970606
E-mail:- cezcgg2-mes@gov.in

HQ Chief Engineer (Coast Guard)
Military Engineer Services
Indian Coast Guard Chicolna
JNC Nagar Road, Bogmalo Post,
Goa– 403 806

83030/ 75 /E8 07 Aug 2025

M/s_____

_____

_____

NAME OF WORK:DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN

Dear Sir (s),

1. Tender documents in respect of above work are uploaded on the site www.deproc.gov.in.The tender is based on single stage two cover e-tendering system. The contents of Cover I & Cover II are specified in NOTICE OF TENDER.

2. Bids will be received online by ACCEPTING OFFICER upto the date and time mentioned in the NOTICE INVITNG TENDER (NIT). No tender/bid will be received in physical form and any tender/bid received in such manner will be treated as non bonafide tender/bid.

3. Bid will be opened on due date fixed for opening as per critical dates given in the portal. Cover-1 will be opened first. Tenderers may see the result of the opening of Cover-1 on the tender portal. Date of opening of Cover-2 shall be decided after technical evaluation of Cover-1 keeping in view the various eligibility criteria given in the NIT and same will be intimated to the tenderers while uploading the technical evaluation on the tender portal.

4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents and/or depute your technical representative for discussion on tender/drawing and to clarify doubts, if any, before the bid submission starts. You are requested not to write piecemeal points and forward your points duly consolidated in one go.

5. Unenlisted contractors are required to submit the scanned copies (in pdf file) of documents required as per eligibility criteria mentioned in instructions for filling the tender documents and Appendix ‘A’ to NIT along with tender fee on e-procurement portal and submit the physical documents in the office of HQ Chief Engineer (Coast Guard) Goa within time limit specified in NIT. Inadequacy/deficiency of documents shall make the bid liable for rejection resulting in disqualification for opening of finance bid.

6. Enlisted contractors of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appx ‘A’ to NIT on e-procurement portal and submit physical documents in the office of HQ Chief Engineer (Coast Guard) Military Engineer Services, Indian Coast Guard Chicona, JNC Nagar Road, Bogmalo Post - Goa, 403806before date and time fixed for this purpose.

7. The contractor must ensure that the tender/bid is uploaded on the tender portal using authorised DSC in time as the Accepting Officer will take no cognizance of any quotations/offer received in any other electronic or physical form like email/fax/by hand/through post from tenderer/bidder even if they are received in time.

8. Keeping in view delays due to system failure or other communication related failures, it is suggested that the tender/bid be uploaded, if necessary, sufficiently in advance of the last due date and time fixed on the portal.

9. General Conditions of Contracts (IAFW-2249) (1989 Print) and errata and amendments thereto, Schedule of minimum fair wages and MES SSR (Part - I and Part - II) are not enclosed with these documents. These are available for perusal in the Office of GE concerned and this office. Tenderer is also requested to keep the copy of these documents with him as same are available in market.

10. **PERFORMANCE SECURITY DEPOSIT**

After acceptance of the tender, the contractor will be required to lodge with the Accepting Officer Performance Security Deposit at 5% of contract amount. The amount is required to be lodged within 28 (twenty eight) days of the receipt by the contractor of notification of acceptance of tender/ bid, failing which action as stipulated in condition No 19.3 (IAFW-2249) (1989 print) shall be taken.

11. Any tenderer, which proposes alterations to any of the condition, specifications laid down in the tender documents or any new condition, whatsoever, is liable to be rejected.

12. Consequent of implementation of GST (wef 01 July 2017 and amended subsequently upto date), the contractors quoted rate shall be deemed to include GST as applicable and applicable percentage will be paid to GST authority by the Contractor at his own arrangement. The liability to pay GST lies with the contractor. Contractor is advised to consider this aspect before quoting the tender. Nothing extra whatsoever shall be admissible on this account.

13. Department may issue amendments /errata in form of CORRIGENDUM to tender/revised BOQ to the tender documents. The Tenderer /bidder is requested to read the tender documents in conjunction with all the errata/amendments/corrigendum if any issued by the Department.

14. In case the BOQ is revised by the Department and the bidder has failed to quote in revised BOQ (i.e. he has quoted in previous BOQ). Such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non-bonafide. In such case the lowest tender shall be determined by the lowest amount amongst the valid/ bonafide bids only. Accepting Officer may decide whether to retender or otherwise. The remark of 'non-bonafide finance bid' against such bidder and copy of CST shall be uploaded along with Finance Bid Opening Summary.

15. Court of the place from where tender has been issued shall alone have jurisdiction to decide any dispute out of or in respect of this tender. After acceptance of tender Condition-72 of Jurisdiction of courts of IAFW-2249 shall be applicable.

Yours faithfully,

Signature of Contractor
Date:

AAD(Contracts)
For Accepting Officer

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER

1. **EARNEST MONEY DEPOSIT (EMD)**

Contractor(s) who are not enlisted with MES/who are enlisted but have not executed the Standing Security Bond shall submit Earnest Money Deposit as detailed in Notice of Tender in one of the following forms, alongwith their tender/bid :-

(a) Deposit at Call Receipt from a Scheduled Bank in favour of GE (I)CG Daman.
(b) Receipted Treasury Challan, the amount being credited to the Revenue Deposit of GE (I) CG Daman

It is advisable that Earnest Money is deposited in the form of deposit call receipt from an approved Schedule Bank for easy refund. In case the tenderer/bidder wants to lodge 'EARNEST MONEY DEPOSIT' in any other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money Deposit shall be submitted in the name of concerned GE.

NOTES: Earnest Money Deposit (EMD) in the form of cheque/Bank Guarantee etc will notbe accepted. NON-SUBMISSION OF EARNEST MONEY DEPOSIT (EMD) (Scanned copy alongwith Technical Bid & hard copy before the date and time fixed for opening of BOQ) WILL RENDER THE BID DISQUALIFIED FOR OPENING OF COVER II (FINANCE BID).

2 **PERFORMANCE SECURITY (Refer Condition 19 of IAFW-2249 General Condition of Contract)**

2.1 After receipt of the letter of Acceptance, the successful contractor shall deliver to the Accepting Officer a Performance Security in any of the forms given below for an amount equivalent to 5% of the contract sum to issue Work Order.

(i) A Bank Guarantee in the prescribed form.
(ii) Government Securities, FDR or any other Government instruments stipulated by the Accepting Officer.
(iii) An Insurance surety bond.

2.2 Work Order No 1 shall be placed only after submission of performance security of adequate value by the contractor.

2.3 If the performance security is provided by the successful Contractor in the form of a Bank Guarantee, it shall be issued by Nationalized / Scheduled Indian Bank but its confirmation shall be done only from the Head Office of the Bank.

2.4 Failure of the successful contractor to comply with the requirements of sub-clause 2.1 and 2.2shall constitute sufficient grounds for cancellation of the award of work and forfeiture for the Earnest Money. In case of MES enlisted contractor, amount equal to the Earnest Money stipulated in the Notice Inviting Tender, shall be notified to the tenderer for depositing the amount through MRO. Issue of tender to such tenderers shall remain suspended till the aforesaid amount equal to the Earnest Money is deposited in Government Treasury.

2.5 All compensation or other sums of money payable by the contractor to the Government under terms of this contract or under any other contract with Government may be deducted from or paid by the sale of a sufficient part of the Performance Security or from the interest arising there from or from any sums which may be due or become due to the contractor by the Government on any account whatsoever and in the event of his Performance Security being reduced by reason of any such deduction, or sale a aforesaid, the contractor shall within ten days thereafter make good in cash or securities, endorsed as aforesaid, and sum or sums which may have been deducted from or realized by the sale of his Performance Security or any part thereof.

2.6 Government shall not be responsible for any loss of securities or any depreciation in the value of securities while in their charge nor for loss of interest thereon.

2.7 In the event of contract being cancelled, under Condition 52, 53&54 of General Conditions of Contract, the Performance Security shall be forfeited in full and shall be credited into Consolidated Fund of India.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER (CONTD…)

3. GENERAL INSTRUCTIONS FOR COMPLIANCE

3.1 The bids received only in the electronic from will be considered. All bids shall be submitted on 'defproc.gov.in' portal. Documents should be scanned and forwarded in 'pdf' form and 'xls' form as indicated.

3.2 Bids shall be uploaded on 'defproc.gov.in' portal on or before the bid closing date mentioned in the tender. No tender/bid in any other electronic or physical form like email / fax / by hand / through post will be considered.

3.3 Bid should be DIGITALLY signed using valid DSC. All pages of tender documents, corrections/alterations shall be signed/ initialled by the lowest bidder after acceptance.

3.4 Drawings, if issued in physical form, must be returned duly initialled by the tenderer/bidder in separate envelope indicating his name and address.

3.5 The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initialled. The Contractor shall initial every page of tender and shall sign all drawings forming part of the tender. Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

3.6 In the technical bid, a scanned copy of Power of Attorney in favour of the person uploading the bid using his/her DSC shall be uploaded. In case the digital signatory himself is the sole proprietor, scanned copy of an affidavit on stamp paper of appropriate value to this effect stating that he has authority to bind the firm in all matters pertaining to contract including the Arbitration Clause, shall be attached in 'pdf' form. In case of partnership concern or a limited company, digital signatory of the bid/tender shall ensure that he is competent to bind the contractor (through) partnership deed, general power of attorney or Memorandum and Articles of Association of the Company) in all the matters pertaining to the contracts with Union of India including arbitration clause. A scanned copy of the documents confirming of such authority shall be attached with the tender/bid in 'pdf' form, if not submitted earlier. The person uploading the bid on behalf of another partner(s) or on behalf of a firm or company using his DSC shall upload with the tender/bid a scanned copy (in 'pdf' form) of Power of Attorney duly executed in his favour by such other or all of the Partner(s) or in accordance with constitution of the company in case of company, stating that he has authority to bind such other person of the firm or the Company, as the case may be, in all matters pertaining to the contract including the Arbitration Clause.

3.7 Even in case of Firms or Companies which have already given Power of Attorney to an individual authorizing him to sign tender in pursuance of which bids are being uploaded by such person as a routine, fresh Power of Attorney duly executed in his favour stating specifically that the said person has authority to bind such partners of the Firm, or the Company as the case may be, including the condition relating to Arbitration Clause, should be uploaded in 'pdf' form with the tender/bid; unless such authority has already been given to him by the Firm or the Company. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Articles of Association.

3.8 Hard copies of all above documents should be sent by the contractor to the Tender issuing authority well in advance to be received before the date and time fixed for the same.

3.9 Bid (Cover 1 & 2) shall be uploaded online well in time.

3.10 The contractor shall employ Indian National after verifying their antecedents and loyalty. Attention is also drawn to special condition 3 referred hereinafter and also conditions 24 & 25 of IAFW 2249 (General conditions of contract).

3.11 Tenderers/ bidders who uploaded their priced tenders/bids and are desirous of being present at the time of opening of the tenders/bids, may do so at the appointed time.

3.12 The tenderer/bidder shall quote his rate on the BOQ file only. No alteration to the format will be accepted, else the bid will be disqualified and summarily rejected.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER (CONTD…)

3.13 In case the tenderer/bidder has to revise/modify the rates quoted in the BOQ (excel sheet) he can do so only in the BOQ, through defproc.gov.in site only before the bid closing time and date.

4. **REVOKATION/ REVISION OF OFFER UPWARD/ OFFERING VOLUNTARY REDUCTION, AFTER CLOSING OF BID SUBMISSION DATE & TIME**.

In the event of lowest tenderer/bidder revoking his offer or revising his rate upward/ offering voluntary reduction, after closing of bid submission date & time, his offer will be treated as revoked and the Earnest Money deposited by him shall be forfeited. In case of MES enlisted Contractors, the amount equal to the Earnest Money stipulated in the Notice of tender, shall be notified to the tenderer/bidder for depositing the amount through MRO. Bids of such Contractors/bidders shall not be opened till the aforesaid amount equal to the earnest money is deposited by him in Govt Treasury. In addition, bids of such tenderer/bidder and his related firm shall not be opened in second call or subsequent calls. Reduction offered by the tenderer/bidder on the freak high rates referred for review shall not be treated as voluntary reduction.

5. **CPM (Critical Path Method)**

5.1 The project planning for work covered in the scope of tender is based on CPM.

5.2 The tenderer/bidder is expected to be fully conversant with the CPM technique and employ technical staff who can use the technique in sufficient details. Sufficient books and other literature on the subject are widely available in the market which the tenderer/bidder may make use of.

5.3 The tenderer's /bidder's attention is drawn to special condition of the tender regarding preparation of the detailed network analysis and time schedule for the work and his liability for employing sufficient resources to adhere to this schedule. Any inability on the part of the tenderer/bidder in using the technique will be taken as his technical inefficiency and will affect his class of enlistment and future prospect/invitation to tenders for future works.

5.4. Department may issue amendments/errata in form of CORRIGENDUM to tender/revised BOQ to the tender documents. The tenderer/bidder is requested to read the tender documents in conjunction with all the errata/amendments/corrigendum, if any, issued by the department.

6. In case the BOQ is revised by the Department and the bidder has failed to quote in revised BOQ (i.e he has quoted in previous BOQ), such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non-bonafide. In such cases the lowest tender shall be determined by the lowest amount amongst the valid/bonafide bids only. Accepting Officer may decide whether to retender or otherwise. The remark of 'non bonafide financial bid' against such bidder and copy of this CST shall be uploaded along with Financial Bid Opening Summary.

7. REFUND OF PERFORMANCE SECURITY

(Refer Condition 19 of IAFW 2249 General Condition of Contract):- The Performance security Deposit mentioned hereinbefore may be refunded to the contractor after expiration of the defects liability period (ref Sch A notes) by the GE provided always that the contractor shall first have been paid the final bill and rendered a No-Demand Certificate (IAFW-451). Any recovery if outstanding against the contractor shall be effected from the performance security as stipulated in IAFW-2249 (as amended).

8. These instructions shall form part of the contract documents.

Signature of Contractor **AAD(Contracts)**
Dated: **for Accepting Officer**

MILITARY ENGINEER SERVICES
NOTICE INVITING TENDER (NIT)

(IN LIEU OF IAFW-2162 (REVISED 1960)

1. A tender is invited for the work as mentioned in Appendix 'A' to this NOTICE INVITING TENDER (NIT).

2. The work is estimated to cost as indicated in aforesaid Appendix 'A'. This estimate, however, is not a guarantee and is merely given as a rough guide and if the work costs more or less, a tenderer / bidder will have no claim on this account. The tender shall be based on as mentioned in aforesaidAppendix 'A'.

3. The work is to be completed within the period as indicated in the aforesaid Appendix 'A' in accordance with the phasing, if any, indicated in the tender from the date of handing over site, which will be on or about two weeks after the date of Acceptance of tender.

4. Contractors whose names are on the MES approved list and within whose financial category the estimated amount would fall and un-enlisted contractors may submit tender/ bid subject to the other criteria mentioned in Appendix-A. However in case of term contracts, enlisted contractors of class SS to E may submit tender. Not more than one tender shall be submitted/ uploaded by one contractor/ firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/ bid for the same contract as separate competitors. Two firms shall be deemed to have business dealing if any of the partners/proprietor/ director is common among both of them. A breach of this condition will render the tenders/ bids of both the parties liable for rejection.

5. The Office of Chief Engineer (Coast Guard) Goa will be the Accepting Officer here-in-after referred to as such for the purpose of the contract.

6. Not more than one tender/ bvid shall be submitted/ uploaded by one bidder firm. Under no circumstances will a father and his son(s) or other close relations who have business dealing with one another be allowed to tender/ bid for the same tender as separate competitors. A breach of this condition will render the tenders/ bids of both the parties liable for rejection.

7. The Technical Bid and Financial bid (Cover-1 and Cover-2) shall be uploaded by the tenderer/bidder on or before the date & time mentioned in **NIT**. A scanned copy of DD with enlistment details/ documents shall be uploaded as packet 1/ cover-1 ('T' bid) of the tender / bid on e-tendering portal. DD is refundable in case T bid is not accepted resulting in non-opening of 'Q' bid. The applicant contractor shall bear the cost of bank charges for procuring and encashing the DD including revalidation of DDs and shall not have any claim from Government whatsoever on this account.

8. Tender forms and conditions of contracts and other necessary documents shall be available on **defproc.gov.in**/ eprocure.gov.in site for download and shall form part of contract agreement incase tender/bid is accepted.

9. In case of MES enlisted contractor who has not executed the Standing Security Bond and unenlisted contractor, the Cover-1 shall be accompanied by Earnest Money for the amount mentioned in Appendix 'A' in the form of deposit at call receipt in favour of concerned CE (See appendix 'A') by a scheduled Bank or in received treasury Challan the amount being credited to the revenue deposit of concerned Chief Engineer. The Chief Engineer will return the Earnest money, whatever applicable, to all unsuccessful tenderers/ bidders by endorsing an authority on the deposit at call receipt for its refund, on receipt of intimation from Accepting Officer to do that.

10. In case successful contractor i.e the lowest contractor having submitted the EMD, he shall have the option of converting the EMD instrument nto part of the Performance Security to be deposited by him within 28 days from the receipt of intimation of acceptance of tender from Accepting Officer.

NOTICE INVITING TENDER (NIT) (contd…)

11. Sample of materials and stores to be supplied by the contractor will also be available for inspection by the bidder at the office of concerned GE/ GE(I)/ AGE(I)/ Project Manager during working hours. The bidder is advised to visit the site of work by making prior appointment with concerned GE/ GE(I)/ AGE(I)/ Project Manager, who is the executing agency of the work (see Appendix-A0. The bidder shall be deemed to have full knowleddge of all relevant documents, samples, site etc whether he has inspected them or not.

12.Any bid which proposes any alteration to any of the conditions laid down or proposes any other new condition whatsoever, is liable to be rejected.

13. The uploading of bid implies that bidder has read this notice and the Conditions of Contract and has made himself aware of the scope and specifications of work to be done and of the conditions and rates at which stores, tools and plants etc will be issued to him and local conditions and other factors having bearing on the execution of the work.

14. Tenderers/bidders must be in possession of a copy of the MES Standard Schedule of Rates (see Appendix 'Á') including errata and amendments there to.

15. The Accepting Officer does not bind himself to accept the lowest or any tender bid or to give any reason for not doing so.

16. The Accepting Officer reserves the right to accept a tender submitted by a Public Undertaking/ Small & medium Enterprises (SMEs), giving a price preference over other tender(s)/bids which may be lower, as are admissible under the Government Policy. No claim for any compensation or otherwise shall be admissible from such tenderer/bidder whose tender/bid is rejected.

17. This Notice Inviting Tender (NIT) including Appendix 'Á' and annexures thereto if any, shall form part of the contract agreement.

Signature of Contractor **AAD (Contracts)**
Dated: **for Accepting Officer**

APPENDIX 'A' TO NOTICE OF E-TENDER

1.	Name of Work:	:	**DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN**
2.	Estimated Cost:	:	**Rs. 150.00 lakhs(Rupees One Hundred Fifty lakhs only) (At par market)**
3.	Period of completion:	:	**12 (Twelve) months**
4.	Cost of tender documents	:	**Rs. 1000/-**(Rupees One Thousand only) in the form of Demand Draft /Bankers Cheque from any Scheduled Bank in favour of **GE (I) (CG) Daman**payable at **Daman** (Note:- In case of retendering, the contractor who had quoted in the previous call is not required to submit the cost of tender).
5.	Website/Portal Address	:	https://defproc.gov.in
6.	Type of contract	:	The tender shall be based on drawings and specifications (IAFW-2159) and GCC (IAFW-2249) with schedule 'A' (list of items of work) to be priced by tenderer. The tenderers are required to quote their lumpsum amounts for part of schedule 'A' and quote rates against items of other parts of schedule 'A'.
7.	Timeline Details (a) Bid submission start date (b) Bid submission end date (c) Date of bid opening	:	Refer critical dates on the website.
8.	Eligibility Criteria: (A) For MES enlisted contractor	:	Contractors shall be enlisted with MES in **Class 'B'** and **category 'a(i)**' subject to satisfactory remarks wrt performance in respect of works in hand as reflected in workload return (WLR) or any other report circulated by competent Engineer authority.
	(B) For contractors not enlisted with MES	:	(a) Contractors not enlisted with MES should meet the enlistment criteria of **'B' class** and **'a(i)**' **category**contractor with regards to satisfactorily completion of requisite value works with Central/ State Government/ Central/ State PSUs/ AWHO/ AFNHB/ CGEWHO/ DGMAP, Annual turnover, Bank Solvency, working capital and other requirements given in Para 1.4 & 1.5 of Section 1 of MES Manual on Contracts-2020 as available in all MES formations as well as MES website (www.mes.gov.in) (b) Not carrying adverse remarks in work load report (WLR) or any other similar report circulated by any competent authority, if already working in MES. (c) Not suspended/debarred/blacklisted (either permanently or temporarily) from participating in any bid or for business dealings by any Central/State Government department or any Central/State Government PSU or any Autonomous Body under Central/State Government or any Local Body as on the bid submission end date.
	(C) Technical PQC Criteria	:	NA
9.	Tender issuing and Accepting Officer	:	**Chief Engineer (CG) Goa** Address:-Military Engineer Services,Coast Guard Air Enclave Dabolim, Goa– 403 806 Phone No:- 0832 2970606 Email:- cezcgg2-mes@gov.in
10.	Executing agency	:	**GE (I) (CG) Daman**
11.	Earnest Money Deposit	:	**Rs. 2,25,000/- (Rupees Two Lakh Twenty Five Thousand Only)** (In favour of **GE (I) (CG) Daman** in the form of Deposit at Call Receipt from Scheduled Bank)
12	Integrity Pact	:	NA

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT) (Contd…)

NOTES:-

1. After opening of Cover 1, the number of MES enlisted contractors of eligible class as well as eligible un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT are less than 7 (Seven), applications in respect of contractors **One class below** eligible class shall also be considered subject to fulfilment of other eligibility criteria given in NIT. Therefore MES contractor's One class below may also bid for this tender. Such contractors of one class below the eligible class shall not be considered in case their present residual workload in hand is more than five times their present tendering limit. However, in case such contractors fulfil the criteria of upgradation to the stipulated eligible class based on past experience (05 years) of completed works (Individual work experience and/or average annual turnover, as applicable) and financial soundness (Solvency/financial soundness and working capital), the ceiling of present residual work will not apply and they will be considered for issue of tender. Such bidders shall upload in their Cover-1 bid details of works in hand showing names of work, name of Accepting Officers, Contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date. Such contractors, if claim to fulfil the criteria of upgradation shall also upload the requisite information/documents in support of upgradation. These details shall be verified by the Tender Issuing Authority from concerned formations in case bids of such contractors are considered for evaluation.

2. In case after opening Cover 1, the number of MES enlisted contractors of eligible class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT, are 7 (Seven) or more, applications of only those One class below the eligible class bidders shall be considered, who have previously completed similar works satisfactorily and are meeting the criteria of up gradation in respect of past experience of completed works (individual work experience and/or average annual turn over as applicable) and financial soundness (solvency/financial soundness and working capital) as per details given in manual on contracts. Therefore such contractors shall upload the requisite information/documents in the Cover-1.

3. Un-enlisted contractor shall be considered provided he meets the criteria. Foreign firms shall not be eligible for this tender. However Indian firms having foreign national/Indian nationals staying abroad / Indian national having taken foreign citizenship, as director(s) shall be considered subject to security clearance from concerned authorities.

4. Contractors enlisted with MES will upload following documents in Cover-1 for checking eligibility :-
   (a) Application for the tender on Firm's letter head.
   (b) Enlistment letter issued by the Registering Authority duly renewed for the cyclic period in vogue.
   (c) Scanned copy of DD / Bankers cheque towards cost of tender and EMD instrument in case SSD bond is not signed at the time of registration.
   (d) GST and EPF registration
   (e) Any other document required as described in this Appendix.

5. Contractors not enlisted with MES will required to upload following documents in Cover-1 for checking eligibility:-
   (a) Application for the tender on Firm's letter head.
   (b) Scanned copy of DD / Bankers cheque towards cost of tender and EMD instrument
   (c) Copy of Police Verification Report / Police clearance Certificate / Character Certificate from the Police Authority of the area (not less than one year old)where the registered office of the firm is located ornotarized copy of valid passport of Proprietor /each Partner/each Director.
   (d) All documents required for enlistment in MES for the class mentioned in Para 8(B) above as per Para 1.5 of Section 1 of MES Manual on Contracts 2020.
   (e) Details of works being executed in MES, if any.
   (f) GST and EPF registration
   (g) Any other document required as described in this Appendix.

6. Tenders not accompanied by scanned copies of requisite DD/Bankers cheque towards cost of tender and earnest money (as applicable) in Cover-1 shall not be considered for validation of 'T' bid and their Financial Bids will not be opened.

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT) (Contd…)

7. Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD) instrument (as applicable) reach the office of Accepting Officer within 05 days of bid submission end date failing which following action shall be taken.

   (a) In case of tenders from an enlisted contractor of MES. Where scanned copies of requisite DD/ Bankers cheque towards cost of tender have been uploaded in Cover 1, but physical copies are not received within the stipulated period, their financial bids (Cover-2) will be opened. However non-submission of physical copies of cost of tender shall be considered as wilful negligence of the tenderer with ulterior motives and such tenderer shall be banned from bidding for a period of six months commencing from the date of opening of Financial Bid (cover-2).
   (b) In case of tenders from unenlisted Contractor, where scanned copies of requisite DD / Bankers Cheque towards cost of tender have been uploaded in Cover-1 but physical copies are not received within the stipulated period, their financial bids (Cover-2) will not be opened. Name of such contractors along wih complete address shall be circulated for not opening of their bids for a period of six months commencing from the date of opening of Financial bid (Cover 2).
   (c) In case of tenders from enlisted and un enlisted contractors, where scanned copies of instrument for Earnest money (as applicable) have been uploaded in Cover 1 but the same are not received in physical form within stipulated period, such tenders shall not qualify for opening of financial bid (Cover 2).

8. Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/another firm on his behalf. However a contractor can execute the work through power of attorney to Sons/Daughters/Spouse of proprietor /Partner/Director and firm's own employees, director, project manager provided they are not having a separate enlisted firm in MES in their name as Proprietor/Partner/Director.

9. After opening of Cover -1 and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderers as per NIT, a communication in the form of e-mail/SMS/Speed post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from date of communication failing which their financial bid (Cover 2) shall not be opened and contractor shall not have any claim on the same.

10. Invitation for e-tender does not constitute any guarantee for validation of Technical bid and subsequent opening of financial bid of any applicant/bidder merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the Technical bid and not to open the financial bid of any applicant/bidder. Technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given in this Appendix. Tenderer/bidder will be informed regarding non validation of his Technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the next higher engineer authority (NHEA) viz **HQ Additional Director General (Projects), Military Engineer Services C/o Chief Engineer Chennai Zone Island Ground, PallavanSalai Chennai-600009** on email id **adgpchn-mes@nic.in**with copy to the Accepting Officer on email before the scheduled date of opening of Cover 2. NHEA shall decide the matter within a period of seven working days from the date of receipt of appeal. The decision of the NHEA shall be final and binding. The tenderer/ bidder shall not be entitled for any compensation whatsoever for rejection of his bid.

11. In case un enlisted contractor is already executing works in MES, he shall not be considered eligible for the subject tender if the total value of such works is more than twice the tendering limit of the MES Class of contractor for which it is eligible. For this purpose, details of the work being executed by such contractor shall be uploaded in the Cover 1 of the bid and shall be checked/verified by the Accepting Officer.

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT) (Contd…)

12. In case the BOQ is revised through the corrigendum and the bidder has failed to quote on revised BOQ (ie, he has quoted on pre revised BOQ), such bid shall be treated as willfull negligence by the bidder and his quotation shall be considered non bonafide. In such cases the lowest tender shall be determined from amongst the valid/bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance.

13. Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tenderer after opening cover 2 shall be considered as a wilful default. For this default a penalty of an amount equal to Earnest Money shall be levied. In case of an unenlisted tenderer, Earnest money deposited by him shall be forfeited. In case of MES enlisted tenderer having deposited the standing security bond, an amount equal to the earnest money stipulated in the NIT, shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the aforementioned amount is deposited in the Government Treasury. No other disciplinary/administrative action shall be taken against such tenderers. In such situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be resorted to in a transparent and fair manner and the defaulting tenderer and his related firm if any, shall not be eligible for this tender in second call or subsequent calls.

13(A) Irrespective of whatever is mentioned in condition 19.3 of IAFW 2249 with regard to suspension of tenders on accont of non-submission of Performance Security, issue of tenders to such tenderers shall remain suspended for a period of six months from the date of cancellation of contract under condition 19.3 of IAFW 2249 in case of unenlisted contractors. In case of MES enlisted contractor, issue of tenders shall remain suspended till deposit of EMD or six months from date of cancellation whichever is later.

14. Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/One or more Partner/Directors are common. Decision of Accepting Officer on issue/deny the tender to any one of the related firms shall be final and binding.

15. In case of any document uploaded by bidder is found to be false or concealed some relevant information before acceptance, in that case his EMD shall be forfeited and action will be initiated against the defaulting bidder as per law.

16. In case after acceptance of contract it is observed that contractor has submitted false document or has concealed some information, in that case Contract will be cancelled and Performance security will be forfeited, also action will be initiated against the contractor as per law.

Signature of Contractor

AAD (Contracts)
for Chief Engineer

83030/ 75 /E8 07 Aug 2025

HQ Chief Engineer (Coast Guard)
Military Engineer Services
Indian Coast Guard Chicolna
JNC Nagar Road, Bogmalo Post,
Goa– 403 806

(In lieu of IAFW-1779A (R-1947)

(TO BE USED IN CONJUNCTION WITH GENERAL CONDITIONS OF CONTRACTS IAFW-2249)

MILITARY ENGINEER SERVICES

Telephone: 0832-253 1035

HQ Chief Engineer (Coast Guard)
Military Engineer Services
Indian Coast Guard Chicolna
JNC Nagar Road, Bogmalo Post,
Goa– 403 801

83030/ 75 /E8 07 Aug 2025

LUMPSUMTENDER AND CONTRACT FOR WORKS REQUIRED IN THE EXECUTION OF:DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN

M/s _____ is hereby authorised to tender for the above work. The technical and price bid (Cover 1 & 2) to be uploaded in the web site **www.defproc.gov.in** on **date and time as per critical dates given in the tender documents** and Bids (Cover 1 & 2) will be opened on **date and time as per critical dates given in the tender documents**.

Any correspondence concerning this tender should be addressed as indicated at the top of this sheet quoting reference as given.

"THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANY TENDERS."

Signature of Officer issuing documents
Appointment: AAD (Contracts)

Date: Aug 2025

_____
Signature of contractor

Dated: _____

SCHEDULE `A' NOTES
(List of Works and Prices)

NAME OF WORK: DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN

NOTES:

1. SCOPE AND PARTS OF BOQ (SCHEDULE 'A')

1.1 This BOQ (Schedule `A') has been divided into different parts described as under:-

i.	BUILDINGS/ STRUCTURES	:	Schedule `A' Part-I (BOQ item No.1.00)
ii.	EXCAVATION, EARTHWORK, SITE CLEARANCE & SITE DEVELOPMENT	:	Schedule `A' Part-II (BOQ item No.2.00)
iii.	DEMOLITION AND DISMANTLING		Schedule `A' Part III (BOQ item No.3.00)
iv.	INTERNAL WATER SUPPLY	:	Schedule `A' Part IV (BOQ item No.4.00)
v.	INTERNAL ELECTRIC SUPPLY	:	Schedule `A' Part V (BOQ item No.5.00)
vi.	SEWAGE DISPOSAL		Schedule `A' Part VI (BOQ item No.6.00)
vii.	ROAD/PATH/CULVERTS/HARDSTANDING	:	Schedule `A' Part VII (BOQ item No.7.00)
viii.	EXTERNAL WATER SUPPLY	:	Schedule `A' Part VIII (BOQ item No.8.00)
ix.	EXTERNAL ELECTRIFICATION	:	Schedule `A' Part IX (BOQ item No.9.00)
x.	ARBORICULTURE	:	Schedule `A' Part X (BOQ item No.10.00)
xi.	RAIN WATER HARVESTING		Schedule `A' Part XI (BOQ item No.11.00)
xii.	REFRIGERATION/AC & WATER COOLER	:	Schedule `A' Part XII (BOQ item No.12.00)
xiii.	MISCELLANEOUS ITEMS	:	Schedule `A' Part XIII (BOQ item No.13.00)

1.2 The description of building works and services given in the various parts of BOQ (Schedule `A') are in brief and include for complete materials and labours unless otherwise mentioned. These shall be amplified and read in conjunction with Special Conditions, Particular Specifications, Drawings forming part of the contract including notes thereon, Specification for materials and workmanship and conditions in relevant sections of MES Standard Schedule of Rates 2009 (Part I) and relevant preambles to various trade sections in MES Schedule Part II (2020). Words "all as specified and / or shown on drawings" shall be deemed to be included in all items of BOQ (Schedule `A') whether specifically mentioned or not.

1.3 The typical drawings (TD drawings) attached with the tender shall be supplemented to the main drawings and only relevant details shall be followed. In case of any discrepancy between details in main drawings and TD drawings, the details in main drawings shall take precedence over the typical drawings unless specifically indicated.

2.0 PERIOD OF COMPLETION:

2.1 The entire work under this contract comprising building works & services described in Schedule 'A' Part I to XIII shall be completed within **12(Twelve) Months** from the date of handing over of site as per work order No 01 placed by GE.

3. Site for execution of work will be available as soon as the work is awarded. In case, it is not possible for Department to make the entire site available on the award of work, the contractor will have to arrange his working program accordingly. No claim whatsoever for not giving the entire site on award of work and for giving the site gradually will be tenable.

SCHEDULE `A' NOTES (Contd.)

4. **UNIT RATES**

4.1 The unit rates inserted in Column 5 of BOQ (Schedule `A') Part- I to XII are deemed to be at par with rates contained in MES Schedule of Rates 2020 or rates analogous thereto. The contractor is required to quote his lump sum price in Column 6 of BOQ (Schedule `A') Part- I to XII and the percentage representing his lump sum price against relevant Parts of these BOQ (Schedule `A') on or off the total cost inserted by MES will be automatically indicated in Column 9 of BOQ (Schedule `A') of tender documents. The percentages and the amounts tendered by the contractor shall be deemed to have been calculated in the manner set out in Condition 6A(B) of General Conditions of Contracts IAFW-2249.

4.2 The contractor is required to quote his unit rates against each item of work given under BOQ (Schedule `A') Part - XIII. The rates quoted shall include supply of material and labour and fixing etc. complete.

4.3 Unless otherwise specified in the description of items "Rate/Unit" (inserted by MES) of each items of works includes supply of all materials and labours and fixing, fabricating, erecting, laying, testing, commissioning etc. as required for complete execution and completion of the work.

4.4 M&L or S&F wherever occurring shall be read as “Material and Labour” or “Supply and Fix” respectively.

5. All quantities given under column 3 of Schedule ‘A’ Part-II to XII& quantities given under column 3of BOQ (Schedule ‘A’) for Schedule ‘A’ Part –XIII are provisional. Probable layout of variousitems are indicated in drawings. The layout shown therein is intended for guidance only and may be varied where necessary at the discretion of the Engineer-in-Charge. The contractor shall not be entitled for any claim on account of such varied alignments.

6. **SCHEDULE ‘A’ PART I**

6.1 Unit rate quoted by the tenderer against items of Schedule `A' Part I shall be deemed to include for entire completion of the work and all relevant items of works as shown on drawings including notes thereon except works covered in Sch `A' Part II to XII.

6.2 The cost of following items shall also be deemed to be included in the unit rates quoted by the tenderer against items of Schedule `A' Part I:-

(i) Foundation for walls, columns, footing, PCC Block walls under all plinth beam, all the structures above ground level including superstructure including plinth and complete excavation and earth work, filling under floors and sides of walls/ foundations wherever required, DPC, Hard core, Sub base to floor, flooring, window cills and items not covered under internal water supply and internal electrification for connecting and commissioning unless mentioned otherwise in the particular schedule.

(ii) Cutting chases, leaving / forming holes in walls, floors and concrete etc. as required for embedding concealed pipe/conduits/strips and making good to match with the adjoining surfaces in connection with the works included in other parts of Schedule ‘A’ except specifically mentioned otherwise under particular items of other Schedules. No price adjustment shall be made for any increase/ decrease in works of cutting / chase / forming holes etc. consequent to variation in quantities of items included in other parts of Sch ‘A”.

(iii) Complete Water proofing treatment to RCC roof slabs and over top of RCC canopies as specified.

(iv) All sanitary apparatus, appliances, accessories, toilet fittings, Canteen Tiffin& Janitor fitings, mirrors etc., Gully traps including PCC encasing, CI floor traps/ Nahani traps with CP grating whether shown on drawings or not, plumbing work, UPVC Soil/waste/vent pipes,1stmanhole and RCC NP-2 pipes from Gully traps to 1st manhole. All soil/waste pipes shall be taken vertically below GL to a depth as required and shall be provided with a heel rest bends at the lower end and soil pipes shall be connected upto 1st manhole with required slope. Irrespective of whatever shown on drawings, the first manhole shall be constructed at a distance of 3.00 mtr from the respective external face of the wall except in cases where 1st manhole are shown in shafts. In case of any variation in distance of manhole, the same shall be regularized through DO.

SCHEDULE `A' NOTES (Contd.)

(v) Looking mirror drapery rod with finial on either side and windows, peg set fittings and fixtures, niches and boxes to house switch box and the like shall be provided to all windows.

(vi) Pre-constructional anti-termite treatment shall be provided to all buildings at of schedule 'A' Part-I. In buildings having stilt in the ground floor, ATT treatment shall also be provided.

(vii) Overhead water tank as shown on drawings. Rotational moulded HDPE 'ISI' marked white coloured, double walled overhead tanks of capacity as shown on drawings including CP brass float valve of required size, 20 mm bore GI medium grade overflow pipe, 40 mm bore GI flush pipe with plug, 15 mm bore GI vent pipe and RCC staging as shown on drawings. The length of overflow pipe shall be brought upto nearest sump/ outlet of rain water spouts / holes and anti-mosquito rose be provided with overflow & vent pipe.

(viii) Fan hook with CI/ pressed steel fan boxes with hooks treated with 2-coats of synthetic enameled paint over a coat of red oxide primer on exposed surfaces and tarred on surfaces in contact with concrete.

(ix) Numbering to blocks.

(x) Strengthening measures for buildings for required seismic zone.

(xi) Over head water tank, steps, ramps, plinth protection, platforms, RCC shelves etc. as shown on drawings.

(xii) Coping over parapet walls, drapery rods, rain water pipe, rain water spouts & splash stones.

(xiii) Over Head water tank and below counter Cupboardsandover head cupboard in Janitor room, loft doors, glass shelves, stainless steel plate racks and stainless steel sinks with drainage board, letter box, kitchen cabinets etc as shown on drawings.

(xiv) Vent pipe, upto the height of 0.30 m above parapet on roof slab.

(xv) Excavation & earthwork, preparatory works, surface excavation and surface dressing.

(xvi) Aluminium Grills, RCC jali, RCC shelves, opening for exhaust fans, arrangements for drying clothes.

(xvii) The ready made steel walk board shall be provided by the contractor for movement of labours and materials on each slab while concreting. No pedestrian/ machinery movement shall be allowed on bare reinforcement. The cost of this provision shall deemed to be included in lump sum quoted rates against Sch 'A' Part I.

(xviii) The work is required to be carried out as indicated in tender. Cover blocks for slabs, beams and columns shall be of factory made as approved by GE. The cost of this provision shall deemed to be included in lump sum quoted rates against relevant items of Sch 'A' Part I.

(xix) Provision of stainless steel railing to stair case and ladder of grade SS-316 all as shown on drawing.

7.0 SCHEDULE 'A' PART II to PART XIII

7.1 All quantities given under column 3 of Schedule 'A' Part-II to XII& quantities given under column 3 of BOQ (Schedule 'A') for Schedule 'A' Part – XIII are provisional.

7.2 All excavation and earthwork required for BOQ (Schedule 'A') Part III to Part XII shall be measured and paid under respective items of Schedule 'A' Part II except where the description of BOQ (Schedule 'A') Part III to Part XII items specifically include work of excavation and earth work. However excavation & earth work in strata as specified hereinafter for schedule 'A' Part I shall be included in lump sum quoted.

SCHEDULE `A' NOTES (Contd.)

8.0 **VALUATION OF DEVIATION**

The percentage addition / deduction of MES Schedule of Rates for the purpose of pricing deviation vide sub clause C (c) or C (d) of Condition 62 of IAFW – 2249 shall be as follows:-

a)	For works covered under BOQ (Schedule 'A') Part- I to XII	Percentage above / below as automatically indicated inColumn 9 of BOQ (Schedule 'A') of tender documentsof respective schedule,comparing with the lump sum quoted by tenderer to theamount inserted by MES as stipulated in Condition 62 ofIAFW – 2249.
b)	For works covered under BOQ (Schedule 'A') Part - XIII	As per Condition 62 of IAFW 2249 (General Conditions ofContracts)

9. **MINOR DETAILS:**

9.1 Lump sum quoted for the rate quoted for a particular item by the tenderer shall be deemed to include for all minor details / items of work and / or constructions which are obviously and fairly intended and which may not have been included in these documents but which are essential for the execution and entire completion of work and services in workman like manner and sound construction.

9.2 In case of difference of opinion between the contractor and Garrison Engineer as to whether or not certain item of work constitutes `Minor Constructional Details' which is deemed to have been included in the contractor's quoted lump sum, the decision of the Accepting Officer shall be final, conclusive and binding.

9.3 However some of the minor details / items which shall be deemed to be essential for execution and entire completion of work are described as under for guidance

(i) Reinforcement for any RCC member not indicated in the drawing but is a structural requirement.

(ii) Dwarf wall in situations like verandah, passage etc. not indicated in drawing.

(iii) Lintel over doors, windows and or opening not shown in drawing.

(iv) Builder's hardware for doors / windows etc. though not indicated on drawing but essential for usage.

9.4 In all the above and similar cases, the details indicated elsewhere in the drawings which are similar or near similar to missed out items or works shall be followed. In the absence of any other similar or near similar details, minimum essential requirement for completion of the work from structural, architectural and utility point of view shall be deemed to be included in the lump sum quoted. In the event of any dispute, decision of the Accepting Officer shall be final, conclusive and binding.

10. (a) For structural details, structural drawings shall only be referred. If there is any discrepancy between architectural and structural drawings with regard to structural details, details shown on structural drawings shall prevail. Similarly, if there is discrepancy between architectural and structural drawings with regards to architectural details, details shown on architectural drawings shall prevail. The decision of the Accepting Officer as to what constitutes structural or architectural details shall be final, conclusive and binding.

(b) For missing reinforcement details, if any, of RCC work, minimum reinforcement as required as per IS shall be adopted.

(c) In case where type and size of beam, slab and column etc. are not indicated these shall be provided as decided by the Accepting Officer.

SCHEDULE `A' NOTES (Contd.)

(d) If there is any discrepancy regarding general notes on RCC works in TD (typical details) drawings and structural drawings, structural drawings shall be followed. Similarly, details in main drawings shall be followed in case of discrepancy in main drawing and TD drawings.

(e) Nothing extra shall be admissible on account of work executed as stated above and the contractor shall be deemed to have taken into consideration the above provisions before quoting his lump sum cost and submitting the tender.

11. The lump sum quoted by the tenderer in BOQ for all the items of Schedule 'A' Part-I shall also include for making of block number (at two places in a particular block) as well as making number for all blocks with numbered ceramic tiles of size 8" x 8" for each alphabet / number separately for numbering of each block and of size 6" x 6" for each alphabet / number separately for numbering of each over 15 mm thick cement plaster in CM (1:4) as directed by the GE / Engineer-in-Charge. The number of tiles required for numbering of each block shall be different as the same will depend on quantum of alphabet / numbers required for numbering of any particular block.However, no adjustment shall be made for such variations and the unit rate / lump sum quoted in BOQ (Schedule 'A') Part-I shall be deemed to include for such variations and nothing extra shall be admissible on this account. Similarly the lump sum quoted by the tenderer in BOQ (Schedule 'A') Part- I shall also include for marking of block / building numbers as mentioned here-in-before.

12. Contractor shall prepare a sample board of approved samples fixtures of internal finishes and external finishes as instructed by GE/EIC during the execution of the project without any extra cost to Govt. and that sample board shall be displayed in the GE/EIC office.

13. The Rate quoted by the contractor shall be deemed to be inclusive of all taxes (including Sales Tax / Vat on materials, Sales Tax/Vat on Works Contracts Service Tax,Turnover Tax, Labour Welfare Cess / Tax etc.), duties, Royalties, Octroi, GST& other levies payable under the respective Statues.

14.0 LAYOUT

14.1Layout of buildings indicated on the site plan is tentative. No adjustment in price shall be done on account of minor changes/ modifications in the final approved layout within the site plan area

14.2Probable distributions of various items of internal / external services are indicated on drawings. These are tentative and likely to be varied where necessary at the discretion of the Engineer-in-Charge. The tenderer shall not be entitled for any claim whatsoever on account of such varied alignment of services.

15.0 DEFECT LIABILITY PERIOD:-In term of condition 46 of IAFW 2249 General Conditions of contract (as amended vide amendment No 48 of the defect liability period for the subject work shall be **24 (Twenty four) calendar months** after the work has been handed over to Government.

16.0 All wooden door fittings/ mongery shall be of stainless steel of grade SS-316.

17.0 All windows shall be of Aluminium powder coatedand to be fabricated in workshopand to be installed at site by authorized fabricators only. All windows to have aluminium powder coated grill.

18.0 The bidders shall note that, if two identical items exists in BOQ, same rate shall be quoted for such identical items. If two different rate are found quoted, the lowest rate shall be considered for such items for payment which contractor shall have no claim whatsoever.

19.0 FIXING OF SPECIAL (STAR) RATES

(a) In case of any deviation, mode of pricing shall be decided by Accepting Officer in terms of Condition 62 of IAFW-2249.

SCHEDULE `A' NOTES (Contd.)

(b) In the event of a deviation order involving fixation of Special (Star) Rate, DraftRate shall be prepared by GE (within a maximum period of 30 days) while initiating the proposal for deviation seeking approval of Accepting Officer and notified to contractor. While notifying the Draft Rate, it will be clearly stipulated that the same is merely an estimated rate and firm rate shall be fixed based on actual and receipt of supporting documents from contractor such as vouchers/literature of product/test certificates etc (as applicable) on completion of the work involvingStar Rate. Any objection to the method of fixing Star Rate will be dealt as per Condition 7 of IAFW-2249.

(c) Draft Star Rate shall be made based on market enquiry through telephonic enquiry/ quotations/ email/rate lists/internet-based sources, material & labour constants available in various Civil Engineering books and record available in respect of Star Rates approved in the past for similar items of work etc. Contractor may also assist GE's office in preparation of draft Star Rate.

(d) The Draft Star Rate shall be purely a draft rate and shall not be used for claiming final payment during execution of work. However GE shallallowpart payment to the tune of 80% during execution to avoid any financial hardship to contractor.

(e) After completion of the item of work involving Star Rate, contractor shall submitthe vouchers/literature of product/test certificates (as applicable, decision of GE being final in case of any disagreement) for nserviceabl of Star Rate. The Star Rate shall be technically checked by DCWE©/Director© depending upon the financial effect & approved by competent authority within a period of one month from submission of the relevant documents by contractor as mentioned above.

(f) The StarRateasapproved bycompetentauthorityafter technicalcheckbyDCWE©/Director© depending upon the financial effect shall be referred as "the rate decided by GE" under Para 62(G) of IAFW-2249.

21. After completion of work contractor shall prepare and submit PSMBs, Record drawings of building and external services to Engr –in –Charge before submission of final bill.

Signature of Contractor **AAD (Contract)**
Date: **for Accepting Officer**

SCHEDULE OF CREDIT

DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN

Notes:

1. The materials enumerated below shall become the property of the contractor. These materials shall be allowed to be removed from the site only after the credit amount is recovered from immediate interim/final bill (which ever submitted earlier)
2. The rate fixed by department is firm. The contractor shall have no claim whatsoever against the department if actual rate turns out to be different from the rates inserted by the department.
3. The amount of credit in respect of items shown shall be provisional and actual quantities shall be measured in MB against each item and the same shall be considered for arriving total amount of credit.

Sr	Description of items	Unit	Qty	Rate	Amount
	B/ R items				
1	Old wooden nservic & frames exc 1.5 Sqm but not exc 4 Sqm each including iron mongeries (doors)	Each	36.0	500.0	18000.00
2	Old Aluminium nservic & frames not exc 1.5 Sqm each including mongeries.(Windows)	Each	72.0	1000.0	72000.00
3	Old wrought iron (structural of angle/channel/ beams) each including iron mongeries.	Kgs	200.0	15.0	3000.00
4	Old rusted/ unsv TMT steel bars (@1% of RCC weight per Cum) Note:- Irrespective of actual quantity received the measurement shall be done as percentage of RCC only.	Kgs	4155.69	20.0	83113.80
5	Brick bats obtained from demolition.	Cum	209.00	200.0	41800.00
6	Old Rain water pipes	Rm	31.50	5.00	157.50
7	Old steel tubing 15/20mm	Rm	200.0	20.00	4000.00
8	Old sanitary fittings and water connections and taps like WHB, WC, Pillar tap, bib tap, stop cock, angle cock, shower rose like etc	Each	64.00	10.00	640.00
9	Old UNSV water storage tanks	Each	8.00	500.0	4000.00
	E/M Items				
10	Old unserviceable point wiring including accessories such as switches,socket,regulator, ceiling rose,holder,bell/buzzer, bell push of any type	Point	182.0	49.00	8918.00
11	Old unserviceable light fitting of any type	Each	90.0	25.0	2250.00
12	Old unserviceable DBs/ with MCCB's of any type	Each	7.0	50.0	350.00
13	Old unserviceable LT panel	Each	1.0	1000.0	1000.00
14	Old nserviceable GI pipe	RM	35.0	4.00	140.00
			Total Amount		**2,39,369.30**
			C/O to General Summary		

Signature of contractor AAD(Contracts)

For Accepting Officer

DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN

GENERAL SUMMARY

1	**Total Amount quoted brought forward from BOQ Ser Page No.**	**Rs:**	
2	**Schedule of Credit**	**Rs:**	**(-) 2,39,369.30**
3			
4	**NET CONTRACT SUM**	**Rs:**	

(Rupess_____

_____only)

Signature of Contractor
Dated:

AAD (Contracts)
For Accepting Officer

DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN

SCHEDULE-`B'

(ISSUE OF MATERIALS TO CONTRACTOR)
(See condition 10 of IAFW-2249)

Sl No	Particulars	Unit	Rate at which Materi al shall be issued to the Contractor	Place of issue of material	Remarks
1	2	3	4	5	6
----NIL----					

S C H E D U L E - `C'

List of Tools and plant (other than transport) which will be hired to the Contractor (See conditions 15, 34 and 35 of IAFW 2249)

Sl no	Quantity	Particulars	Details of MES crew supplied	Hire charges per unit per working day	Standby charges per unit per off day	Place of issue (by name)	Remarks
1	2	3	4	5	6	7	8
---------- Nil ----------							

S C H E D U L E - `D'

(TRANSPORT TO BE HIRED TO THE CONTRACTOR)
(See conditions 16 and 35 of IAFW 2249)

Sl no	Quantity	Particulars	Rate per unit per working day	Place of issue (by name)	Remarks
1	2	3	4	7	8
----------Nil----------					

Signature of Contractor
Dated:

AAD (Contracts)
for Accepting Officer

DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN

T E N D E R

To,
The President of India
Having examined and perused the following documents:

1. Specifications & Special Conditions signed by the AAD (Contracts) for the Chief Engineer (Coast Guard) Goa -403806
2. Drawings detailed in the specifications.
3. BOQ and Schedule ‘A', ‘B', ‘C', and ‘D' attached hereto.
4. MES Standard Schedule of Rates Part-I 2009 Specifications (including amendment Nos. 01 to 03) and MES Standard Schedule of Rates 2020 Part-II Rates (including amendment No. 01 to 122 & section 30) {hereinafter and in IAFW-2249 referred to as the MES Schedule}.
5. General Conditions of Contracts IAFW-2249 [1989 Print] together with errata 01 to 20 and amendments 01 to 49 and the Schedule of Minimum Wages.
6. Water Condition 31 of IAFW-2249 [General Conditions of Contracts].
   WATER WILL NOT BE SUPPLIED BY MES.
7. Should this tender be accepted, * I/WE AGREE
   ***(a)** "That the sum of **Rs 2,25,000/- (Rupees Two Lakh twenty five thousand Only)** forwarded as Earnest Moneywhich shall either be retained as a part of Security Deposit or be refunded by the Government on receipt of an appropriate amount of the Security Deposit, within the time specified as per condition 22 of IAFW-2249.

   (b) To execute all the works referred to in the said documents upon the terms and conditions contained or referred to thereupon as detailed in the General Summary and to carry out such deviations as may be ordered vide Condition 7 of IAFW-2249 upto a maximum of 10% (**TEN PERCENT)** and further agree to refer all disputes as required by the condition 70 to Arbitration and Arbitrator to be appointed by the Engineer-in-chief, New Delhi-11 or in his absence the Officer officiating as Engineer-in-Chief, New Delhi-11 or Dir Gen of Wks, if specifically delegated in writing by the Engineer-in-Chief, AHQs, New Delhi-11 whose decision shall be final conclusive and binding.

.

**Delete whichever not applicable*.

In lieu of IAFW-1779A(Revised 1947)

DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN

SIGNATURE _____ IN THE CAPACITY OF_____ _____ duly authorized to sign the tender for and on behalf of _____.
(IN CAPITAL LETTERS)

(WITNESS)

_____

_____

_____

NAME & POSTAL ADDRESS
OF SIGNATORY OF DOCUMENT

Name _____

Address_____

_____

_____ Alterations have been made in this document and as evidence that these alterations were made before the execution of the contract agreement these have been initialled by the contractor and Shri. _____

The said officer is hereby authorised to sign and initial on my behalf the documents forming part of this contract.

The above tender was accepted by me on behalf of the President of India for the Contract Sum of Rs_____ (Rupees_____ _____ Only) on the _____ day of_____2025.

Signature _____ dated this _____day of_____2025.

Appointment

Chief Engineer (CG) Goa
Military Engineering Services
Chicolna, JNC Nagar Road
Bogmalo Post, Goa– 403 806
(For and on behalf of the President of India)

DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN

GENERAL CONDITIONS OF CONTRACTS (IAFW-2249,1989 PRINT) FOR LUMP SUM CONTRACTS (IAFW-2159),TERM CONTRACTS FOR ARTIFICERS WORKS (IAFW-1821) AND MEASUREMENT CONTRACTS (IAFW-1779 & 1779 A)

1. It is hereby agreed by me/us* that the General Conditions of Contracts, including condition 70 & 71 pertaining to the settlement of disputes by Arbitration (IAFW-2249 -1989 Print with errata 1 to 20 and amendments 1 to 49 shall form an integral part of the tender documents.

2. This tender submitted by * me/us is subject to the aforesaid General Conditions of Contracts in IAFW-2249, an amended copy of which has been supplied to *me/us and is in my/our * possession and which I/We* have read and fully understood before submission of this tender.

3. The existing condition 70 of IAFW – 2249 is hereby substituted with revised condition 70 as per enclosed 'Annexure – I' (Ser Page No 35 &36). Condition 70 of IAFW - 2249 wherever referred in these tender documents shall be the revised condition as per Annexure –I'.

4. *My/our signature there under is in token of *our having accepted the aforesaid amended General Conditions of Contracts together with errata and amendments (IAFW 2249-1989 Print) and the clause relating to arbitration forming an integral part of this tender.

* Delete whichever is not applicable.

Signature of Contractor **AAD (Contracts)**
Dated: **for Accepting Officer**

Annexure 'I'

The existing description of Condition 70 of IAFW - 2249 shall be substituted by the revised description as under:-

70. Arbitration

(a) **Arbitration Where Applicability of Section 12(5) of the Arbitration and Conciliation Act Has Been Waived Off** :-

All disputes between the parties to the Contract (other than those for which the decision of the CWE or any other person is by the Contract expressed to be final and binding) shall, after written notice by either party to the Contract to the other of them, be referred to the Arbitral Tribunal of a Sole Arbitrator (in case of contract sum less than or equal to Rs. 100 crore) or to Arbitral Tribunal of three Arbitrators (in case of contract sum exceeding Rs. 100 Crore) from MoD Panel of Arbitrators. The officers so considered for appointment of Arbitrator, either as sole Arbitrator or for Arbitral Tribunal, shall be having degree in Engineering or equivalent or having passed Final/Direct Final Examination of Sub-Division II of Institution of Surveyor (India) or similar other Institutes recognised by the Government of India.

In case of arbitration by Sole Arbitrator, the Arbitrator shall be appointed by the Authority mentioned in the contract document within a period of thirty days of having received the notice from any of the parties to Contract, out of MoD Panel of Arbitrators. In case of Arbitral Tribunal consisting of panel of three Arbitrators, both the parties willbe asked by the AppointingAuthorityto suggest at least two names out of MoD Panelof Arbitrators within thirty days.The Appointing Authority will appoint two Arbitrators, one Arbitratoreachout of the suggested names.The two Arbitrators soappointed willselect one Arbitrator from the MoD Panel of Arbitrators who will be the 'Presiding Arbitrator'. The Serving Officer(s) so appointed as Arbitrator(s), either as Sole Arbitrator or as one of the here Arbitrators in the Arbitral Tribunal, can continue as Arbitrator even after retirement, provided both the parties to the Contract give written consent to this effect. In such case, however, the Arbitrator shall not be entitled for any fee even after retirement.

(a) Arbitration where applicability of Section 12 (5) of Arbitration & Conciliation Act Has Not Been Waived Off

All disputes, between the parties to the Contract (other than those for which the decision of the CWE or any other person is by the Contract expressed to be final and binding) shall, after written notice by either party to the Contract to the other of them, be referred to the Arbitral Tribunal of a Sole Arbitrator (in case of contract sum less than or equal to Rs. 100 crore) or to Arbitral Tribunal of three Arbitrators (in case of contract sum exceeding Rs 100 crore). The Officers so considered for appointment as Arbitrator, either as Sole Arbitrator or for Arbitral Tribunal, shall be having degree in Engineering or equivalent or having passed Final/Direct Final Examination of Sub Division II of Institution of Surveyor (India) or similar other Institutes recognised by the Government of India.

In case of arbitration by sole Arbitrator, the Arbitrator shall be appointed by the authority mentioned in the contract document from the MoD Panel of Arbitrators within a period of thirty days of having received the notice from any of the parties to Contract. In case of Arbitral Tribunal consisting of panel of three Arbitrators, both the parties shall be asked by the Appointing Authority to suggest at least two names out of MoD Panel of Arbitrators within thirty days. The Appointing Authority will appoint two Arbitrators, one Arbitrator each out of the suggested names. The two Arbitrators so appointed shall select one Arbitrator from the MoD Panel of Arbitrators who will be the 'Presiding Arbitrator'.

(b) Common For All Arbitration

Unless both parties agree in writing, such reference shall not take place until after the completion or alleged completion of the works or termination or determination of the contract under Condition Nos 55, 56 and 57 hereof.

Annexure 'I' (Contd/---)

Provided that in the event of abandonment of the works or cancellation of the Contract under Condition No. 52, 53 or 54 hereof, such reference shall not take place until alternative arrangements have been finalised by the Government to get the works completed by or through any other Contractor or Contractors or Agency or Agencies.

Provided always that commencement or continuance of any arbitration proceeding hereunder or otherwise shall not in any manner militate against the Government's right of recovery from the Contractor as provided in Condition 67 hereof.

If the sole Arbitrator or one or more Arbitrators of the Arbitral Tribunal so appointed resign(s) from his/her appointment or vacate(s) his/her office or is unable or unwilling to act due to any reason whatsoever, the Authority appointing him/her will appoint a substitute Arbitrator to act in his/her place in the manner specified hereinabove.In case the Arbitrator resigning in this manner is the Presiding Arbitrator, the other two Arbitrators of the Arbitral Tribunal shall appoint the substitute Presiding Arbitrator.

The Arbitral Tribunal may proceed with the arbitration, exparte, if either party, in spite of a notice from the arbitrator fails to take part in the proceedings.

The Arbitral Tribunal may from time to time with the consent of the parties, enlarge the time for making and publishing the award subject to the limit laid down in the Arbitration & Conciliation Act 1996 as amended upto the date on which arbitration proceedings commence.

The Arbitral Tribunal shall make the award within the period as provided in the Arbitration & Conciliation Act 1996 (as amended upto the date on which arbitration proceedings commence) from the date of entering on the reference or within the extended period as the case may be on all matters referred to it and shall indicate findings alongwith sums awarded separately on each individual item of dispute. The Arbitral Tribunal shall give reason for the award in each and every case irrespective of the value of claims or counter claims.

The venue of Arbitration shall be such place or places as may by fixed by the Arbitral Tribunal in its sole discretion.

The award of the Arbitral Tribunal shall be final and binding on both parties to the Contract.

Signature of contractor **AAD (Contracts)**
Date: _____ **For Accepting Officer**

DEMOLITION AND RECONSTRUCTION OF JCOS/SOS IN-LIVING ACCOMODATION (BLDG NO P-17) AT ICGAS DAMAN

PAYABLE BY CONTRACTORS UNDER FAIR WAGES
(See Condition 58 of IAFW-2249)

It is hereby agreed by me/us that the "Schedule of minimum fair wages".(SMWF) Containing as published vide Government of India Ministry of Labour letter No 1(2)/92-IS-II Dated 10 Mar 1992 revised up-to-date of opening of tender comprising 10 (Ten) pages, Sl Page Nos 37 to 46 shall form part of these tender documents.

* My/our signature hereunder is deemed to me/us* having signed the foresaid Schedule of minimum fair wages forming part of the tender.

Signature of Contractor
Dated:

AAD (Contracts)
for Accepting Officer

SPECIAL CONDITIONS (Contd…)

1. **GENERAL**

These special conditions shall be read in conjunction with the general conditions of contracts (IAFW-2249) and IAFW-2159 including errata and amendments thereto. Any provisions in these special conditions if at variance with the provision in the above mentioned documents shall be deemed to take precedence there over.

2. INSPECTION OF SITE BY THE TENDERER

(i) The tenderer is advised to contact the Garrison Engineer for the purpose of inspection of site (s) and relevant documents other than those sent herewith, who will give reasonable facilities for the purpose. The tenderer shall also make themselves familiar with the working conditions, accessibility of site (s), availability of materials and other cogent conditions, which may effect the entire completion of work under this contract.

(ii) The tenderer shall be deemed to have inspected the site (s) and made themselves familiar with the working conditions, whether they have actually inspected the site (s) or not.

3. CO-ORDINATION WITH OTHER AGENCIES: The contractor shall permit free access and afford normal facilities and usual convenience to other agencies or departmental workmen to carry out connected works or services under separate arrangements. The contractor will not be allowed any extra payment on this account and no compensation shall be admissible to the contractor on this account.

4. CONTRACTORS REPRESENTATIVE, AGENTS, SERVANTS AND WORKMEN:-

(a) Contractor shall employ only Indian Nationals as his representatives, agents, servants and workmen after verifying their antecedents and loyalty. He shall ensure that no person of doubtful antecedents and nationality is in any way associated with the works. If for reasons of technical collaboration or other considerations the employment of foreign national(s) is unavoidable, the contractor shall furnish full particulars to this effect to the Accepting Officer at the time of submission of the tender. As a proof that the contractor has employed only Indian National, he shall render a certificate to GE within one month from the date of acceptance of tender to this effect. In case the GE desires, the contractor will have the police verification done of personnel employed by him. If for reasons of technical collaboration or other considerations the employment of foreign national(s) is unavoidable, the contractor shall furnish full particulars to this effect to the Accepting Officer at the time of submission of the tender.

(b) The GE shall have full powers and without giving reasons; to order the contractor immediately cease to employ, in connection with this contract, any agent, servant, or employee whose continued employment is, in his opinion, is undesirable. The contractor shall not be allowed any compensation on this account.

(c) The contractor's attention is also drawn to condition 25 of IAFW-2249 in this connection.

5. SECURITY OF CLASSIFIED DOCUMENTS

Contractor's special attention is drawn to condition 2-A and 3 of General Conditions of Contracts (IAFW-2249). The contractor shall not communicate any classified information regarding works either to sub contractors or others without prior approval of the Engineer-in-Charge. The contractor shall also not make copies of the design/drawings and other documents furnished to him in respect of works and he shall return all documents on completion of the works or on earlier determination of the contract. The contractor shall along with final bill attach a receipt from the Engineer-in-Charge in respect of his having returned the classified documents as per condition 3 of General Conditions of Contracts (IAFW-2249).

6. ACCEPTANCE OF TENDER

The tender shall remain open for acceptance for a period of **90 days** from the **next day subsequent to the bid submission end date.**

7. CRITICAL PATH METHOD CPM NETWORK

7.1 The time and progress chart to be prepared as per condition 11 of General Conditions of Contracts (IAFW-2249) shall consist of detailed network analysis and a time schedule. The critical path network will be drawn jointly by the GE and contractor soon after acceptance of tender. The time scheduling of the activities will be done by the contractor so as to finish the work within the stipulated time. On completion of the time schedule a firm calendar date schedule will be prepared and submitted by the contractor to GE, who will approve it after due scrutiny. The schedule will be submitted in four copies within two weeks from the date of handing over site.

SPECIAL CONDITIONS (Contd…)

7.2 During currency of the work the contractor is expected to adhere to the time schedule and this adherence will be a part of the contractor's performance under the contract. During the execution of work contractor is expected to participate in the reviews and updating of the network undertaken by the GE. These reviews may be undertaken at the discretion of the GE either as a periodic appraisal measure or when the quantum of work ordered on the contractor is substantially changed through deviation orders or amendments. Any revision of the schedule as a result of the review will be submitted by the contractor to the GE within a week who will approve it after due scrutiny. The contractor will adhere to the revised schedule thereafter. In case of contractor disagreeing with the revised schedule the same will be referred to the Accepting Officer whose decision will be final, conclusive and binding. GE's approval to the revised schedule resulting in a completion date beyond the stipulated date of completion shall not automatically amount to a grant of extension of time. Extension of time shall be considered and decided by the appropriate authority mentioned in condition 11 of (IAFW-2249) and separately regulated.

7.3 The Contractor shall mobilise and employ sufficient resources to achieve the detailed schedule within the broad frame work of the accepted methods of working and safety.

7.4 No additional payment will be made to the contractor for any multiple shift work or other incentive methods contemplated by him in his work schedule even though the time schedule is approved by the Department.

7.5. **PRECAUTION AGAINST RISKS**

7.5.1 The Contractor shall be responsible at his own expense in taking precaution to prevent any damage from what so ever cause arising, other than out of accepted risks and to minimise the amount of any such loss or damage and for adoption of necessary protective measures required for the purpose in compliance with Condition 38 of IAFW-2249 and Rule 5 of the MES SAFETY CODE code vide (Annexure `B') of IAFW-2249 until the works have been handed over duly completed to the Engineer-in-charge.

8. SCHEDULE OF FINISHES:- In the event of a discrepancy between schedule of finishes and other drawings forming part of the tender, the provisions in the schedule of finishes shall take precedence over the provisions in the other drawings.

9. MATERIALS AND SAMPLES

9.1 Refer condition 10 of IAFW-2249.

9.2 The materials and articles, which have been specified from certain makes/ manufacturers, shall be of makes / manufactures as specified. If the manufacturers specified in tender documents make both ISI marked and conforming to ISI, the materials / articles shall be ISI marked.

9.3 The materials and articles, which have not been specified in tender documents by makes / manufacturers, shall be as under:-

(a) If ISI marked materials are being manufactured the same shall be ISI marked. For list of ISI marked manufacturers refer website of BIS i.e. www/bis.org.in.

(b) If ISI marked materials are not being manufactured the same shall be conforming to IS specifications.

9.4 Materials of local origin shall be as specified and conforming to samples kept in GE's office. The tenderer is advised to inspect sample of materials which are displayed in the office of GE, before submitting his tender. The tenderer shall be deemed to have inspected the samples and satisfied himself as to the nature and quality of materials, he is required to incorporate in the work irrespective of whether he has actually inspected them or not.

9.5 The contractor shall not procure materials and articles unless the samples are first got approved by the GE.

SPECIAL CONDITIONS (Contd…)

10. RECORD OF MATERIALS

10.1 (A) Original vouchers / invoice for the following items (as incorporated / required to be executed) in this work shall invariably be produced by the contractor:-

(i) Cement,
(ii) Steel
(iii) LT /HT/ APFC Panel Board, lights, Electric poles,
(iv) HT/LT Cables, High mast pole & light, Light fittings,
(v) APP modified polymeric membrane, primer
(vi) Proprietary materials like: - Water proofing compound, chemical for Anti termite Treatment, Oil bound distemper, Cement based paint, Synthetic enamel paint, primer
(vii) Bitumen for water proofing and Bitumen for road work.
(viii) Flooring tiles, Sanitary fittings, Water supply pipes fittings & fixtures,
(ix) Major E/M equipments such as Transformer, DG Set , Pumps, Motors, AC Plant & fittings, Lift, Firefighting equipments etc.
(x) Any other material as asked by Engineer-in-Charge.

(B) In addition to above, original vouchers / invoice for the items for which payment is claimed in RAR as "Material lying at site" shall also be produced. However, for the items of local origin except perishable items payment in RAR may be allowed based on market enquiry / verification.

10.2 The vouchers/invoice shall be from the manufacturers and /or from their approved agents for full quantity of the respective materials required for the work under the contract.

10.3 The quantity of proprietary articles/materials brought to site shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work.

10.4 proprietary materials referred above brought at site shall be stored as directed by the Engineer-in-Charge and those already recorded in measurement book shall be suitably marked for identification.

10.5 The contractor shall ensure that the materials are brought to site in original sealed container in packing bearing manufacturers marking.

10.6 The vouchers shall be certified by the Engineer-in-Charge as soon as materials are brought to site by the contractor and suitable endorsement made on the face of original paid vouchers indicating the contract No and other particulars of work for which used. Certified true copies of all such vouchers shall be carefully maintained by the Engineer-in-Charge.

10.7 Bitumen shall be purchased directly from main manufacturers only. In case of small quantity, the Accepting Officer may allow procurement of Bitumen from authorised dealers of main manufacturers. Purchase vouchers for the same shall be defaced by the Engineer-in-Charge, indicating reference to the Contract number under his dated signature and photocopies thereof shall be kept on record so as to avoid their being used again.

10.8 The vouchers will clearly indicate the Contract number and the IS No and specific alternative to which the material conforms in case of various alternatives in IS. In case the contractor procures the Bitumen mix from Hot Mix Plant, the copy of voucher of Bitumen provided by the main manufacturer to the Hot Mix Plant shall be submitted by the contractor.

11. QUARRIES:-Reference to condition 14 of IAFW-2249. Quarries are not available on Defence land for use by the contractor.

12. WORKING CONDITIONS IN RESTRICTED AREA

Condition for working in Restricted Area: All works lie in **Restricted Area**. The restrictions for entry to work site and conditions of working in restricted area shall be as under:-

SPECIAL CONDITIONS (Contd…)

12.1 **Entry And Exit**:- The contractor /his agents / representatives / workmen etc. and his materials, carts, trucks or other means of transport etc. will be allowed to enter through and leave from only such gate or gates and at such times as the GE or authorities in charge of the restricted area may at their sole discretion permit to be used. Contractor's authorised representative is required to be present at the places of entry and exit for purpose of identifying his carts, trucks, etc to the person in charge of the security of the restricted area.

12.2 **Identity of workmen**:
(i) Every workman shall be in possession of an identity card. The identity card shall be issued after a thorough investigation of antecedents of the labourers by the contractor and attested by Officer-in-Charge of the unit concerned in accordance with the standing rules and regulations of the unit. Contractor shall be responsible for conduct of his workmen, agents or representatives.
(ii) Identity Card or Passes: The contractor, his agents and representatives are required individually to be in possession of an identity card or pass which will be examined by the security staff at the time of entry into or exit from the restricted area at any time or number of times inside restricted area.
12.3 **Search** : Through search of all persons and transport shall be carried out at each gate and for as many times the gate is used for entry or exit and may also be carried out any number of times at the site within the restricted area.
12.4 **Working Hours**: The units controlling restricted area, usually work during six days in a week and remain closed on the 7th day. The working hours available to the Contractor's labour and staff are however appreciatively reduced because of the time of entry and exit during working hours. The exact working hours, working days and number of working days observed for these restricted area (s) where works are to be carried out shall be deemed to have been ascertained by contractor before submitting his tender. The tenderer's attention is invited to the fact that number of working hours for a unit are prescribed in regulations and that they cannot be increased by the Garrison Engineer or authorities controlling the restricted area. The definition of "working days" as given under condition 1 (t) of IAFW-2249 does not apply in case where the works are carried out in restricted area. However, presently the working hours allowed are as under. Monday to Friday – 8.00 AM to 4.00 PM. Saturday – 8.00AM to 01.00 P.M.

12.5 **Working on Holidays**: The contractor shall not carry out any work on gazetted holiday, weekly holidays and other non working days except when he is specially authorised in writing to do so by the Garrison Engineer. The GE may at his sole discretion declare any day as holiday or non working day without assigning any reasons for such declaration.
12.6 **Fire Precautions**:-
(i) The contractor, his agents, representatives, workmen etc shall strictly observe the order pertaining to fire precautions prevailing within the restricted area.
(i) Motor transport vehicle, if allowed by the authorities to enter the restricted area must be fitted with the serviceable fire extinguisher.
12.7 **Female Searcher**: If the contractor desires to employ female labour or works to be carried out inside the area of factory, depot, park etc, and a female searcher is not borne on the authorised strength of the factory, depot park etc at the time of submission of tender, he shall be deemed to have allowed in his tender for pay and allowances etc for a female searcher (Class IV servant) calculated for the period female labour is employed by him inside that area. If more than one contractor employs female labour during any month and female searcher (s) has / have to be employed in addition to the authorised strength of the factory, depot, park etc, the salary and allowances paid to the additional female searcher (s) shall be distributed on equitable basis between the contractors employing female labour taking into consideration the values and periods of completion of their contracts. The GE's decision in regard to the amount payable on this account by any contractor shall be final and binding.

13. **MINIMUM WAGES PAYABLE**: -
(a) Refer condition 58 of IAFW-2249. The Contractor shall not pay wages lower than minimum wages for labour as fixed by the Government of India / State Govt / Union territory, whichever is higher.
(b) Contractor's attention is also drawn, amongst other things to the 'explanations' to the schedule of minimum wages referred to above.
(c) The fair wages referred to in condition 58 of IAFW-2249 will be deemed to be the same as the minimum wages, referred to above as upto date from time to time.

SPECIAL CONDITIONS (Contd…)

(d) Schedule of minimum wages are not enclosed alongwith tender documents. However contractor shall be deemed to have verified the minimum fair wages payable as on the last due date of receipt of tender.

(e) The Contractor shall have no claim whatsoever, if on account of local factor and or regulations, he is required to pay the wages in excess of minimum wages as described above during the execution of work.

14. HANDING OVER OF SITE: Site for execution of work will be available as soon as the work is awarded. In case it is not possible to make the entire site available on the award of work, the contractor will have to arrange his working programme accordingly. No claim whatsoever, for not giving entire site on award of work and for giving site gradually, will be tenable.

15. ADVANCE ON ACCOUNT AGAINST MATERIALS (APPLICABLE TO CONTRACTS OF VALUE RS 50 LAKHS AND ABOVE)

Add the following in continuation to Para 8 of condition 64 of IAFW-2249:-

'Provided further the contractor may be paid advance on account to the full value of under mentioned materials only brought on site on his furnishing guarantee bonds from schedule bank for the amount of the retention money which should otherwise be recoverable from him under the contract.

(a) Electric fittings & Equipments, Electrical cables/ wires/ fittings/ fixtures, conductors, poles , panels & D G Set
(b) Builders hardware fittings (iron mongery)
(c) Water supply pipe & fittings. Valves etc.
(d) Steel
(e) Cement
(f) Sanitary fittings
(g) Factory made Aluminium windows/ ventilators, Doors, Windows & frames, shutters
(h) All other non-perishable materials which will not loose its identity as decided by the GE.

The Bank Guarantee bonds shall be executed for a period and on a form as directed by the Accepting Officer. The contractor shall further arrange to extend the period of guarantee bond (s) if and when necessary as directed by the Accepting Officer or shall furnish fresh guarantee bond for similar value in lieu.

16 BANK GUARANTEE BOND AGAINST PERFORMANCE SECURITY

16.1 Condition 19.1 of IAFW-2249 and condition 14A.1 of IAFW-1815Z provides for submission of Performance Security by the successful contractor in the form of Bank Guarantee Bonds of Govt Securities, FDR or any other form of deposit stipulated by the Accepting Officer.

16.2 The full amount of Performance Security for each contract shall be given by the contractor in any one of the forms stipulated in Para 16.1 above. Mix of forms for Performance Security for same contract shall not be allowed

16.3 If the Performance Security is provided by the successful contractor in the form of a Bank Guarantee, it shall be issued by Nationalized/Scheduled Indian Bank but its confirmation shall be done only from the Head office of the Bank.

16.4 Form for Bank Guarantee Bond against Performance Security Deposit shall be same as Appendix 2.1 of MES Manual on Contracts 2020 as also available on www.mes.gov.in.

16.5 The period of validity of the Bank Guarantee Bond against Performance Security shall be upto and including the scheduled date of expiry of Defects Liability Period.

16.6 Bank Guarantee Bond in lieu of Performance Security is due for discharge on expiry of Defect Liability Period. If any recovery is outstanding against the contractor, release of Bank Guarantee is subject to compliance of the procedure for effecting recovery/withholding the due amount as stipulated in Condition 67 of IAFW-2249 as amended.

SPECIAL CONDITIONS (Contd…)

17. AVAILABILITY OF LAND FOR ACCOMMODATION OF LABOUR AND CANTEEN :(Refer condition 24 of IAFW-2249)
17.1 No land will be allotted for labour camp and canteen

18. AVAILABILITY OF LAND FOR STORAGE OF MATERIALS/TEMPORARY WORKSHOP/ OFFICE(Refer condition 24 of IAFW-2249):-Delete the following in Para 1 of condition 24 of IAFW-2249, 'In the event of ---- of areas of land allotted to him' and following shall be read in conjunction with condition 24 of IAFW-2249.
"The contractor shall be permitted to store his materials including erection of temporary sheds thereof and also to erect his temporary workshop and the like at the area of land marked on the site plan free of rent if it is on class A1 land. For other types of land he shall be charged a nominal licence fee of Rs 1/- per year or part thereof in respect of each and every separate area of land allotted to him".

19. OFFICIAL SECRET ACT: The contractor shall be bound by the official secret act 1923.

20.0 SCOPE OF CONCILIATION PERTAIN TO CONTRACTS OF VALUE LESS THAN 10.00 CRORE .

20.1 The scope of conciliation shall be restricted to the following types of disputes with financial limits as indicated therein:-
(a) Disputes relating to levy of compensation for delay in completion-actual amount of compensation.

(b) Disputes relating to technical examination of works.

(c) Disputes relating to interpretation of the provisions of the contract with reference to their application to parties.

(d) Disputes relating to non return of Schedule 'B' stores over-issued to the contractor.

(e) Any other dispute having fair chances of being resolved by conciliation and considered fit to be referred to conciliation by the parties.

For item (b), (c), (d) and (e) as stated above, the financial limit shall be rupees two Lakhs or one percent of the contract amount whichever is less for each item.

20.2 Commencement of Conciliation Proceedings.
(a) The party initiating conciliation shall send to the other party a written invitation to conciliate, briefly identifying the subject of the dispute.
(b) Conciliation proceedings shall commence when the other party accepts in writing the invitation to conciliate.
(c) If the other party rejects the invitation, there will be no conciliation proceedings. If the party initiating conciliation does not receive a reply within 30 days from the date on which he sends or within such other periods of time as specified in the invitation, he may elect to treat this as a rejection of the invitation to conciliate and if he so elects, he shall inform in writing the other party accordingly.

20.3 Number of Conciliators. There shall be a Sole Conciliator.

20.4 Appointment of Conciliator. All disputes brought out in Para 42 (a) to (e) above shall be referred to the Sole Conciliator viz. Serving officer not below the rank of Superintending Engineer/Superintending Engineer (QS&C) having degree in engineering or equivalent or having passed final/direct final examination of Sub Division II of Institution of Surveyors (India) to be appointed by the Engineer-in-Chief, Army Headquarters, New Delhi or in his absence the officer officiating as Engineer-in-Chief or Director General of Works if specifically delegated by the Engineer-in-Chief in writing.

20.5 Status of Effect of Settlement Agreement. The settlement agreement signed by the parties as a result of conciliation proceedings shall have the same status and effect as an arbitral award on agreed terms.

SPECIAL CONDITIONS (Contd…)

21. ELECTRIC SUPPLY

(a) **In the case the contractor desires to buy electricity from MES**, he shall be charged for the electric energy consumed at **Rs. 19.65 perkilowatt** hours for KWH lighting and power each.

(b) The above rates are exclusive of duty/tax, if any levied by the local state Govt on the electric energy. Duty and taxes, if any levied by the State Govt and/or electricity under taking and the like on the electricity charges, will be borne by the department.

(c) Electric supply required for works shall be made available by MES. However exact location shall be shown by the GE. The main switch and KWH meter to register the electric energy supplied shall be provided and installed by MES. Contractor shall provide all necessary connections, cables, fittings etc from the main switch in order to ensure a proper and suitable supply of electricity for the execution of work.

(d) MES do not guarantee continuity of supply and no compensation whatsoever shall be allowed for supply becoming intermittent or for breakdown in the system.

(e) The supply shall be AC 3 phase, 3 wire, 400 to 440 volts (within permissible limits) 50 cycles up to 10 KW at point of supply.

(f) GE or his representative shall be free to inspect all the power consuming devices or any electric lines provided by the contractor. Any device or electric lines provided by the contractor, which is not to the satisfaction of the GE, shall be discontinued from the supply if so desired by him.

22.WATER :- (REFER CONDITION 31 OF IAFW-2249)

22.1 Water shall not be supplied by the MES. Contractor shall make his own arrangement for the water at his own cost from the sources outside the defence land, alternatively contractor shall be permitted to bore the tube wells on defence land and no recovery on this account shall be made from the contractor. The contractor's quoted lump sum shall be deemed to include the cost of installation of tube wells. The tube well bore (Except pump) shall be property of Government and the same shall be handed over to the department after completion of work without any extra cost to Govt.

22.2 It will be the responsibility of the contractor to ensure that the water brought to site meets the requirement of clause 5.4 of IS 456. Each source of water shall be got approved from the GE, well in advance of the requirement of water for construction purpose. GE will first approve each source of water before commencement of any concreting/masonry/plaster work. Thereafter, the water from each source will be tested at least at an interval of every three months. The sample of water from the sites will be got collected by the GE and sealed by him. All the testing charges including the arrangement for forwarding the sample to the Recognised/Approved Laboratory/ NABL accredited lab will be borne by the contractor.

22.3 Work done with water arranged by the contractor and not meeting the requirement of IS 456 is liable for rejection.

23. QUALIFIED TRADESMEN (APPLICABLE FOR WORKS COSTING RUPEES ONE CRORE OR MORE):-

In compliance with the condition 26 of IAFW-2249 (General Conditions of Contracts), the contractor shall employ skilled/semi skilled tradesmen who are qualified and possessing certificate in particular trade from Industrial Training Institute (ITI)/National Institute of Construction Management and research (NICMAR)/ National Academy of Construction (NAC) Hyderabad/ Construction Industry Development Council (CIDC)/ Similar reputed and recognized Institutes by State/Central Government, to execute the works of their respective trade. The number of such qualified tradesmen shall not be less then 25% of total skilled/semi skilled tradesmen required in each trade. The contractor shall submit the list of such tradesmen along with requisite certificates to Garrison Engineer for verification and approval. Notwithstanding the approval of such tradesmen by GE, if the tradesmen are found to have inadequate skill to execute the work of their trades, leading to un-satisfactory workmanship, the contractor shall remove such tradesmen within a week after written notice to this effect by the GE and shall engage other qualified tradesmen after prior approval of GE. GE's decision whether a particular tradesmen possess requisite qualification, skill and expertise commensurate with nature of work shall be final and binding. No compensation whatsoever on this account shall be admissible.

SPECIAL CONDITIONS (Contd…)

24. **TESTING:-** The cost of all tests shall be borne by the contractor. In case any implement and equipments for the testing at site of work are required, the same shall also be provided by the contractor without any extra cost to the Govt.

25. RETENTION MONEY/COMPENSATION FOR DELAY:-

(a) Refer condition 64 and 50 of IAFW-2249.

(b) For the purpose of calculating retention money under condition 64 of IAFW-2249, and compensation for delay under condition 50 of IAFW-2249, the value of contract as revised by price variation in terms of special condition mentioned herein tender under modified condition 63 of IAFW-2249 shall be taken into account.

26. **LOCATION OF BUILDING AND WORKS** : There may be some changes in locating / siting of building shown in site (layout) plan(s) to suit local conditions and / or departmental requirements. The contractor shall have no claim what-so-ever consequent to such changes in the location / siting of works.

27. **WATCH / LIGHTING** : The contractor shall at his own cost take all possible precautions to ensure safety of life and property by providing necessary fencing, barrier, light, watchmen etc., during the progress of work and as directed by the Engineer-in-Charge.

28. RE-IMBURSEMENT/REFUND ON VARIATION IN TAXES DIRECTLY RELATED TO CONTRACT VALUE :-

(a) The rates quoted by the Contractor shall be deemed to be inclusive of all taxes (including GST on materials, GST on Works Contracts, Labour welfare cess/tax etc.), duties, Royalties, Octroi & other levies payable under the respective statutes. No reimbursement/refund for variation in rates of taxes, duties, Royalties, Octroi & other levies and/or imposition/abolition of any new/existing taxes, duties, Royalties, Octroi & other levies shall be made except as provided in sub para (b)(i) here-in-below.

(b) (i) The taxes which are levied by Government at certain percentage rates of Contract Sum/Amount shall be termed as “taxes directly related to Contract value” such as GST on materials, GST on Works Contracts, Labour welfare cess/tax and the like but excluding Income tax. The tendered rates shall be deemed to be inclusive of all “taxes directly related to Contract value” with existing percentage rates prevailing on the bid submission end date. Any increase in percentage rates of “taxes directly related to Contract value’ with reference to prevailing rates on the bid submission end date shall be reimbursed to the Contractor and any decrease in percentage rates of “taxes directly related to Contract value” with reference to prevailing rates on the bid submission end date shall be refunded by the Contractor to the Govt/deducted by the Government from any payments due to the Contractor. Similarly imposition of any new “taxes directly related to Contract value” after the bid submission end date shall be reimbursed to the contractor and abolition of any “taxes directly related to Contract value” prevailing on the bid submission end date shall be refunded by the Contractor to the Govt/deducted by the Government from the payments due to the Contractor.

(ii) The Contractor shall within a reasonable time of his becoming aware of variation in percentage rates and/or imposition of any further “taxes directly related to Contract value” give written notice thereof to the GE stating that the same is given pursuant to this Special Condition, together with all information relating thereto which he may be in a position to supply. The Contractors shall also submit documentary proof/information as the GE may require.

(iii) The Contractor shall, for the purpose of this condition, keep such books of account and other documents as are necessary and shall allow inspection of the same by a duly authorized representative of Government, and shall further, at the request of the GE furnish, verified in such a manner as the GE may require any documents so kept and such other information as the GE may require.

(iv) Reimbursement for increase in percentage rates/imposition of “taxes directly related to contract value” shall be made only if Contractor necessarily and properly pays additional “taxes directly related to contract value” to the Government, without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Government Authority and submits documentary proof for the same as the GE may require.

SPECIAL CONDITIONS (Contd…)

29. APPROACHES : The Contractor shall make arrangements for and provide at his own cost all temporary approaches, if required to the site (s), after obtaining approval in writing of the GE to the layout of such approaches.

30. TESTING OF MATERIALS:-

(a) **(i) A LEVEL TESTS FOR WORKS COSTING 100 LAKHS ANDABOVE** :-
The contractor shall set up site laboratory for testing of materials (except Sch 'B' materials) for 'A' level tests as listed in Appendix - B hereto. The contractor shall arrange all equipment / machines for the tests specified in Appendix - B as A level tests at his own cost with prior approval of GE. This cost shall be included in the lump sum costs quoted by the contractor. The contractor shall employ a competent technical representative as approved by the GE for the purpose of testing and all such tests shall be carried out in the presence of Engineer-in-Charge. The successful tests result thereof shall be recorded and signed jointly by the contractor and the Engineer-in-Charge. The charges for these test i.e. A level tests carried out in site laboratory of the contractor shall not be recovered. In case, the contractor has not set up the site laboratory and the tests are carried out in Zonal or any other laboratory setup / approved by GE, the recovery shall be made at rates applicable i.e. as given in Appendix - B.

(ii) A LEVEL TESTS FOR WORKS COSTING UPTO 100 LAKHS : The contractor may set up site laboratory at his option for works costing up to 100 lakh. The other stipulations will be same as specified in preceding para (aa) (i). However in case the contractor has not set up the site laboratory and tests are carried out in Zonal or any other laboratory approved / setup by GE, the recovery shall be made at rates applicable i.e. as given in Appendix - B.

(b) **B & C LEVEL TESTS** : For tests of 'B' and 'C' level as indicated in Appendix - B, the contractor shall provide all facilities for testing of materials at Zonal laboratory / Govt approved laboratories or test house / Engg colleges at his own cost. The lump sum / rates quoted by the contractor shall deemed to be inclusive of these tests. The rates of various tests conducted in Laboratory of MES are indicated in Appendix - B. The contractor shall bear the actual charges of 'C' level tests (to be done in labs other than MES labs) irrespective of rates indicated in Appendix - B. Wherever it is convenient to get 'B' level test done at approved test house / Engg college, the same can be done at the cost of the contractor and no separate recoveries will be made by the department for the same.

(c) The recoveries on account of testing charges wherever applicable shall be effected from the running account payments due to the contractor payable after completion of the respective tests or whenever the test is due whichever is earlier.

31.DAMAGE TO EXISTING STRUCTURES: Any damage to the existing structures, any existing road etc, during the execution of work shall be made good by the contractor at his
own expense. Rectification, replacement, making good and touching up etc. shall be carried out, conforming to the materials and workmanship originally provided and to the satisfaction of the Engineer-in-Charge. In case of any dispute on this account, the decision of the GE shall be final, binding any conclusive.

32. DEFECT LIABILITY PERIOD
Refer Condition 46 of IAFW 2249 General Condition of Contract. The defect liability period for this work is 24(Twenty-four) calendar months after the work have been handed over to Government.

Signature ofContractor **AAD(Contracts)**
Dated: **ForAccepting Officer**

PARTICULAR SPECIFICATIONS

1. GENERAL

1.1 The following specifications shall be read in conjunction with the General Conditions of Contract (IAFW-2249) and IAFW-2159 including Errata/Amendments thereto. If any provision in the Specifications is at variance with that of the aforesaid documents, the former shall be deemed to take precedence there over.

1.2 The work under this contract shall be carried out in accordance with Schedule 'A', the Particular Specifications, drawings, General specifications & other provisions in MES SSR Part-I of 2009 & Part-II of 2020 (herein after called MES Schedule) read in conjunction with each other.

1.3 The term 'General Specifications' referred to herein before as well as referred to in IAFW-2249 (General conditions of contracts) shall mean the specifications contained in the MES Schedule.

1.4 General Rules, specifications, special conditions and all preambles in the MES Schedule shall be deemed to apply to the work under this contract, unless mentioned otherwise in these documents, in which case, the provision in these documents shall take precedence over the aforesaid provisions in the MES Schedule. Where specifications for any item of work are not given in MES Schedule or in these particular specifications, specifications as given in relevant Indian standard or code of practice shall be followed.

1.5 (a) Rate quoted for a particular item and/or lump sum quoted by the tenderer shall be deemed to include for any minor details/ items of work and /or construction which are obviously and fairly intended and which may not have been included in these documents but which are essential for the execution and entire completion of the work.

(b) Decision of the Accepting Officer as to whether any minor details of work and/or construction is obviously and fairly intended to be included in the contract or not shall be final, conclusive and binding. However, some minor details, items which shall be deemed to be essential for execution and entire completion of work are detailed as under:-

(i) Reinforcement for any RCC member not indicated in the drawings but required as per structural or code requirement.

(ii) Dwarf wall in situations like verandah, passage, ramp, platform etc not indicated in drawings.

(iii) Lintels over doors, windows ward robes/cup board and opening not indicated in drawings.

(iv) Fittings to doors, windows, ward robe/cup board and such other built-in furniture items not shown.

(v) PCC benching at the junction of chajjas and walls not shown in drawings.

(vi) Coving at junctions of wall plaster with dado/skirting.

(c) In all the above and similar cases, the details indicated else where in the drawings which are similar or near to the missed out item of work shall be followed. In the absence of any other similar or near similar details, minimum essential requirement for completion of the work from structural and utility point of view shall be deemed to be included in the lump sum quoted. In the event of any dispute, decision of the Accepting Officer thereon shall be final, conclusive and binding.

2. SCOPE OF WORK

(a) The contract includes for the full, final and entire completion of all works described in BOQ (Schedule 'A') included in the General Summary specified in these particular specifications and shown in the drawings forming part of this tender.

PARTICULAR SPECIFICATIONS (Contd…)

(b) Probable sources of local materials shall be ascertained by the contractor before quoting. However, the samples shall be got approved by the contractor from the GE before bulk procurement of the material. The contractor shall bring material conforming to the specifications from any available source.

(c) In appendices/Annexure to particular specifications, list of manufacturers of items have been given. Contractor shall obtain these items from one of the approved manufacturer only. However, it will not absolve the contractor from ensuring that the products supplied are conforming to the relevant IS in all respects before procuring the same for approval by the GE and before procuring bulk product for incorporation in work.

3. EXCAVATION AND EARTH WORK

3.1 EXCAVATION IN GENERAL

3.1.1 **PREPARATORY WORKS:** Before setting out layout of buildings and commencing the excavation and other works the contractor shall carryout preparatory works such as removal of grass, vegetation etc, trimming surface dressing of the area as per clause 3.6 and 3.10 of MES Schedule Part-I to the entire satisfaction of Engineers-in-Charge. The 'Area' referred to implies the entire building plot extending upto 3m all around the outer edge of plinth protection/ramps/aprons of the building. The lump sum buildings under Schedule 'A' Part-I shall be deemed to include cost of surface excavation.

3.1.2 **EXCAVATION-GENERAL**

(a) Excavation and earth work involved in Schedule 'A' Part-I has been priced at the rate for Earth work in excavation by mechanical means (Hydraulic excavator) in **soft soil.** If during excavation of work any other type of strata is met with the same shall be brought to the notice of GE in writing, who shall after verification, order the same through deviation.

(b) The measurement of excavation shall be as given in MES Schedule (Part-II). The lumpsum quoted by the contractor for Sch A, Part –I shall be considering the site as levelled as per details given on drawings. Any extra earthwork required to level the site to confirm to drawings shall be measured and paid extra under Sch A, Part-II.

(c) Boulders and stones obtained from excavation shall be sorted out and neatly stacked as directed by Engineer-in-Charge, without any extra cost to the Govt and shall become contractor's property for which he will be charged at **Rs. 1192.13** per Cum as per the method of measurements described vide preamble 3.1 of MES Schedule Part III 2020 (Earthwork section).

(d) In case hard rock is met with, the contractor shall immediately notify the same to GE under intimation to Accepting Officer and prior approval of Accepting Officer shall be obtained by the GE before ordering work of excavation in hard rock on the contractor. Where hard rock is met with during the excavation, foundation design is likely to be changed which may entail delay in finalisation of drawings. The contractor shall mobilise his resources accordingly in consultation with GE and no claims of what so ever nature for delay in finalisation of revised details shall be entertained. Decision of the Accepting Officer in this regard will be final and binding.

(e) In case timbering to excavation is required and specifically ordered by the GE in writing, it shall be paid as deviation.

(f) Bailing/pumping of water where required shall be carried out as described in clause 3.17 of MES Schedule. The cost of such work as may be necessary shall be deemed to be included in the contractor's lump sum.

PARTICULAR SPECIFICATIONS (Contd…)

(g) Surface excavation shall not be more than 30cm deep and averaging 15cm deep and getting out. The GL (Ground Level) marked in drawings shall be that obtained after surface excavations.

(h) During excavation in case **HARD ROCK** is encountered; the quantity shall be ascertained through a BOO ordered by the Accepting officer. Hard rock met with during excavation shall be entered in the measurement book duly signed by the GE and the contractor and irrespective of the quality of rock whether approved for re-use or not, the complete quantity of hard rock as obtained in excavation shall become the property of contractor for which recovery at the rate of **Rs.1500.57 per cubic metre** (Solid contents) shall be made from contractor's advance on account payment.

(j) The hard rock so obtained may be allowed to be used in the work by the GE provided it meets the specified requirements. Location (s) etc where it is to be used (if allowed to be used) shall also be as decided by the GE whose decision shall be final and binding. If hard rock obtained from excavation is not fit to be used in the work the contractor shall remove the same outside the MD land without any extra cost to the Government.

3.1.3 Surplus excavated soil if any (i.e. unused soil left over after filing in trenches, plinth, under floors and formation of earthen traverse) shall be removed to a distance exceeding 250 metres but not exceeding 500 metres deposited (unless specific lead is mentioned in Sch A Notes/BOQ for particular item), spread and levelled as directed. Lump sum tendered by the contractor shall include for above lead for removal of surplus soil.

3.1.4 Necessary removal of hard rock and stacking exceeding 250 meters but not exceeding 500 meters distance as directed by Engr-in-Charge shall be deemed to be included in the lump sum cost.

3.2 SITE CLEARANCE AND SURFACE DRESSING

After completion of the work and before handing over any building, the area around as defined in clause 3.1.1 to 3.1.3 hereinbefore shall be dressed and cleared as directed to the entire satisfaction of the Engineer-in-Charge.

3.3 EXCAVATION IN TRENCHES AND OVER AREAS

3.3.1 Excavation in trenches and over areas shall be carried out as described in Para 3.13 of MES schedule Part-I. Bailing and pumping of water, if required will be done as described in Para 3.17 of MES schedule Part-I. No extra payment shall be admissible for the above-mentioned operation. If however, an inflow of water in to excavation is caused by powerful springs or river seepage, broken water mains or drains and the like (other than those broken through the contractor's negligence) the contractor shall inform the GE. If in the opinion of the GE the inflow is due to any of these causes, he shall issue instructions in writing as to the method to be employed in clearing the excavation of water and additional payment shall be allowed as deviation order.

3.3.2 Depth of foundation shown on drawing is after surface dressing/made up ground level achieved through site development.

3.4 FILLING IN TRENCHES, PLINTH AND UNDER FLOORS

3.4.1 Filling in foundation trenches, plinth, under floors, and in other situations shall be done with approved earth. It shall be ensured that turf, peat vegetable and other organic and deleterious matter is removed from the soil used for filing. Black cotton soil shall not be used for filling. Filling in general shall be done as specified in clause 3.19 & 3.21 of MES schedule Part-I, and filling in trenches for pipes, drains, cables etc shall be done as per clause 3.20 of MES schedule Part-I and shall be all as specified in drawings.

3.4.2 Earth obtained from excavation (Except soil obtained from surface dressing) shall be used for filling only after approval of engineer in charge. Soil not approved for filling surplus to the requirement shall be removed to a distance exceeding 250 metres and not exceeding 500 metres spread & levelled.

PARTICULAR SPECIFICATIONS (Contd…)

3.4.3 No charge shall be levied against contractor for use of approved soil obtained from excavation.

3.4.4 If the soil obtained is not adequate for filling in trenches, plinth and under floor etc, additional approved earth required for filling shall be obtained by the contractor from outside the MOD land at his own cost and no extra payment on this account shall be admissible. However for the lump sum quoted the entire soil obtained from excavation shall be considered suitable for filling. In case black cotton soil is met with the same shall be ordered as deviation and black cotton soil shall be removed as specified hereinbefore and approved earth for filling shall be obtained from outside MD land.

3.4.5 In case of deviation, approved earth shall be brought by Contractor from outside MD land and shall be priced at the rate for rough excavation in soft/loose soil not exceeding 1.5 metres in depth and getting out and removal to a distance exceeding 1.5 Km but not exceeding 5 Km adjusted by the contractor's percentage irrespective of actual lead/cost involved.

3.5 **HARD CORE**

3.5.1 The material for hardcore shall be of granite best quality and as per sample kept in GE's office. These materials may be from any one of the quarries approved by GE.

3.5.2 Hard core shall be of stone aggregates broken to gauge not exceeding 63mm, deposited, spread and levelled in layers not exceeding 15cm thick and well watered, rammed to a true surface and compacted. The thickness of the hard core specified or as indicated in the drawings, is the thickness after consolidation and where not specifically mentioned, it shall be 100mm consolidated thickness. After consolidation, the top surface shall be blinded with a thin layer of approved earth (Moorum) to fill interstices, watered and well rammed. Cost of moorum to be brought from outside MD land shall also be deemed included in lump sum.

4.00 PRECONSTRUCTION ANTITERMITE CHEMICAL TREATMENT:-

Antitermite treatment shall be provided all as specified in clause 3.26 of MES SSR Part-I except that termite mound treatment if required to be provided shall be ordered as deviation.

4.1 The work of A.T.T. shall be carried out through a specialist firm. The firm shall meet the following requirements to be approved by the GE before antitermite treatment work is taken in hand:-

(a) The firm executing the work should have valid licence for carrying antitermite treatment as per clause 13 of insecticide Act 1968 (as amended from time to time)

(b) The firm shall be a member of IPCA or should have a satisfactory record of performance

(c) The person carrying out antitermite treatment should be qualified as per rule 10 promulgated under Insecticide Rule 1971.

4.2 Daily record shall be maintained duly signed by the contractor and Engineer-in-Charge giving the following details:-

(i) Location of work done.

(ii) Name of chemical and its concentration used.

(iii) Items of the treatment work done (bottom and sides of excavation of masonry wall), back fill, under plinth protection, top of filling and along external perimeter of building/plinth protection) and area covered under each item.

(iv) Quantity of chemical required for work done during the day and quantity of chemical actually used.

4.3 The contractor shall give a service guarantee of 10 (TEN) years for the antitermite treatment. The aforesaid guarantee period of 10 year shall reckon from the certified date of completion of work.

PARTICULAR SPECIFICATIONS (Contd…)

4.4 The amount calculated as per table given below shall be retained out of the contractor's bill as security deposit for the guarantee period of 10 years which shall be refunded to him after expiry of the guarantee period satisfactorily. Alternatively, contractor may give a separate interest bearing security deposit or bank guarantee bond to GE valid for 10 years for the sum mentioned above. Security deposit/bank guarantee shall be released after expiry of defect liability period of 10 years as specified. The contractor may at his discretion furnish fixed deposit receipt for the above amount in lieu of the security for guarantee period mentioned:-

Amount of anti-termite treatmentat contract rate		**Amount to be retained from contractor's dues rounded to nearest thousand Rupees**
(i)	Up to Rs 50.00 lakh	2% of amount subject to minimum of Rs 5000/- and enhanced by 25%
(ii)	Over Rs 50.00 lakh and up to Rs 100.00 lakh	Rs 1.00 lakh + 1.50% of amount exceeding Rs 50.00 lakh and enhanced by 25%
(iii)	Over Rs 100.00 lakh and up to Rs 500 lakh	Rs 1.75 lakh + 1% of amount exceeding Rs 100.00 lakh and enhanced by 25%

4.5 Should the GE at any time during construction or prior to the expiration of the guarantee period of 10 years find that the buildings have been infected with termites, the contractor shall on demand in writing from GE, specifying the building(s) complained of, notwithstanding that same may have been inadvertently passed, certified and paid for, forthwith undertake to carryout such treatment as may be necessary to render the said building(s) free from termite infection at his own expenses for the guarantee period of 10 years from the certified date of completion, and in event of his failing to do so within a period to be specified by the GE in his demand aforesaid, the GE may undertake such treatment at the risk and expense, in all respect, of the contractor. The liability of the contractor under this condition shall not extend beyond the guarantee period of 10 years unless the GE had previously given notice to the contractor. The security deposit referred to in condition 4.4 herein before may be refunded to the contractor after the expiry of the period of ten years from the certified date of completion by the GE provided always that the contractor shall first have been paid the final bill and have rendered a no demand certificate (IAFW-451). Condition 10.46 and 68 of the General Conditions of Contracts (IAFW-2249) shall be deemed to be amended to the extent mentioned above.

4.6 Chemicals for antitermite treatment shall be purchased by the contractor from the manufacturers or their authorised agents only and brought to site in manufacturers sealed containers. Name of the authorised agent shall be verified from the manufacturer by the GE before approval. Only ISI marked chemicals shall be approved for use.

4.7 Regarding chemical for antitermite treatment, provisions of special condition No. 11 (Record of materials) shall be strictly complied with.

4.8 The antitermite treatment chemical when brought to site shall be inspected by the Engineer-in-Charge and the quantity brought to site shall be recorded in the measurement book and signed by the contractor and the Engineer-in-Charge as a check to ensure that the required quantity has been brought to site for incorporation in the work.

4.9 The chemical brought to site shall be stored as directed by the Engineer-in-Charge and those already recorded in the MBs shall be marked for identification.

4.10 Reference clause 3.26.1 of MES SSR Part-I. Chemical for antitermite treatment shall be chloropyriphos (20% EC).

PARTICULAR SPECIFICATIONS (Contd…)

4.11 Chemicals shall be stored carefully at site, seals of containers should be broken only in the presence of the Engineer-in-Charge (EIC). Empty containers should be got removed off the site promptly. If on any particular day the contents of one full container could not be used in the work, the container should be got sealed at the end of the day in the presence of the EIC and opened when required, also in the presence of the EIC.

4.12 The Engineer-in-Charge (EIC) should ensure that paid vouchers are produced by the agency executing the work, for the full quantity of the chemical required and brought to site and a record of such vouchers should be kept by the EIC.

4.13 Tests may be carried out in a recognised laboratory or test house / NABL accredited labat the discretion of the GE, of the chemical brought by the agency executing the work, to satisfy that spurious materials are not being used.

4.14 The EIC should work out on his own cost, the total requirements of chemicals required for the work as per specifications and ensure that full quantity is brought to site and used in the work, for this purpose, entries should be made in the Measurement Book (not for payment-not to be abstracted) indicating the brand name, quantity brought etc and signed by the EIC as well as representative of the agency executing the work.

4.15 A copy of IS-6313 (Part-II) should be available at site with every Engineer-in-Charge who gets anti termite treatment works executed.

5.00 CONCRETE WORK

5.1 **TYPE OF CEMENT:-** Cement shall be Ordinary Portland Cement (OPC) Grade 43 conforming to IS 8112-1989 (Latest revision) or Portland Pozzolana Cement (PPC) confirming to IS-1489-1991 Part-I. If the type of cement is mentioned in structural drawings then same shall be used irrespective of type of cement specified in these specifications.

5.2 The following checks and procedures shall be followed before the cement supplied by the contractor is accepted and is approved for incorporation in the works.

5.2.1 **PROCUREMENT**:

(a) The cement supplied by the contractor will be procured from main producers of cement. However where estimated requirements of cement is less than 1200 Bags and in all term contracts, contractor can procure cement from authorised distributors / dealers of the approved firms but the contractor will have to submit test certificates of the batch issued by the main producers. The particulars of the manufactures of cement along with the date of manufacture shall be obtained from the contractor for every lot of cement separately. The documents in support of the purchases of cement shall be verified by the site staff and GE. **The main producers of cement are given in Appendix A.** In addition to above manufacturers cement can also be procured from manufacturer as approved by E-in-C's branch even after the date of submission of bid.

(b) The contractor shall furnish the particulars of the manufacturers of cement alongwith the date of manufacture to the GE for every lot of cement separately. The documents in support of the purchases of cement shall be verified by the site staff and GE. The cement so brought shall be fresh and in no case older than 60 days from the date of manufacture. Before placing the order for cement by the contractor, he shall obtain written approval from the GE regarding name of manufacture, quantity of cement etc. Cement shall be procured for requirement of not more than two months at a time. The cement shall be consumed in the work within three months after receipt. Cement shall confirm to the requirements of IS specification and each bag of cement shall bear relevant ISI Mark. The weight of eachconsignment shall be verified by the GE and recorded. The content of cement shall be checked at random to verify the actual weight of cement per bag.

PARTICULAR SPECIFICATIONS (Contd…)

5.2.2 USE OF PPC: For using PPC, the following requirements should be met:-

(a) PPC should meet the strength criteria of 43 Grade OPC as laid down in IS- 8112-1989

(b) The minimum period for curing and striking off form work shall be as under in lieu of that specified in clause 4.11.6.3 of MES Schedule Pt-I :-

(i) Curing:- Structural RCC work, PCC work, Plastering etc. : 14 days
(ii) Striking off Formwork from Walls, columns and vertical sides of beams. : 3 to 4 days
(iii) Slabs (Props left under) : 8 days
(iv) Beams soffits (Props left under) : 9 days
(v) Removal of props to slab/beam
(a) Spanning upto 6 M span : 16 days
(b) Spanning beyond 6 M span : 22 days
(vi) For cantilever portion of slab form work with support shall be retained until the completion of the casting of the entire frame work of the building.

(c) Mixing of OPC & PPC shall not be allowed in the same building except for plaster and mortar. However with strict control of the ground executives, different buildings can have different types of cement.

(d) While procuring PPC, the following requirements are to be ensured and certificate to the effect is to be obtained by the contractor for each batch from the manufacturer and the same shall be submitted to the GE for approval to procure PPC:-

(i) The quality of fly ash is strictly as per IS-1489 (Part-I) - 2015.

(ii) Fly ash is inter-ground with clinker not mixed with clinker.

(iii) Dry fly ash is transported in closed containers and stored in silos. Only pneumatic pumping should be used.

(iv) Thefly ash received from thermal power plants using high temperature combustion above 1000oC should be used.

(e) For using PPC non availability of OPC shall not be insisted upon. Also no price adjustment shall be made for use of PPC.

(f) In cold climate reasons, where temperature is lower than 15^0C, only OPC shall be used. (However, during working seasons in cold climate regions, where OPC is not available use of PPC may be permitted provided it shall be ensured that while using PPC atmospheric temperature should be more than 15^0C.

5.3 **TESTING OF CEMENT**

5.3.1 The contractor shall submit the manufacturer's test certificate in original alongwith test sheets giving the results of each physical test as stipulated in accordance with relevant IS provision and the chemical composition of cement or authenticated copy thereof duly signed by the manufacturer with each consignment, as per the following IS provision :-

(a) Method of sampling hydraulic cement as per IS-3535-1986.
(b) Methods of physical test for hydraulic cements as per IS-4031.
(c) Method of chemical analysis of hydraulic cement as per IS-4032-1985.

The test certificate and test sheet shall be furnished with each batch by manufacture. The Engineer-in-Charge shall record these details in cement acceptance register to be maintained by him which will be signed by Junior Engineer (Civil), Engineer-in-Charge, Garrison Engineer and the contractors as given in the format hereinafter for verification.

PARTICULAR SPECIFICATIONS (Contd…)

5.3.2 The GE shall carryout independent testing as per the tests mentioned in the 'CEMENT SUPPLY / ACCEPTANCE FORM' of random samples of cement drawn from various lots. The testing shall be carried out through national test house, SEMT wing of CME, Regional Research Laboratories, Govt approved Laboratories, Zonal Laboratories, NABL accredited lab as per IS-3535-1986 (method of sampling hydraulic cement) and IS-4031 (method of physical test for hydraulic cement) and IS-4032-1985 (method of chemical analysis of hydraulic cement) referred to above. The cement shall be tested both for physical and chemical properties as per relevant IS codes. In case the cement is not of requisite standard despite manufacturer's test certificate, the contractor shall remove the total consignment from the site at his own cost after written rejection order of the consignment by the GE.

5.3.3 The random samples as per relevant IS shall be selected by GE in accordance with IS 3535-1986 for independent or additional tests before carrying out testing. The cement shall be tested within one week on supply but before in-corporation of the same in works. Cost of testing transportation and materials used in testing etc shall be borne by the contractor and no extra claim whatsoever will be admissible. The record of such samples selected by the GE for testing shall be properly maintained in the 'Cement Testing Register' giving cross reference to relevant consignment of cement and quantity received etc.

5.3.4 Cost of transportation of samples to the approved laboratory/test house and all testing charges including cost of sample shall be borne by the contractor Testing of cement shall be carried out in Govt approved lab and accepted only on satisfactory test results of the tests. All the expenditure for the testing of cement in Govt approved laboratory shall be borne by the contractor.

5.3.5 The contractor shall be required to set up adequate testing facilities at site to the entire satisfaction of Garrison Engineer for conducting the tests as per IS codes referred to hereinbefore for the samples collected from the lot brought at site. These tests shall be carried out within 7 days of receipt of cement at site. The tests can alternatively be carried out at the Zonal/Project laboratory, NABL accredited labor any other recognized laboratory so designated by GE at Contractor's expense.

5.3.6 The contractor shall submit original purchase vouchers for the total quantity of cement supplied under each consignment to be incorporated in the work. All consignments received at the work site shall be inspected by the GE alongwith the relevant documents to ensure the requirements as mentioned hereinbefore, before acceptance. The original purchase vouchers and the test certificates shall be verified for subject contract and defaced by the Engineer-in-Charge and kept on record in the office of the Garrison Engineer duly authenticated and with cross reference to the consignment/control number recorded in the Cement Acceptance Register. The cement acceptance register shall be signed by the JE (Civil), Engineer-in-Charge, GE and the contractor. The contractor shall maintain schedule of supply of cement for each consignment.

5.3.7 The Accepting Officer may order a board of officers at different stages of work (6 stages as fixed by GE) for conducting random check of cement and verification of connected documents during the currency of contract. The cost of samples, transportation and tests, if any, shall be borne by the contractor. The GE will intimate to the Accepting Office as and when the consignment of cement is received at site of work for ordering the requisite Board of Officers. However, it should be kept in mind that it should not hamper the progress of works at any stage.

5.4 **STORAGE/ACCOUNTING/PRESERVATION OF CEMENT**.

5.4.1 Cement shall be stored in covered godown over dry platform at least 20 cm high in such a manner as to prevent deterioration due to moisture or intrusion of foreign matter. In case of store room, the stack should be atleast 20 cm away from floors and walls. The stacking of cement shall be done as specified in relevant IS. The storage, accounting and preservation of cement supplied by the contractor shall be done as per standard engineering practice till the same is incorporated in the work and the cost of the same shall be deemed to be included in the unit rate/amount quoted by the tenderer. The Engineer-in-Charge shall inspect once a day to verify that cement lying at site is stored, accounted, preserved and maintained as per the norms.

PARTICULAR SPECIFICATIONS (Contd…)

5.4.2 Stacking of cement shall be done as per relevant IS and as under :-

(a) Each cement consignment shall be stacked separately and removal shall be made on the basis of `First in First out'.

(b) Adequate top cover will be provided.

(c) Stacks in no case shall be higher than 12 bags. The maximum width of each stack shall be 3.00 m. If the stack is to be more than 7 or 8 bags high, the bags shall be arranged in header and stretcher fashion; i.e alternatively lengthwise and cross wise so as to tie the piles together and avoid danger of topping over.

(d) Adequate space shall be kept between two stacks.

(e) Tested and untested cement are segregated and stored with distinct identification.

5.4.3 Cement godown shall be provided with two locks on each door. The key of one lock at each door shall remain with the Engineer-in-Charge or his representative and that of the other lock with the contractor's authorised agent at site of works so that cement is removed from the godown only according to daily requirement with the knowledge of both the parties. During the period of storage, if any cement bag(s) found to be in damaged condition due to whatsoever reason, the same shall be removed from the cement godown on written orders of the GE and suitable replacement for the cement bag(s) so removed shall be made and no claim, whatsoever, shall be admissible on this account.

5.4.4Cement shall be removed from the store only according to daily requirement with the knowledge of both parties and daily consumption of cement shall be recorded in cement consumption register which shall be signed by the Engineer-in-Charge and the contractor.

5.4.5 In case the consumption of cement as per cement consumption register is found to be more than the estimated quantity of cement due to whatsoever reason, the contractor shall not have any claim whatsoever for such excess consumption of cement.

5.5 **SCHEDULE OF SUPPLY**: The GE in consultation with the contractor will work out the schedule of procurement in accordance with agreed CPM chart and ensure that it is adhered to. This schedule shall be vetted by the CE's office from time to time. The contractor shall procure the cement timely. The contractor will forfeit his right to demand extension of time if the supply of cement got delayed due to his failure in placing order in time to the manufacturer.

5.6 **MEASUREMENTS AND PAYMENT OF CEMENT.**

5.6.1 The entire quantity of cement shall also be suitably recorded in the Measurement Book for record purposes as "Not to be abstracted" before incorporation in the work and shall be signed by the Engineer-in-Charge and the contractor duly checked by GE.

5.6.2 The payment shall only be allowed after production of original purchase vouchers, certified copies of test certificates from manufacturer for each consignment and results of testing carried out in laboratory on receipt of cement (7 days compressive test) are found satisfactory after testing as specified hereinbefore. Cement shall be paid as material lying at site as per condition 64 of IAFW-2249. Rate of cement given in MES schedule shall be applicable for cement irrespective of grade of cement specified for use in the work.

PARTICULAR SPECIFICATIONS (Contd…)

CEMENT SUPPLY & ACCEPTANCE REGISTER

1. CA No. & Name of Work
2. Control No.*
3. Name of Manufacturer/Brand Name/Gde of Cement (a) Manufacture …….. (b) Brand …… (c) Grade …….
4. Quantity of Cement & Lot No./Week No. (in bags): Qty …… (b) Lt No./Week No. ……
5. Manufacturer's Test Certificate No. ……
6. Random Test Details (a) Physical test Report from ……. Vide their letter No. ……. (Name of approval Lab/Engg College)

   (c) Chemical test report from …….. vide their letter No. ……. (Name of approval Lab/Engg College)
7. Details of Physical & Chemical properties.

	Physical Requirement (As per IS 4031)										Chemical Requirements (As per IS 4032)							
	Specific Surface Area (M^2/kg)	Soundness by Le Chatellar	Soundness by Auto Clave	Initial Setting Time (Minutes)	Final Setting Time (Minutes)	Compressive Strengths (Mpa)			Temp during testing ^0C	Standard Consistency (%)	Lime Saturation Factor (Ratio)	Aclumina Iron Ratio (Ratio)	Insoluble Resdue (%)	Magnesium (%)	Sulphuric Anhydride (%)	Loss on ignition (%)	Alkailes (%)	Chlorides (%)
						03 Days	07 Days	28 Days										
As per relevant IS																		
As per manufacturer's test certificate																		
As per random test certificate																		

Remarks with Signature Accepted / Rejected

Contractor Junior Engineer Engineer-in-charge Garrison Engineer

Remarks of BOO / Inspecting Officer / CWE

**** To be allotted serially by GE Consignment wise***

PARTICULAR SPECIFICATIONS (Contd…)

5.7 **FINE AGGREGATE**: Fine aggregate for concrete work shall be river sand conforming to materials specifications and grading within the limits of grading zones I to III all as specified in Clause 4.4.1 to 4.4.7 of MES schedule Part I. Sand shall be obtained from source as approved by GE. In the alternative crushed stone sand conforming to grading and other characteristics Zone I to III and other requirement in all respect may also be used without any price adjustment. The crushed stone sand shall be from clean hard stone / aggregate to required grading and shall be free from all deleterious materials. Sand obtained from quarry dust by washing and screening method shall not be used.

5.8 **COARSE AGGREGATE**: Coarse aggregate for all concrete work ie PCC/RCC shall be crushed or broken stone(granite) aggregate and shall be as specified in Clause 4.4.1 to 4.4.7 of MES schedule Part I. The same shall be obtained from approved quarries.

5.9 **GRADING OF COARSE AGGREGATE**: Graded aggregate of nominal sizes given hereunder, shall be used, unless specified otherwise.

(a) Reinforced cement concrete

(i) For structural elements of depth/thickness more than and including 100mm-20mm

(ii) For structural elements of depth/thickness less than 100mm-12.5mm

Note: However, in no case the nominal size of Aggregate shall be greater than one fourth the minimum size of the member.

(b) Plain concrete

(i)	Under 30mm thickness	:12.5mm
(ii)	30 to 80mm thickness	:20mm
(iii)	Exceeding 80mm thickness	:40mm

5.10 **HANDLING OF AGGREGATES**

5.10.1 Stockpiles shall be made immediately on receipt of aggregates. Aggregates shall be stacked separately according to be nominal sizes of the single sized coarse aggregates. For fine aggregates separate stacks should be made for different grading zones, if required.

5.10.2 Aggregates shall be stacked on a hard surface so as to exclude the possibility of soil or grass being mixed up. When stacks are in close proximity the stockpiles shall be separated by the baulk heads to prevent the aggregates from mixing together. Special care will be taken to clean and wash the last layer of aggregate in contact with ground surface before use.

5.10.3 Before batching, the aggregates shall have been stockpiled for at least 24 hours to allow for draining of water, if any. Contractor shall make adequate provisions for stock piling aggregates to an extent sufficient to meet the needs of the work taking into account the availability of supplies and rate of delivery, etc, and he shall include in his tender for necessary double handling and transportation of materials from stock piles to mixing plant etc.

5.10.A **WATER**: Water shall conform to the requirements stipulated in IS-456-2000 and as per clause 4.9 of MES Schedule.

5.10.A.1 **CURING** : The contractor shall take necessary arrangement to maintain an environment of high humidity around the freshly placed concrete till it attains reasonably good strength and proper curing. Contractor shall follow any of the following method of curing while execution of work based on the site conditions:-

(a) Ponding or immersion of water

(b) Fogging or spraying of water

(b) Moist fabric covers

(c) Covering by plastic sheets

5.10.A.2 Recovery @ 0.25 % of total cost of concrete at contract rate shall be recovered as penalty in case of improper curing and cracks if any are noticed by GE or by the Dept official at any stage.

PARTICULAR SPECIFICATIONS (Contd…)

5.11 **Mix of concrete:** The proportion and type of cement concrete and its grading in any particular building/location shall be all as shown in structural drawings or notes thereon. In case of varied provisions as in notes on drawings and sketches, provision as in notes shall take precedence. The proportion and type of mix and its grading, if not shown/described otherwise in drawings, it shall be as under:-

LOCATION		TYPE OF MIX
(a)	Foundation concrete for masonry walls/pillars	PCC (1:4:8) type D-2 using 40 mm graded crushed stone aggregate
(b)	Lean concrete below RCC column footing/ plinth beam	PCC (1:4:8) type D-2 using 40 mm graded crushed stone aggregate
(c)	Sub base to floor	PCC (1:4:8) type D-2 using 40 mm graded crushed stone aggregate. Thickness of sub base to floor shall be as given in respective schedule of finishes. However, in case where thickness is not shown, 75 mm thickness shall be provided.
(d)	RCC in all situations except minor articles such as chajjahs, cills, fins, lintel bands etc as decided by GE	M-30 Design Mix (RMC) from any of the manufacturers listed in the list of manufacturer hereinafter.
(d)	RCC in minor articles such as chajjahs, cills, fins, lintel bands etc as decided by GE	M-30 Design Mix (RMC)
(f)	PCC in floors and other plain cement concrete work shall be as follows :-	
(i)	40 mm thick and above	PCC (1:2:4) type B-1 using 20 mm graded crushed stone aggregate.
(ii)	Thickness less than 40 mm	PCC (1:2:4) type B-0 using 12.5 mm graded crushed stone aggregate.
(g)	PCC bed blocks	PCC (1:2:4) type B-1 using 20 mm graded crushed stone aggregate.
(h)	PCC coping and benching	PCC (1:3:6) type C-1 using 20 mm graded crushed stone aggregate.
(j)	Surroundings to surface of gully traps, padding under nahani traps/floor traps/ deep seal traps/ surrounding of WC seats	Lean concrete PCC (1:5:10) type E-2 using 40 mm graded crushed stone aggregate

5.12 **IMPORTANT REQUIREMENT OF CONCRETE**: -

5.12.1 The plain cement concrete shall be mixed in mechanical mixer of Hopper type. The mixing procedure supplying, conveying, placing etc shall comply with the specification mentioned under para 4.12.6 of MES Schedule Part-I. For smaller quantity up to 2 cum of concrete per day i.e. for concreting to cills, bed blocks etc hand mixing may be carried out with written permission from GE. The mixing of concrete shall be as per clause10.3 of IS 456 2000". Where hard mixing is permitted, it shall be carried out on a water tight platform and care shall be taken to ensure that mixing is continued until the concrete is of uniform colour and consistency.

5.12.2 The Engineer-in-Charge shall maintain a register for actual consumption of cement (Other than cement register) as mentioned in Special Condition duly signed by Engineer-in-Charge, and the contractor for every change in quantity/source of material.

5.12.2.1**DESIGN MIX CONCRETE:**Wherever design mix concrete is specified/shown on drawing, the same shall be provided as per the following:-

PARTICULAR SPECIFICATIONS (Contd…)

(a) Design mix concrete shall have requisite workability and 28 days characteristic compressive strength and target mean strength shall be as per relevant IS. The mix shall be designed as per IS for **Severe** condition "as per IS 456 – 2000 (Table V) from any of IIT / NIT / Govt Engg College (Degree level) SEMT / Zonal (MES) / Command (MES) laboratory/NABL accredited lab. Adequate number of trial mixes shall be carried out at the commencement of the work to decide the proportion for the grade of the concrete and yield. The initial mix design particulars alongwith all the connected documents shall be submitted to GE for approval by GE. The work on ground shall not be progressed until the approval of design mix is accorded. Subsequent design mix due to change in source/ grading of aggregates or source/batch of cement shall also be approved by GE. Cost of the design mix shall be borne by contractor & rates quoted shall be deemed to be inclusive of the same.

(b) Adequate number of tests at regular intervals shall be carried out to ascertain the properties of materials. In case of variation in grading of aggregate noticed at any point of time the same shall be made good by suitably blending different sizes of aggregates, wherever, possible. If this is not feasible the mix design shall be changed immediately before carrying out further work.

(c) To ensure adequate control and consistency of the properties of materials incorporated in concrete as per initial mix design, the materials shall be got tested at lab approved by GE at the intervals as decided by GE. Costs of all such testing charges are to be borne by the Contractor. Cost of the design mix & any other testing as required shall be borne by contractor & rates quoted shall be deemed to be inclusive of the same.

5.12.3 **BATCHING OF CONCRETE:** As per IS 456, in proportioning the plain cement concrete of any grade, the quantity of cement, aggregate and water shall be determined by mass using weigh batcher. The equipment shall consist of one or two weighing buckets connected through a system of lever to spring loaded dials which indicate the load. The accuracy of mass measurement shall be within ± 2 %. The mass of water may be determined by storing water in tank which shall be filled up for each batching. The system shall be designed to have control to admit any desired quantity of water for each batching. Alternatively water meter can be fitted in pipe feeding water to concrete mixer to allow exact quantity of water to go in each batch.

5.12.4 Testing : Tests of concrete cubes shall be carried out as laid down in IS 456-2000. The cost of all materials and labour for casting and curing the requisite number of cubes, cost of testing in laboratory approved by the GE including transportation of cubes to approved laboratory, testing fees and other expenses etc shall be borne by the contractor.

5.12.5 All concrete for RCC work shall be consolidated/compacted by mechanical vibrators of approved type. Precast/cast-in-situ, plain cement concrete shall, however, be rammed and consolidated by tamping and rodding as specified in page 59 of MES Schedule 2009 (Part- I).

5.12.6 In the event of breakdown of mechanical mixer and vibrator, the contractor must make arrangement for standby mechanical mixer and vibrators.

5.12.7 Water shall be used as per IS and shall be approved by GE before incorporation.

5.13. **READY MIX CONCRETE (RMC)**

5.13.1 **DESIGN MIX CONCRETE**: Wherever design mix concrete is specified/ shown on drawing, RMC shall be provided as per following:-

(a) The RMC shall be of grade as specified.
(b) The slump of the concrete shall be as per IS 456-2000
(c) The contractor shall engage any one of the following manufacturer for manufacture and supply of RMC :-
  (i) Lafarge India Pvt Ltd
  (ii) ACC Ready Mix concrete
  (iii) RMC, Ready Mix (India) Pvt. Ltd,
  (iv) Ultratech RMC Plant
  (v) M/s Mary Matha Infrastructure Pvt Ltd.
  (vi) Any other plant confirming to IS 4926 (2003) as approved by E-in-Cs Branch.

PARTICULAR SPECIFICATIONS (Contd…)

(vii) Any other plant confirming to IS 4926 (2003) as approved by Accepting Officer(This option shall only be exercised in case of non availability of RMC from sources as indicated above)

(d) It is the responsibility of the contractor to make payments to the RMC supplier independently by the contractor and the department is not responsible for any dispute between contractor and RMC manufacturer for non payment or delayed payment.

5.13.2 Immediately on commencement of the work the Contractor shall intimate the name of manufacturer of RMC to whom he proposes to engage. The contractor shall make arrangements for a visit of GE to the manufacturer yard for inspection of following to enable GE to accord approval:-

(a) Infrastructure in terms of storage of cement and aggregates.
(b) Arrangements for washing of aggregates.
(c) Source of supply of aggregates and water.
(d) Adequacy of required testing facilities for cement, water aggregates, at the manufacturing yard of manufacturer.
(e) Batching and weighing arrangements including arrangements for adding of water.
(f) Mixing mechanism.
(g) Quality Engineering, supervision and technical skill of workmen.
(h) Records maintained to ensure the quality of materials.
(i) Data showing proof of certificate by approved testing laboratories for accuracy of calibration.
(j) Details of batching showing quantity of input materials.
(l) Type of cement to be used.

5.13.3 Even though the firm for manufacturing and supply of RMC to site of work is approved by the GE, the responsibility to maintain quality and grade of concrete fully rest with the contractor.

5.13.4 Maximum size, type of aggregate and its strength determined from samples of plastic concrete taken at the place and time of delivery in accordance with the requirements of IS-456 -2000(4th revision).

5.13.5 Minimum acceptable compressive strength determined from samples of plastic concrete taken at the place and time of delivery in accordance with the requirements of IS-456 - 2000(4th revision).

5.13.6 **MATERIALS**

(A) CEMENT:- Cement for RMC shall be OPC 43 Gde / OPC 53 Gde /PPC. The OPC cement shall conform to specifications as per IS 269-1989, IS 8112-1989 and 12269-1987 and the PPC cement shall conform to specifications as per IS 1489 (Part-I) -1991, however mixing of OPC and PPC shall not be permitted in the entire project till completion.

(B) FINE AGGREGATES: Fine aggregates shall be as specified here-in-before and as per IS 383-1970.

(C) COARSE AGGREGATES : The coarse aggregate shall be as specified here-in-before and as per IS 383-1970

(D) WATER: Water shall conform to the requirements stipulated in IS-456-2000 and as per clause 4.9 of MES Schedule.

(E) ADMIXTURE :Admixture shall conform to IS 9103 and of approved manufacturers such as M/s FOSROC Chemicals, India (P) Ltd, Hafeefa chamber 2nd floor,111/3 KH Road, Bangalore – 27, Roffe Construction chemicals (P) Ltd,12C, Vikas Centre, SV Road Santacruz (W), Mumbai-54, STP specialty chemicals limited, 16, NGN Vaidya Marg, Mumbai-23, M/s Choksey Chemical Pvt Ltd. or any other make as approved by E-in-Cs Branch

PARTICULAR SPECIFICATIONS (Contd…)

5.13.7 Mix design shall be got carried out by the contractor from the approved RMC manufacturer incorporating materials complying with the requirements as stated above for exposure condition as stated hereinbefore for mix design as per IS-456-2000(Table-V). The mix design shall be in properly bound booklet form shall be submitted to GE in quadruplicate for approval by GE. The RMC incorporated in the work shall be in accordance with the approved mix design. The CE/GE/Engineer-in-Charge/JE (Civil) who are connected with administration and execution and other operations connected with the execution of this work or any other officer from CE (CG) Goa, shall have access to inspect/check the quality of materials used for manufacture of RMC in RMC manufacturer's yard as well as the quality/grade of RMC supplied by the manufacturer. The contractor shall make all arrangements for the aforesaid inspection and check. Any change in source of materials including the change in the brand of cement shall be notified to the EIC/GE and the mix shall be redesigned and approved by GE. Alternatively contractor may be permitted to incorporate mix design already approved by GE within one year provided source of materials and RMC manufacturer remains same to expedite the progress of work.

5.13.8 After approval of RMC mix design by GE, the contractor shall make cubes of trial mixes as per approved mix design at site Lab in presence of EIC using same ingredients as adopted for design mix, prior to commencement of concreting and get them tested in presence of EIC for 7 days and 28 days. For each design mix, a set of 6 cubes shall be prepared and from each set, three cubes shall be tested at the age of 7 days and three cubes at the age of 28 days. The cubes shall be made cured transported and tested strictly as per IS requirement. The average strength of three cubes at the age of 28 days shall exceed the specified target mean strength for which design mix has been approved.

5.13.9: Contractor shall obtain a certificate from RMC manufacturer for the RMC supplied for each day to the effect that materials used for manufacture of RMC complies strictly as per mix design requirements and the materials incorporated are as per para 5.7 to 5.9 hereinbefore. In addition the contractor shall collect samples of materials for each days concreting in the presence of Engineer-in-Charge and representative of RMC manufacturer and shall be tested to ensure that materials used are as per requirements specified herein before. Such test for each material for various kinds of test shall be made as per relevant BIS requirements and shall be entered in register of test results. Register shall be signed by the contractor and Engineer-in-Charge.

5.13.10: A register shall be maintained by the contractor duly signed by the Engineer-in-Charge showing the following details of RMC.

(a) Date of delivery
(b) Time of mixing
(c) Time of delivery at work site
(d) Quantity of RMC delivery in each transport vehicle
(e) Location where used and reference to cube test register.

5.13.11 If the condition of RMC delivered at site is not acceptable to the Engineer-in-Charge it shall be taken back and removed from site by the contractor. The decision of the GE with regard to non acceptability of RMC shall be final and binding. Some of the conditions under which RMC can be rejected are given below:-

i. Initial setting due to delay in transit. RMC shall be delivered at site of work within two and half hours of mixing.
ii. Segregation of aggregate due to excessive rotation of mixer during transit.
iii. The RMC supplied does not give required slump.
iv. In addition to above, GE may further decide the rejection of RMC on any other grounds which he decides necessary. Further, the payment made by the contractor to RMC supplier is independent of the payment to be made by the Department to contractor for carrying out the work.

5.13.12 Acceptance of concrete: Acceptance criteria for the RMC shall be as per IS-456-2000. In case the RMC supplied and incorporated fails to meet the strength requirements as per IS 456-2000 work done shall be rejected by GE and contractor shall demolish the rejected work and re-do the same without extra payments so as to produce the work complying with the strength requirements as per IS 456.

PARTICULAR SPECIFICATIONS (Contd…)

5.13.13 Conveying equipment's for concrete shall be water tight, well maintained and thoroughly cleaned before commencement of concrete mixing. Concrete shall be transported by transit mixers.

5.13.14 Water shall be used as per IS and shall be approved by GE before incorporation.

5.13.15 Concrete shall not be dropped from a height, thrown or otherwise treated so that segregation, undesirable finish, or defective structural quality results.

5.13.16 No extra water shall be added to concrete mix after it has left the batching plant.

5.13.17 The contractor shall take adequate precautions to protect concrete in transit from the effects of the weather.

5.13.18 In case of rejection of supply of RMC supplied by the manufacturer at site, the contractor shall be responsible for making good the same without extra cost to the Govt. The GE/Engineer-in-Charge & other officers authorized by the Accepting Officer shall have access to check on the quality of production in RMC suppliers manufacturing yard as and when they desire so. The contractor shall make all arrangements for such inspections. Payment of RAR/ deviation pertaining to RCC works involving RMC, rates given in MES schedule Part-II 2020 for corresponding grade of concrete (Design mix 20mm graded stone aggregate) subjected to contractor's quoted percentage for BOQ Sch 'A' Part I shall be applicable. No advance payment will be made towards payment made by contractor to RMC manufacturer and no advance on account for cement, coarse aggregate and fine aggregate stored in the yard of RMC manufacturer shall be payable. Type and quality of cement used in RMC shall be in accordance with the specifications as already specified for design mix in the contract agreement here-in-before. It will be the responsibility of the contractor to ensure that the RMC Manufacturer uses in manufacturing RMC the cement/aggregates of grade and quality as described here-in-before. He shall also be responsible to produce necessary paid vouchers and test certificate for the above materials as and when called for by GE for verification to ascertain the correctness of type and quality of materials used by the RMC manufacturer. Paid voucher of RMC shall also indicate details of make/brand of cement used besides batch No, consignment details and test certificate details of procurement of cement

5.13.19 Certified photo copy of paid vouchers and test certificate of cement purchased shall be also enclosed along with paid vouchers of RMC for claiming payment. The contractor shall be responsible to ensure that the cement used in RMC is of type as specified and of approved make/brand as specified in the tender.

5.13.20 Requisite facilities for verification and check of the raw material shall be made available to GE by the RMC manufacturer. GE will have easy access to the RMC plant to check quality of concrete being produced at the plant at intervals as decided by GE. The GE shall independently carry out the following tests at random at site lab established by the contractor.

5.13.21 Compaction factor test, flexural strength test, slump test and cube tests shall be carried out as per IS-456-2000(4th Revision) at site from samples collected by the contractor during placing of the concrete. Cost of the collection of samples and testing shall be deemed to be included in the quoted rate. Acceptance criteria for the RMC shall be all as specified in the IS-456-2000 (4th Revision). In Case RMC supplied by the approved firm fails to meet the strength requirement as per acceptance criteria laid down in IS-456-2000(4th Revision) and rejected by GE, the Contractor shall make necessary arrangement for removal of such rejected quantity of RMC and will redo the work at his own cost.

5.13.22 Contractor shall take every precaution to strengthen the shuttering as required to withstand the additional pressure that may be created due to pumping of concrete. Workability of concrete shall be as specified in IS-456-2000.

5.13.23 If certain admixtures are required to be added to RMC either in plant or at place of delivery to improve the workability the same shall be added as per manufacturer's instructions with prior approval of Garrison Engineer. Quantity of admixture at any stage should not exceed the maximum limit as specified by manufacturer. Admixture if any to be used should be compatible with cement.

PARTICULAR SPECIFICATIONS (Contd…)

5.13.24 Contractor if desires to use admixture shall conduct the test for cement, admixture etc., in an independent testing lab accredited by NABL and approved by Garrison Engineer without any extra cost to the Govt.

5.13.25 Revision of design mix due to change in aggregate proportions due to change in size and relative density/bulk density of aggregate shall be immediately brought to the notice of the Garrison Engineer and in all cases provisions contained in IS-456-2000(4th Revision) for mix considerations/acceptance criteria shall be complied with.

5.13.26 The time between mixing of concrete at manufacturer's yard and transportation and delivery at site shall not be more than 2½ hours. In case longer time is unavoidable the same can be permitted by Garrison Engineer adopting using admixtures all as permissible as per IS without any extra cost to the Govt.

(a) One set of 9 cubes of every 50 cum of concrete production or every 50 batches whichever is greater or as per the discretion of Engineer-in-Charge shall be cast for testing 28 days and 7 days cube strength and balance 3 cubes for future tests if required at later stage.

(b) Delivery ticket for truck should show cement content used by the manufacturer type of cement, admixture used and locations of concreting etc.

(c) RMC shall be supplied as per IS 4926-2003 and concrete cube shall be casted as per IS-456-2000 (4th Revision).

(d) Minimum quantity of cement shall be as per provision of IS: 456-2000 (4th Revision) meeting the requirements from durability point of view and the details regarding proper training and works quality control shall be in accordance with IS:456-2000(4th Revision) and IS:4926-2003. However if extra quantity of cement is used for whatever reasons, the same shall be at no extra cost basis to Government.

(e) Pumping operation whenever commences shall proceed continuously so as to prevent "cold" joints between placed sections. For concrete less than 15 Cubic metres may be deposited manually.

(f) The delivery line of the pump shall be 100 mm dia or greater and pump shall be capable of pumping concrete containing a 20 mm normal size aggregate.

(g) The pump shall have receiving hopper and pumping chamber shall be capable of replacing atleast 15 Cum of concrete per hour against horizontal delivery head of atleast 90 m and/or a vertical delivery head of 20 m.

(h) Pumping lines shall be of approved metallic type laid to avoid bends.

(i) The joints in pumping lines shall be sealed tight to prevent leakage.

(j) All equipments pump chamber, hoppers, lines and rubber hoses shall be kept clean at all times. Any build ups in the lines of materials from previous operations shall be cleaned out prior to pumping.

(k) In the event of breakdown in the equipment causing delay not exceeding 20 minutes, which time concrete cannot be replaced, the following procedure shall be adopted :-

(l) With the approval of Engr-in-Charge, the concrete already placed shall have the "wet edge" and vibrated into mass. Where atmospheric temperature exceeds 30 degree centigrade the receiving hopper and lines shall be cleaned out and concrete contained therein discarded and immediately removed from the site. The concrete shall be discarded if initial setting of the concrete has begun in the hopper or discharge lines. All lines shall be cleaned free of concrete prior to resumption of pumping after each break down. Concrete in the lines shall be pumped approximately 8 minutes intervals to ensure the concrete in the line is live, whenever delivery of concrete in the pump is delayed. This pumping interval shall be reduced to 5 minutes during extra hot weather conditions. Delivery lines were exposed to hot sun shall be protected by covering with bags, wet hessian or other approved means.

(m) Due to mechanical malfunctioning, if concreting is required to be stopped, necessary precautionary measures shall be taken by the contractor. Cost of any additional work caused due to these stoppages shall be contractor's responsibility.

(n) Other preparatory works required are completed, inspected and approved by the Engr-in-Charge.

PARTICULAR SPECIFICATIONS (Contd…)

(o) The contractor shall take adequate precautions and strengthening measures to strengthen the shuttering as required to withstand the pressure that will be created due to pumping of concrete.

5.13.27 The contractor may use fully automatic batching plant with print facility of capacity 10-15 cum/hr in lieu of RMC. The concrete shall meet criteria of RMC. Pour card register shall be maintained.

5.13.28 CONSOLIDATION OF CONCRETE: - Consolidation shall be done by mechanical vibrators, plate type for slab and needle type for other locations. 10

5.13.29 CURING: Curing shall be done as specified hereinbefore.

5.13.30 All concrete for RCC work shall be consolidated/ compacted by mechanical vibrators of approved type. Precast/cast-in-situ, plain cement concrete shall, however, be rammed and consolidated by tamping and rodding as specified in para 4.11.11 of MES Sch 2009 (Part I)

5.13.31 In the event of breakdown of mechanical mixer and vibrator, the contractor must make arrangement for standby mechanical mixer and vibrators.

5.13.32 **TESTING OF BUILDING MATERIALS** : Test for building materials shall be conducted as per IS requirement as directed by the GE. The Contractor shall provide necessary samples, make adequate arrangement for transportation and testing of the same. Testing charges shall be paid by the Contractor. Recovery on account of testing charges shall be made from the Contractor for various tests conducted in MES Zonal lab/Command test lab as per Annexure 'I' to Special Conditions enclosed. The test for which the facility is not available in MES lab shall be got done by contractor in approved lab outside MES and the testing charges as charged by the concerned authorities shall be borne by the contractor as provided in condition 10A of IAFW-2249, General Conditions of Contracts and no claims, whatsoever, shall be entertained on this accounts.

5.14 **FORM WORK**

5.14.1 Form work shall comply with requirements of clause 4.11.6.1 to 4.11.6.5 and 7.15.1 to 7.15.10 of MES Schedule (Part I). Formwork to be used in the work shall be of steel only.

5.14.2 **GENERAL**

(a) The work under these specifications consists of furnishing all labour, equipments and materials required for form work including all supporting structure for all works of cast in situ concrete as required by drawing and specification of contract.

(b) The contractor shall submit form and centering layout to Engineer-in-Charge for checking and final approval by Garrison Engineer. The complete final approval of the plan will be obtained by the contractor before such work is started on ground. The procedure of erection and removal of forms will be decided at this stage itself. These approvals shall not relive the contractor of any responsibility for correct and complete performance of all works included in the contract. The design and engineering of the formwork as well as its construction shall be the responsibility of the contractor.

(c) Form work shall be adequately designed, fabricated, erected , supported, braced and maintained to safely support all vertical and lateral loads that might be applied until such loads can be supported safely by the concrete structure. Such loads will include the following items:-

(i) Dead weight of forms
(ii) Dead weight of freshly placed concrete
(iii) 100% impact for dumping of fresh concrete on forms
(iv) General construction live load
(v) Lateral wind of earthquake force
(vi) Forces due to operation of construction equipment
(vii) Lateral pressure of fresh concrete
(viii) Other site specific forces such as cable tensions, lateral earth or water forces.

PARTICULAR SPECIFICATIONS (Contd…)

(d) Form work shall be designed, constructed and maintained to provide that all parts and members of reinforced concrete will be true and correct in size, dimensions, shape, alignment, elevation and position. Vertical alignment from floor to floor or floor to roof shall be maintained straight and plump, and horizontal alignment level and straight on each floor.

5.14.3**MATERIALS AND ACCESSORIES**

(a) Forms for all exposed reinforced work during execution shall be of steel sheet suitably stiffened at edges with angle irons or like and shall be new or in new condition free from kinks. Form joints shall not permit leakages and shall be Morton tight neatly fitted and fastened.

(b) Form lining if used shall be such that it does not discolour the concrete. All nuts and bolts shall be counter sunk and well grounded to provide a smooth plain surface to finished concrete, When concrete is required to have rounded edges, moulded edges and like, the provision will be made in the form itself,. Opening for fan clamps and other fittings connected with the work shall be provided in the form work (shuttering) as directed by Engineer-in-Charge.

(c) The forms should be held together using clamps or other fastening devices so that the form does not bulge and the fastener does not protrude into the concrete mass. Forms shall be thoroughly cleaned of all dust, dirt, wood shavings and other matter by washing with water. The interior, surfaces of the form likely to come in contact with concrete shall be coated with non –oil base material that will not stain or cause imperfections on the concrete surface; before placing of reinforcement.

(d) A thorough inspection of tubular steel centering support structure is necessary before erection and members showing evidence of excessive resting, kinks, dents or damaged welds are to be discarded. Buckled or broken members will also be rejected and removed from site. All locking devices should be in good working order and the coupling pins are effectively aligned to the frames.

(e) Manufactured Vertical steel tubing shores shall be set in plumb at not more than 90cm center to center under forms for all beams and girders, and under forms for slabs on all supporting stringers and purlins. They shall be adjustable in height in a simple manner and without releasing their load. The shores shall be firmly seated on a base plate not less than 200 mm wide. The manufactures instructions regarding placing of shores will be always followed. All shores shall be cross braced adequately Vertical adjustment shall be carried out by screw jack, wedges or other approved methods. All slab forms should be fastened together firmly so as to prevent vertical displacement of joints.

(f) All forms and supporting structure must be braced laterally by cross bracing and securing with the previously constructed unyielding portion of structure like columns and walls. The supporting structure so provided should became a rigid and solid unit. The properties of soil should be always kept in view as the base of dry clay becomes plastic when it comes in contact with water either due to rains, or accidental leakages from forms or during curing. Adequate care is to be taken in this regard.

(g) After erecting the form work for beams and slabs, the form work will be placed by contractor for inspection of Engineer-in-Charge who after checking will propose the same for approval of Garrison Engineer. Only after approval by Garrison Engineer the reinforcement will be placed as per Drgs and specification.

(h) All centering shall be finally inspected to ensure that :-
   (i) Footings or support under every post of the centering are sound.
   (ii) All lower adjustment screws or wedges are snug against legs of panels
   (iii) All upper adjustment screws or heads of jacks are in full contract with form.
   (iv) Panels are plumb in both directions.
   (v) All cross braces are in place and locking devices are closed and secured in position

PARTICULAR SPECIFICATIONS (Contd…)

(vi) In case of chajjahs, balconies and other building elements which are projected out, the props are adequately secured and placed firmly to transfer the load to the supporting point.

(vii) The sequence of casting or pouring of concrete also adds to the total stability of the entire form work. The pouring should proceed from supports to the center in case of simply supported or continuous beams /slabs and from support to free end in case of projections.

(viii) During casting and pouring of concrete the centering shall be closely inspected and strengthened, and if required the wedges below the vertical supports tightened. The adjustment screws will be properly adjusted if necessary. The workmen should be wearing the hard hats and other protective gear without fail.

(ix) Forms shall not be removed earlier than laid down as per good practice and until it is certain that the concrete has developed sufficient strength to support itself and all loads that will be imposed on it,. Only workmen, actually engaged in removal of form are allowed in the area during these operations. These workers will have protective gear like helmets, gloves, heavy soled shoes and safety belts. While cutting any wire or rope in tension adequate care to be taken to prevent backlash which might hit the worker.

(x) The particular sequence in which supports are to be removed or dismantled should be given by Engineer-in-Charge as per sound engineering practice. In case continuous or simply supported slabs the removal of formwork should start from center of span to support and in case of cantilevers from free end to support.

(j) In case of deviation the rates for formwork shall be applied for wooden formwork for fair finished surfaces based on MES schedule Part II 2020.

5.15 **EXPOSED SURFACE OF CONCRETE**

5.15.1 All exposed surface of concrete which are ultimately required to be finished by application of cement based paint, white/colour washing, distempering, painting etc shall be finished with a thin layer of cement and sand mortar 1:3, 5mm thick finished fair and even as specified in para 4:11:16,2(b) of MES Schedule Part I.

5.15.2 Exposed surface of lintels, beams, column bands etc which are continuous with plastered surface of walls shall be plastered, in the manner as specified for the walls.

5.15.3 Junction of RCC with PCC solid block wall shall be plastered after reinforced with chicken wire netting strip of GI wire of 0.91mm dia and 12 mesh. The width of wire netting shall be 30cm. The wire net shall be properly secured with nails / staples as directed by the Engineer-in-Charge.

5.16 **PRECAST CONCRETE ARTICLES**: Cement concrete lintels with or without integrally cast chajjas up to 1.5 m clear span, shelves, bed blocks/plates, cover slabs fencing posts, fins and jalli and the like may either be precast or cast in situ at the contractor's option unless otherwise shown on drawings. If precast, these shall be set in cement mortar (1:3). In case of deviation involving these items, pricing shall be done on the basis of cast-in-situ work as specified here-in-before.

5.17 **RCC CHAJJAS / FACIAS**

5.17.1 RCC chajjas/facias shall be provided as per details shown on drawings.

5.17.2 Upper surface of RCC chajja/ facia shall be plastered with water proofing plaster 5mm thick in CM (1:3) & surface finished even & smooth without using extra cement ,mixed with water proofing compound at the rate recommended by the manufacturer finished even and smooth without using extra cement while concrete is green.

5.17.3 Internal/External surface of down word facia shall be plastered in CM 1:3, 5 mm and finished even and smooth, without using extra cement, with water proofing compound.

PARTICULAR SPECIFICATIONS (Contd…)

5.18 **THROATING / WEATHERING**: Throating to projections of RCC/PCC beyond external faces of the walls where shown on drawings and where RCC chajjas are not provided with downward facia shall be formed in the concrete while casting by placing a fillet/bar of 12mm diameter in the form work and finished smooth.

5.19 **Polished Granite Stone Cills** :- Irrespective of whatever shown on drawings or specified elsewhere the cills shall be provided with pre polished Granite stone 18mm to 20mm thick single piece over 10mm thick cement mortar 1:4 as screed bed in all windows and ventilator cills. All exposed edges of Granite stone slab shall be rounded and polished. Granite stone cill shall be extended in wall by 25mm on either side.

5.20 **PCC CURB**:- PCC curb shall be provided all as shown on drawings.

5.21 **RCC SHELVES**: RCC shelves shall be provided all as shown on drawings where thickness of slab has not been indicted in drawings it shall be 100mm. Edges of shelves shall be rounded off to a minimum radius of 5mm. The underside and edges of slab shall be treated all as specified for adjoining walls.

5.22 **RCC GUTTER**: RCC gutter shall be provided all as shown on drawings.

5.23 **RCC PARAPET WALL**: RCC parapet wall shall be provided all as shown on drawings.

5.24.1 Bearing to all roof/floor slabs on masonry walls shall comprise of 20mm thick bearing plaster in CM(1:4) finished even and smooth with steel trowel using extra cement applied and two layers of laminated water proofing building paper type-I weighing not less than 60 gms/Sqm each layer, laid over it. However, in case floor/slab rests on RCC beams/bands, this provision does not apply.

5.24.2 All RCC beams/lintels / bresummer resting on masonry work shall be provided with PCC bed block cast in situ of twice the width of beam in length covering the entire thickness of wall and of thickness 150mm, if otherwise not shown on the drawings.

5.25 **CONCRETE PADDING**:- Where the required height of walls, openings is not obtained, with adequate size of bricks/PCC blocks, the same shall be obtained by providing PCC 1:2:4 type B1. Bricks shall not be cut to under sized dimensions and used to make the required height in such position.

5.28 **LINTEL BAND**: - Lintel band shall be provided over all walls irrespective of what is shown on drawing. Details of lintel band shall be as shown on drawing.

5.29 **BENCHING**: PCC benching shall be provided in cement concrete 1:2:4 type B1 to a radius of 75mm mixed with integral water proofing compound. PCC benching as specified above shall be provided at the junctions of RCC Chajjas/Canopy/roof slab with the vertical faces of walls/facias.

5.30 **INTEGRAL WATER PROOFING COMPOUND**: Wherever integral water proofing admixture is specified or indicated to be mixed to concrete/mortar, plaster etc. The percentage of integral water proofing compound to be added shall be as per manufactures instructions.(For the purpose of pricing deviations, quantity of integral water proofing compound in powder @3% by weight of cement or liquid form 150 ml shall be taken by weight of cement bag of 50 Kg). Integral water proofing compound shall conform to IS-2645 and shall be ISI marked and shall either be powder or in liquid form.

5.31 **PLINTH PROTECTION.**

5.31.1 Unless otherwise shown on drgs, plinth protection in all situations as shown on drawing shall be provided with 75mm thick PCC (1:3:6) type C1over 100mm thick consolidated bed of hard core over well consolidated earthen surfaces. Unless otherwise specified the width of the plinth protection shall be 75cms. PCC shall be laid in alternate bays (not exceeding 2 Sq meters each) and finished fair on top without using extra cement, 10mm wide, 50mm deep joints shall be provided at 3-meter intervals with mastic joints shall be provided for plinth protection in concrete bays and in between walling and plinth protection. All joints in concrete bays as well as between walling and the concrete in plinth protection shall be filled in with mastic filling comprising 1 part of heated bitumen Blown grade any penetration and 3 parts of sand (all by weight).

PARTICULAR SPECIFICATIONS (Contd…)

5.31.2 Plinth protection shall not be provided at the location where platform/steps/ramp areshown on drawing.

5.32 **RCC JALLI** : Unless otherwise specified elsewhere, Precast reinforced cement concrete Jallies and louvers(1:2:4) type B0 shall be provided where shown on drawings. Details of RCC Jalli shall be followed as per Drawing.

5.33 **PRECAST PCC SOLID BLOCKS FOR MASONRY**

5.33.1 MATERAILS

5.33.1.1 SOLID PCC BLOCKS: - Solid PCC blocks shall be as per IS-2185(Part-I) 2005 locally factory made. Size of PCC solid blocks shall be of 400x200x200mm /300x200x200mm for 200mm thick walls & 400x200x100/300x200x100mm for 100mm thick walls. Min average compressive strength shall be 4 N/ mm^2 and minimum compressive strength of each individual unit shall not be less than 3.2 N/mm^2 . The manufacture of Block, surface texture and finish, physical requirements of blocks, testing of PCC blocks, sampling etc shall be all as per IS 2185-2005 and proper record shall be maintained. Density of block shall be not less than 1800 Kg/Cum . Water absorption shall be not more than 10% by weight of block. The cost of blocks including transportation of block from work site to an approved laboratory for testing and cost of testing shall be borne by the contractor.

5.33.1.2. ----- Blank ------

5.33.1.3 CEMENT:- Cement refer specifications herein before.

5.33.1.4 SAND:- Fine aggregate for PCC block shall be of natural river sand as specified hereinbefore and as specified hereinbefore and as approved by GE. Manufactured sand/manufactured fine aggregate shall be allowed to use if it meet the technical / contractual requirements. Sand for mortar shall be as specified in clause 5.4 of MES Schedule Part I and shall be obtained from sources as mentioned in clause No 4.7 hereinbefore.

5.33.1.5 COARSE AGGREGATE: Coarse aggregate shall be 12.5 mm to 20 mm graded stone aggregate with ratio 3:1 and shall be of broken granite stone.

5.33.2 WORKMANSHIP:-

5.33.2.1 If nothing is shown on drawings then following types of concrete block masonry shall be adopted at different location.
(a) Exceeding 100 thick walls: Solid concrete block masonry in CM 1:6
(b) 100 mm thick walls above plinth level: Solid concrete block masonry in CM 1:4. Concrete block masonry shall be carried out all as specified in clauses 4.23.1 to 4.23.15 of MES Schedule Part I

5.33.2.2 10cm thick solid block masonry wall shall be built from top of plinth beam where plinth beams are provided and where plinth beams are not provided, the same shall be provided from the top of concrete sub base floor without any foundation. 2 Nos of 8 mm dia mild steel high strength corrosion resistant deformed bars shall be provided in 10 cm thick concrete block masonry at every fourth course and anchored 20 cm into the wall.

5.33.3 In case of deviation the precast solid block masonry manufactured out of 16 mm graded aggregate and with 12.5 mm graded aggregate will be considered as costing same.

5.33.4 PCC block shall not be cut to under sized dimensions, and used to make up the required height. Padding under bearing of lintels, slabs and beam etc to make up the height in walls where required shall be done with cement concrete (1:3:6).

5.33.5 Toothing and bonding to all internal walls shall be provided with solid block masonry and no hollow block masonry shall be used for toothing / bonding from external wall to internal wall.

PARTICULAR SPECIFICATIONS (Contd…)

5.33.6 Masonry work shall be kept constantly moist for a minimum of 7 days.

5.33.7 Where cills are not provided below opening, one layer of walling below opening shall be built in solid concrete block. The solid course shall be executed for at least 20 cms beyond the opening of either side.

5.33.8 DIMENSIONS AND TOLERANCE

5.33.8.1 The dimensions of concrete solid blocks shall be as in clause 5.1.1

5.33.8.2 The nominal dimensions of the units are designed to cover the thickness of mortar joints so that they produce wall length which will conform to principle of modular coordination. The maximum variation in the length of units shall not be more than ±5 mm and maximum variation in height and width of unit shall not be more than ±3 mm. Sizes other than specified above may also be used with the approval of GE without any price adjustment.

5.33.9 MANUFACTURING
5.33.9.1 MIXING: All concrete shall be mixed in mechanical mixer.

5.33.9.2 MOULDING OF BLOCKS: The blocks shall be manufactured with vibratory moulding machines. Power operated machine moulds manufactured by M/s Minato Shirke Concrete Machine PVT Ltd, Pune considered to be acceptable standard. However, contractor may use machine moulds of other standard manufacturers if approved by GE. Appropriate consistency of concrete shall be used to enable de-moulding of the blocks immediately after casting.

5.33.9.3 SURFACE FINISH: The blocks shall have sufficiently rough surface to afford a good key to the plaster and joints.

5.33.9.4 CURING: The hardened blocks shall be cured in a water tank in curing yard and shall be kept continuously moist for 14 days. The curing water tank/yard must therefore be of adequate capacity/size. Curing shall be done as per IS-2185.

5.33.9.5 SAMPLING: The required 20 numbers of blocks for testing shall be taken from over by consignment of 5000 blocks or part thereof. The blocks shall be picked up at random form a stack of particular batch. Testing shall be done as per IS. The sample of blocks shall be kept under cover and protected from extreme conditions of temperature, relative humidity and wind until they are put to test. The test shall be under taken as far as practicable immediately after the sample has been taken. The cost of testing shall be borne by the contractor.

5.33.9.6 PCC block masonry with RCC columns shall be anchored to columns with MS flat iron ties of size 25x3 mm and 24.5 cm long at every sixth course starting from bottom.

5.33.9.7 Factory made PCC blocks shall only be allowed to be incorporated in the work. Manufacturer/ contractor to ensure the points as mentioned above during casting of PCC blocks.

5.34 Deviation :- In case of deviation the rates of PCC solid block masonry shall be Rs 7953 00 per cum for 20mm thick waling and Rs. 764.00 per Sqm for 10Cm thick walling subject to percentage quoted by the contractor for Schedule'A' Part –I.

6.00 BRICKS:
6.1 Bricks shall be kiln burnt locally available best quality, and as per samples kept in GE's office. Bricks shall be without frog. Unless specified otherwise in Schedule 'A' Notes or drawings, bricks shall have minimum crushing strength of 35 Kg/Sqcm. Water absorption of bricks shall not exceed 20% when tested in accordance with IS-3495 of 1966. Sampling and testing of bricks shall be carried out as per IS-5454 (1978) and IS- 3495 (1966). The size of brick shall be 220 x 105 x 65mm with a tolerance of ± 8%.

PARTICULAR SPECIFICATIONS (Contd…)

6.2 CEMENT: Refer specifications hereinbefore.

6.3. SAND: - Sand for mortar shall be as specified in clause 5.4 of Section 5 of MES Schedule Part I. Crushed stone sand shall not be used for plastering.

6.4 MASONRY STEPS/RAMP ETC

6.4.1 Wall shall be with brick work with using sub class 'B' bricks built in CM 1:4. Flooring shall be as specified in schedule of finishes. Chequered finish as specified shall be achieved by making impression of expanded metal of mesh as directed by Engineer-in-Charge on surface of concrete while concrete in green. The expanded metal shall be pressed to a depth of 3mm, tamping evenly and removing the same carefully.

6.5 DAMP PROOF COURSE

6.5.1 The damp proof course shall consist of 40mm thick PCC 1:2:4(type B_0) with water proofing compound and laid as specified in para 5.42.1 and 5.42.2 of MES Schedule (Part I).

6.5.2 Water proofing compound for damp proof course, shall be as per IS-2645(1975). It shall be mixed with cement concrete in the proportion and manner as given in manufacturer's instructions. Deviation if any, the WPC shall be priced @ 3% by weight of cement. The compound shall be either in powder or in liquid form and shall be obtained from approved manufacturers.

6.5.3 Damp proof course shall also be provided under door/openings (below floor by giving a vertical drop), irrespective of what is shown on the drawings, PCC/Terrazzo floor where provided shall be extended in doors openings.

7.00 WOOD WORK AND JOINERY

7.1TIMBER: Timber for all wood work shall be second class hard wood unless otherwise specifically mentioned.

7.1.1 Timber for all joinery and wood work shall be as per specifications given in clause 7.3 of the MES Schedule part I and shall be within the permissible limits of defects defined in clauses 7.4 and 7.5 of the MES Schedule Part I. In case of second-class hard wood, it shall be any of the species of Thambakam, Karimatti, Anjili.

7.1.2 Timber shall be well seasoned whether air or kiln dried at the discretion of the contractor but without any price adjustments. The moisture content of timber shall not exceed the limits laid down vide clause 7.7 of the MES Schedule Part I.

7.2 PRESERVATION OF TIMBER

7.2.1 Preservation/ anti termite treatment shall be carried out to all wood work and joinery fabricated by the contractor at site. Factory made ply/boards are not to be treated with any chemical. The timber used for fabricating wood work/joinery shall be pressure treated.

7.2.2 Chemical used for antitermite treatment to wood work and joinery shall be copper NAPTHENATE or any other chemical specified in IS: 401-1982 applied in any one of the manners specified in ibid IS.

7.3 Species of Timber: The species of timber and prefabricated wood products (ie plywood, wooden particle boards etc) shall be as specified below: -

(i) Panelled/wire gauzed door shutter	-	Factory made shutter 2nd Class Hard Wood conforming to IS.
(ii) Wooden frame for doors and, cupboards, glazed shutters of windows ventilators etc where indicated of wood.	-	Second class hard wood conforming to IS.
(iii) Any other wood work not specified elsewhere	-	Second Class Hard wood. conforming to IS.

PARTICULAR SPECIFICATIONS (Contd…)

7.4 BLANK

7.5.Veneered particle board. Veneered particle boards shall be commercial (general purpose) bonded with phenol formaldehyde synthetic resin adhesive and shall conform to exterior grade of IS-3097-1985 with duly marked specifications for veneered particle board. Thickness wherever shown as 19mm shall be amended to read 18mm/19mm.

7.6 PANELLED DOOR SHUTTERS

7.6.1 Panelled shutters of doors (comprising styles, rails and panels) shall be factory made shutters obtained from approved manufacturers listed in appendix here-in-after.

7.6.2 The factory made panelled shutters of all doors except doors of WC/Bath shall comprise of styles and rails of second-class hard wood and panels of 9mm thick marine plywood. Panels shall be fitted in 12mm deep grooves in styles and rails of the shutters. If any variance in drawings, the same shall deemed to have been amended accordingly.

7.6.3 Factory made shutters shall however, conform to size and arrangements of styles rails and panels as per details shown in relevant contract drgs.

7.6.4 Timber used for factory made shutter shall be of kiln seasoned and chemically treated by pressure process as specified in para 5.5 of IS-401.

7.6.5 Other treatment to timber surface such as tarring, painting etc shall be carried out in addition to the preservative treatment.

7.6.6 Factory made panelled shutters shall be brought to site before applying the primer and the shutters shall be got passed by the Engineer-in-Charge before applying the primer.

7.7 FLUSH DOOR SHUTTER:

(a) Flush door shutters shall be of 32 to 38 mm thick solid core melamine faced, three layered prelaminated wood based particle board, exterior grade having shade of choice on both side (PTX) stamped IS-3007-1985 type-I on edges bonded with synthetic resin with second class “Teakwood” E shape or ‘T’ shape beading (fixed with 4 stainless steel hinges of 100 mm size). Flush door shutters of 32 to 38mm thick shall be of factory made core type with black board core and shall confirm to IS 2202 (Part-I)- 1999: specification for wooden flush doors (Solid core type) Part- I plywood face panel, except with regard to the sizes of shutters which shall be as indicated. Flush door shutters shall be melamine finished prelaminated boards, exterior grade having shade of choice on both sides.

(b) Irrespective of what is shown on drawings, provide Factory made flush door shutters to all doors except doors of Toilet and Bathroom

(c) Flush door shutters shall be obtained from approved manufacturer

7.8 SKELETON DOOR SHUTTER

7.8.1 Skeleton door shutter shall be factory made, and shall be made from well-seasoned second-class hard wood. Rails and stiles shall be of size as shown on drawing. Stiles and rails shall be from one piece only. Stiles and rails shall be jointed to each other by tenon and mortice joints, at right angles. Rails more than 18 cm wide shall have two tenons. Wire cloth used shall be stainless steel wire cloth made out of wire 0.45mm dia and mesh shall be 1.4mm aperture. The factory-made skeleton shutters shall be from any one of makes as approved by GE from the list of makes mentioned here in after.

7.9 FIXING OF SHUTTERS

7.9.1 Shutters shall be fixed to the frames all as specified in Clause 8.26 of MES Sch Part I.

7.9.2 Thickness of wooden door shutters where not shown on drawings or specified shall be 35 mm.

PARTICULAR SPECIFICATIONS (Contd…)

7.10 FRP DOOR FRAMES/ SHUTTERS

7.10.1 All door shutters and frames of WCs, Wash and toilets shall be of FRP irrespective of what is shown on drawing. FRP door shall be of make SINTEX, DELUX POLYFAB or ADVANCE MARKETING or SAI SWARNA, FIBRE PLAST, PUNE. The thickness of FRP shutters & thickness of FRP shall be as shown in drawing. Shutters shall be in depressed panel design and colour as approved by GE and shall be include extra reinforcement on edges. The core of Shutter in styles and rails area shall be of polyurethane foam. The panel area shall have ply-FRP sandwich construction having 4mm thick ply to have embedded wooden blocks for taking fixtures. Shutter shall have recesses to take hinge to fix wooden frame. The whole shutter shall have resistant to mild acid /alkali. Aluminium iron mongery to FRP shutter shall be fixed as per manufacturer's instructions. Any tenderer offering FRP shutter with more thickness of FRP will not be paid anything extra.

7.11WORKMANSHIP

7.11.1 Joinery: Timber exposed to view shall be wrought, not exposed to view shall be clean swan. the workmanship and fixing of joinery shall be as per clause 8.11 to 8.23 of MES Schedule Part I.

7.11.2 The dimensions of the various components of joinery (other than block board shutters) shown on drawings, wherever at variance shall supersede the standard dimensions mentioned in clause 8.20 of the MES Schedule. However, for pricing deviations involving any joinery work, the rates in the SSR Part II for the corresponding joinery items shall be applicable.

7.11.3 Unless otherwise specified, all work both carpenter's and joiners shall hold full dimensions shown on drawings except that an allowance of 1.5mm shall be allowed for each wrought face. Wooden beads and fillets shall however, hold the full dimensions as shown on drgs. The contractor shall also maintain the overall sizes of the doors and windows etc as shown on drawings.

7.11.4 Timber members upto 3.0 meters length shall be in one piece.

7.12. ALUMINIUM WINDOWS/VENTILATORS AND DOORS

7.12.1 The types and overall size of aluminium doors/ windows/ventilators/ shall be all as shown in Architectural drawings. Shop drawings shall be prepared by contractor & got approved from GE. Weight of aluminium sections used in manufacturing of frames including shutters of doors, windows and Vents, but excluding all fills likes glazing, particle board, PVC filling, wire mesh etc shall all as shown on respective drawings

7.12.2 In case the weight of aluminium sections for approved drawing is more than the minimum specified here-in-before, the contractor shall not be eligible for any extra payment.

7.12.3 The aluminium sections shall conform to designation 6300 IS-737-1986 and specified in clause No 10:37 of SSR Part I.

7.12.4 Aluminium doors/ windows/ventilators shall be obtained from any of the manufacturer listed hereinafter.

7.12.5 The shutters shall be made out of specially extruded tubular sections where provisions for weather stripping made in the vertical jambs.

7.12.6 The cleats for mechanical horizontal (vertical joints) of the fixed frame and shutters shall be of specially extruded aluminium sections so as to avoid any ply between jointed members.

7.12.7 Single action door shall be fixed by plated brass pivots at top and bottom.

7.12.8 Single action doors shall be provided with overhead door closers. Double action doors shall be provided with floor springs.

7.12.9 Inactive leaf shall be provided with concealed sheet bolt.

7.12.10 Shutters shall be provided with PVC/Neoprene weather stripping.

PARTICULAR SPECIFICATIONS (Contd…)

7.12.11 Neoprene/PVC gasket shall be used in the glazing beads for shutters before fixing glazing, the glass shall be encased in PVC channel so as to avoid metal to glass contact.

7.12.12 The aluminium sections shall be powder coated finish. The thickness of powder coating shall be minimum 50 micron. Colour of powder coating shall be as decided by GE

7.12.13 The glazing beads both on shutters as well as fixed glazing shall be of screw less type.

7.12.14 The glass panel shall be all as specified in IS-1948. Glazing shall be with high quality tinted glass of 5.5 mm for aluminium doors and 4 mm thick for aluminium windows. Ventilators medium weight quality (with mild steel wire pin) as per clause 9.7.2 of MES Schedule Part I and shall be bright with smooth surface. Frosted glass shall be provided to windows/vents of toilet, bath/WC.

7.12.15 All external windows/ventilator shall be provided with aluminium grill except toilet as shown.

7.12.16 Guard Bars/Grills:- Aluminium Guard bars/grills shall be provided all as indicated / shown on drg. Aluminium grill design shall be as per cat No DG-104 (10 mm) of Decogrill Make and all as specified and shown on drawing. The aluminium grills shall be fixed to frames with adequate number of screws.

7.12.17 Aluminium sections shall be obtained from any one of the manufacturer listed list of Makes.

7.13 **Stain less steel wire cloth**: Wire gauge wherever indicated on drawing shall be of Stainless steel and shall confirm to IS. Nominal dia of wire shall be 0.36 mm and average width of aperture shall be 1.40 mm and shall conform to relevant IS.

7.14 UPVC WINDOWS/ VENTILATORS

7.14.1 All windows/ ventilators frames and shutters shall be made up of UPVC extruded hollow profiles all as shown in Architectural /TD drawings. White UPVC extruded hollow profiles used in windows/ ventilators shall conform to EN 12608-1:2016 and surface covered UPVC extruded hollow profiles used in windows/ventilators shall conform to BS 7722.

7.14.2 Fabrication and fixing shall be done as per standards mentioned in UWDMA guidelines.

7.14.3 OPTIMA Series of UPVC sections shall be used for Windows/ ventilators of size upto 1200x1200mm and INVENTA Series of UPVC sections shall be used for Windows/ ventilators of size above 1200x1200mm.

7.14.4 Appropriate hardware fittings, locking arrangements, handles and hinges shall be provided to all windows and ventilators as per manufacturers instructions and as directed by Engineer-in-Charge at site.

7.14.5 The glass panel shall be all as specified in IS-1948. Glazing shall be with high quality float glass of 5.5mm thickness. Ground/figured glass shall be provided to windows/vents of toilet, bath/WC.

7.14.6 Mosquito proof net/ wire mesh to be provided for UPVC windows/ventilators shall be of stainless steel as specified here-in-after.

7.14.7 Factory made Aluminium grills shall be provided with UPVC windows/ventiloators as shown on drawings. Aluminium grills shall be 10mm thick, DG-104 pattern of Make Decogrill or other approved equivalent makes. After fabrication, aluminium grills shall be cleaned in the seven tank cleaning process and powder coated. Minimum thickness of powder coating shall be 50 micron. Shade of powder coating shall be as directed by GE. Necessary openings shall be provided in the grill for smooth operation of shutters as directed by Engineer-in-Charge at site. Edges of openings shall be suitably provided with U-Channel.

PARTICULAR SPECIFICATIONS (Contd…)

8.0 BUILDER'S HARDWARE

8.1 Items and quantities:-Hardware fittings shall be provided according to the scales indicated on the schedule. All items of builders hardware shall be of Stainless steel Grade SS-304 or nearest available grade and shall be of ISI marked on the relevant drawings and in conformity with any note that may be appearing on a particular drawing.

8.2 Sizes: In case the size of particular fitting is not given in the drawing, it shall be of size as directed by the GE.

8.3 All articles of builders hardware shall bear ISI marking. In case ISI marked particular item is not manufactured, it shall conform to the relevant IS and specifications given in MES Schedule.

8.4 Screws used for fixing items of builder's hardware shall be stainless steel screws.

8.5 All items of builders hardware shall be of Stainless steel Grade SS-304 or nearest available grade and shall be of ISI marked.

8.6 All builders hard ware shall conform to relevant IS and shall be as per sample kept in GE office.

8.7 Articles

8.7.1 Butt Hinges: Butt hinges shall be of stainless steel hinges conforming to IS and of quality cold rolled medium weight as approved by GE.

8.7.2 Towel rails: Towel rails shall be tubular type of stainless steel with flanged ends for fixing. It shall be of minimum 19/20 dia mm and 600mm long."

8.7.3 Curtain rail/rods: If nothing is shown on drawings or specified anywhere else then 19mm dia Stainless steel (Gde S-304) curtain rod shall be provided as decorative fittings in all openings. Wall thickness of curtain rod shall not be less than 1.25mm.

9. WATER PROOFING TREATMENT

9.1 The contractor shall carryout water proofing work over roof to ensure the entire roof watertightness during the guarantee period of 10 years and he will be responsible to keep the entire surface of roof water tight for a period of 10 years from certified date of completion of work. The security Deposit at the rate of 2.5% of the waterproofing treatment work done amount towards the guarantee for water proofing treatment shall be retained from the RAR/final bill amount. The amount shall be released to the contractor after the satisfactory completion of TEN YEARS guarantee period. During the guarantee period if any leakage is found out same shall be got rectified by the contractor without any extra cost to the Government. In any case during the guarantee period the contractor who executed the WPT works should inspect and examine the treatment once in every year and make good any defects observed. Defects liability period of one year as laid down under condition 46 of IAFW-2249 is not applicable for water proofing work covered under this contract and shall be deemed to be amended to the extent mentioned above for WPT work. Garrison Engineer may accept Bank Guarantee Bond from Schedule Bank or fixed deposit receipt (pledged in favour of GE) from any approved Bank for the above said amount for a period of 10 years in which case no further amount will be recovered from the final bill on this account.

9.2 Tenderer shall submit 10 years guarantee certificate (as per proforma enclosed herewith) duly undertaking and accepting the responsibility to carryout all required rectifications at his own risk and cost. In case of contractor's failure to carryout the rectifications, the Government shall be at liberty to get all such rectifications carried out at his risk and cost from security amount. Tenderer is required to visit the buildings in which the water proofing treatment is to be done in order to ascertain the exact nature and extent of work involved. His quoted rate in the **Schedule 'A' Part Item No1** against the respective items shall be deemed to include for the following aspects for effective water proofing treatment and no extra claim on whatever account will be entertained from the contractor.

PARTICULAR SPECIFICATIONS (Contd…)

9.3 **WATER PROOFING MEMBRANE**

APP modified water proofing membrane incorporate in the work for WPT shall be only low VOC membranes and shall be obtained from one of the manufacturers listed in Appendix –B herein after

9.4 **PHYSICAL PROPERTIES:** The physical properties of APP modified water proofing membrane shall be as under:-

i)	Atactic Poly Propaneylene (APP) modified bituminous membraneweight	Not less than 3.50 Kg/Sqm
ii)	Single centre core non-woven polyester reinforcement weight	> =160 gms/ sqm.
iii)	Cold flexibility	Minimum (-) 2°C.
iv)	Softening paint of Membrane	Not less than 150°C
v)	Tensile strength	In Longitudinal direction- 650N/ Sqcm& in Transverse direction 350N/Sqcm
vi)	Tear Strength (L/T)	Longitudinal > 300 N,Transverse > 250 N
vii)	Elongation	> 20% in both directions.
viii)	Dimensional stability %change	Maximum 1%.
ix)	Water tightness/ impermeable at	2 kgf/ cm2.
x)	Water absorption	Not more than 1% on mass within 24 hours.

9.5 The water proofing membrane treatment shall be carried out by the approved applicator of the manufacturer as approved by GE.

9.6 **CERTIFICATE BY SPECIALISED AGENCY**

A certificate by the specialized firm, who has executed the work of water proofing treatment under this contract will be rendered to the department stating that the said work has been executed by them and shall be signed jointly with the main building contractor. This certificate however does not absolve the main contractor from his responsibility in respect of specialist work as per contract condition. The main contractor shall be responsible to ensure tat the roof slab shall be water tight/leak proof during guarantee period”

9.7 **APPLICATION AND WATER PROOFING OVER ROOF**

9.7.1 SURFACE PREPARATION: After rectification is completed, top surface of the RCC roof slabs shall be thoroughly cleaned of any dust, loose materials, grease etc wire/coir brushes. The surface shall than be thoroughly washed with fresh water.

9.7.2 **FIVE LAYERED WATER PROOFING TREATMENT WITH ATACTIC POLYPROPYLENE POLYMER MODIFIED PREFABRICATED MEMBRANE**

(i) Atactic Polypropylene Polymer modified prefabricated five-layer water proofing membrane shall be of 3 mm thickness 3.5 Kg/Sqm weight with non-woven polyester matt.

(ii) The prefabricated five layered black finish water proofing membrane comprises of centre core of 170 gsm nonwoven polyester matt sandwiched on both sides by APP polymer modified bitumen which is protected on both sides by 20 micron thermofusible polyethylene sheet. Composite thickness of the membrane including all five layers shall be 3 mm with non-woven polyester matt.

(iii) Primer: Bitumen primer for bitumen membrane shall have density at 25°C in the range of 0.87 - 0.89 kg/litre and viscosity of 70-160 CPS primer shall be applied @ of 0.40 litre/sqm.

(iv) Laying: Over roof concrete, apply 10mm thick rendering in CM 1:3, finished even and smooth whilst concrete is green. Then following treatment shall be applied:-

a) Bitumen primer @ 0.40 lts/sqm shall be applied to the prepared roof, drain and all other surfaces where polymer modified membrane is to be laid.

b) The five layered water proofing membrane shall be laid using Butane torch and sealing all joints and preparing the surface complete.

PARTICULAR SPECIFICATIONS (Contd…)

c) Drain outlets shall be given same treatment as specified for the roof in the description of the item in the manner specified for the flat roof surface.

d) Water proofing treatment shall be carried into the drain pipe or outlets by at least 10 cm.

e) The water proofing treatment laid on the roof surface shall overlap the upper edge of the water proofing treatment in the drain outlets by at least 10 cm.

f) The APP polymer modified prefabricated water proofing membrane shall be cut to the required length.

g) Water proofing membrane shall normally be laid in length in the direction of the slope and laying shall be commenced at the lowest level and worked upto crest.

h) APP water proofing membrane shall be laid in 6 to 8 m lengths

i) The roof surface shall be cleaned and bitumen primer shall be applied in the correct quantity, over this specified water proofing membrane shall be laid with butane torch after allowing 24 hours for primer to dry.

j) Each strip shall overlap the preceding one by at least 10 cm. at the longitudinal edges and 15 cm. at the ends.

k) All overlaps shall be firmly bonded with bitumen primer and levelled by heating the overlap with butane torch.

l) APP water proofing membrane shall be laid as flashing wherever junction of vertical and horizontal surfaces occurs.

m) Longitudinal laps shall be 100 mm.

n) The upper edge of flashing membrane shall be well tucked into the flashing grooves in the parapets, chimney stack etc. to a depth of not less than 6.5 cm; corresponding applications of primer coat shall also be made.

o) The flashing treatment shall be firmly held in the grooves and it shall be sealed with the approved sealant after terminating the membrane.

p) Where parapet walls are of height 45 cm or less APP water proofing membrane flashing shall be provided in the same manner as for splashing in the case of high parapet walls except that upper edge shall be carried out to the full height of the wall and taken right across the top of the parapet and down on the external vertical faces to a minimum distance of 5 cm.

q) Finishing surface with solar reflective bituminous aluminium paint, @ 100 gm/ Sqm. The materials used in water proofing treatment shall be eco-friendly and shall be any one of the make listed in List of approved materials here in after.

(v) **FINISHING LAYER FOR INACCESSIBLE ROOF**: In case of non accessible roof, the APP membrane shall be finished with two coats of solar reflective bituminous aluminium paint @ 0.1kg/sqm, over the above specifications.

(vi) **FINISHING LAYER FOR ACCESSIBLE ROOF AND TERRACES**: In case of accessible roof and terraces, irrespective of what is mentioned in schedule of finish, provide a polypropylene/ polyester, non-woven geotextile fabric of 150 gsm over the membrane, specified here-in-above over 10mm thick rendering in CM 1:3, finished even and smooth whilst concrete is green. Over the geotextile fabric provide 20 mm thick cement screed (1:4) and lay 22 mm thick hydraulically pressed PCC tiles of size 250X250 mm set and jointed over APP membrane. While laying tiles a care shall be taken to see that joints between the tiles are kept minimum (not exceeding 5 to 8 mm). Top of finished surface of tiles shall be laid to the true slope so that no stagnation of water take place.

PARTICULAR SPECIFICATIONS (Contd…)

(vii) FINISHING LAYER FOR RCC WATER TANK: In case of water Tank roof, provide two coats UV Resistant acrylic polymer blended elastomeric coating over APP membrane as above specified.

9.8 Wherever junction of vertical and horizontal structures occur a groove of size 75 x 65mm shall be formed at a height of 15 cm above roof level for tucking in edge of water proofing treatment in case of brick wall etc. The groove shall coincide with the horizontal joint of brick course and shall be shaped with cement and sand mortar (1:3). In case of junction between RCC beam/ parapet etc. PCC 1:2:4 coving with 75mm in radius shall be provided. Cost of aboveprovision shall be deemed to include in the quoted rate against respective Items of Schedule 'A' Part I.

9.9 TESTING OF MEMBRANE

The following tests shall be carried out in approved Government Laboratories/ NABL accredited lab and necessary test reports shall be submitted to Garrison Engineer for the materials before incorporation in the work

a) Thickness of membrane
b) Weight of membrane
c) Tensile strength (Longitudinal/Crosswise)
d) Tear Resistance
e) Softening point
f) Weight of Carrier (Polyester).

Apart from above tests contractor shall also produce manufactures test certificate. Test certificate shall be produced with every consignment of material brought at site by the contractor.

9.10 TEMPLATE (LABEL)

Template (Label) made of 15mm thick plaster in CM (1:3) of size 60 x 60 cm showing the following particulars shall be provided on each bldg before the final completion of work without any extra cost to the Department: -

(a) Name of Contractor
(b) CA Number
(c) Date of completion
(d) Guarantee period

9.11 The above information shall be written with white paint with 5 cm size letters and numbers. The entire quantity of materials such as water proofing membrane bitumen primer, bitumen and bituminous aluminium paint etc required for the whole work shall be brought at site in sealed position and shall be entered in the MB as 'NOT TO BE ABSTRACTED' showing the date, total quantity brought. Each seal shall be opened in front of Engineer-in-Charge only. Paid vouchers from authorized dealer shall be submitted to the Engineer-in-Charge immediately after every consignment and copy of such paid voucher shall be forwarded to Garrison Engineer b y Engineer-in-Charge duly endorsed.

9.12**WATER PROOFING TREATMENT TO SLOPED ROOF (1:3/30 Degree or other sloping roof)**

9.12.1 Concrete for roof slab shall be M-30 design mix having water cement ratio 0.45 (maximum) using approved plasticizer.

9.12.2 (a) After laying the concrete provide 5mm thick (minimum) plaster in CM (1:3) mixed with an approved integral water proofing compound in liquid form (dose and method as per manufacturer's instructions). The above plaster shall be applied when RCC is green so that the plaster is integral part of RCC slab. After curing of plaster clean the entire surface of roof slab thoroughly with wire brush and remove all deleterious matter such as dust, dirt, loose mortar etc, and rectify the undulation if any using cement mortar (1:3) and wash the surface with clean water before application of water proofing treatment.

(b) Prepare the water proofing slurry by adding" DITCHTAMENT' (Polymer modified mineral based powder) into water diluted "NAFUFILLBB2 (acrylic based polymer binding agent). NAFUFILLBB2 liquid polymer shall be diluted with four parts of water. Apply this slurry @ 2Kg/Sqm in two coats over the RCC slab surface. 60 grams of NAFUFILLBB2 per Kg of DICHTAMENT shall be used.

OR

Over the clean surface, apply POLYFLEX (super-C), a two part elastomeric flexible water proofing compound, manufactured by POLYBIT INDUSTRIES LTD. In this two parts system first part is powder working on theory of crystalline penetration and capillary water proofing and the second part is a liquid compound based on styrene Butadiene Rubber (SBR).The surface shall be damp but not wet at the time of application of POLYFLEX. Add powder to the liquid at a ratio of (3:1) slowly and mix using a slow speed drill fitted with suitable paddle.

PARTICULAR SPECIFICATIONS (Contd…)

(c) Do not mix more material that can be used within 45 minutes. Apply the mixed material @ 2 Kg/Sqm on the prepared surface two coats by a brush or roller. There shall be gap of one hour between the two coats.

(d) The above treatment shall be water cured for seven days. Before laying MP tiles, the sloped roof shall be sprinkled with water, using stirrup pump, continuously for 6 hour left for 1 hour then the RCC roof shall be stage passed by GE for efficiency of water proofing treatment.

(e) The chemicals specified above shall be obtained from any one of the manufacturers listed in list of manufactured or through their authorised dealers. All treatment shall be carried out as per manufacturer's instructions and as directed by Engineer-in-Charge. The material shall be obtained in manufacture's sealed drum container and shall be approved by GE before incorporating in the work.

(e) The chemicals shall be applied only by specialist agency/approved applicator of the manufacturer. The specialist agency/approved applicator shall give necessary guarantee for ten years for the effectiveness of the treatment carried out.

9.12.3 After curing, provide mortar bands in fillet in CM (1:4) (size –width 70mm at bottom 25mm at top and 40mm thick) at 30cm centre to centre. The end band shall be of width 90mm at bottom and 25mm at top and 75mm thick. The end band shall be in cement concrete (1:2:4) type B1. Provide 30mm gap at 500mm c/c in the mortar/PCC bands.

9.12.4 Lay Mangalore pattern roof tiles and ridge tiles of approved make over the bands.

9.12.5 MP roofing clay tiles shall comply to requirement of IS-654-1972 and shall be class 'AA' quality double groove type. Ridge tile shall conform IS –1464-1973 and shall be class 'AA' quality. MP tiles and ridge tiles shall be all as per clause 11.5 to 11.6.2 of MES Schedule Part I. MP tiles shall be obtained from M/S Common Wealth Trust India Ltd / M/S Regal Tiles / M/S Thomas Stephen / M/S Cherupushpam / M/S Kalyanam Tiles Co, Pudukkad, Trichur / M/S Scared Heart Tile works, Trichur /

M/S ST Joseph Clay Works, Kalady . Laying of tiles shall be all as per clause 11.25 to 11.25.9 of MES Schedule Part I. MP tiles shall be provided with two coats of antifungal compound such as ROFF HYPLAST(Product code of 221 of ROFF).

9.12.6 The contractor shall furnish a guarantee or a period of 10 years for the effectiveness of water proofing treatment and Security Deposit all as specified in Para 10.2 here in after.

9.12.7 Mini MP tiles shall be laid over RCC chajjah as shown on drawings. Mini MP tiles of size 228x152mm,class 'AA' shall be laid over 15mm thick cement mortar 1:4 screed mixed with integral water proofing compound. Laying of tiles shall be as per 11.2.5 to 11.25.5 of SSR part-I. MP tiles shall be provided with two coats of antifungal compound such as ROFF HYPLAST (Product code of 221 of ROFF).

9.13**Water proofing to floor of toilets/Bathroom/Kitchen**: Water proofing treatment shall be carried out to local sunk of bathroom/toilets and kitchen as specified hereinafter. All the plumbing lines shall be sealed with non shrinking epoxy grout after thorough cleaning and ponding for 48 hours shall be done to check the leakage. If any leakage observed the same shall be made good with pressure grouting using suitable admixture at 225g/bag of cement or as manufactures instruction. Cracks if any noticed shall be filled up with non shrinkable grout. Coving has to be done at all corners at 90 degree with polymer modified mortar as per manufacturer instruction. The product shall be two component acrylic polymer modified cementitious flexible water proof coating which shall be further mixed with water at site as recommended by the manufacturer. The coating shall be applied in two coats to form a total thickness of 1mm. The water proof coating shall be protected with plastering in CM 1:4 using integral waterproofing compound to a minimum thickness of 20mm. The contractor shall give 5 years guarantee for water proofing of these area also. The work shall be got done through authorized applicator of the manufacturer.

PARTICULAR SPECIFICATIONS (Contd…)

9.14 GUARANTEE:

The contractor shall furnish a guarantee for a period of 10 years for the effectiveness of all water proofing treatment carried out as specified here in before for the buildings catered under Sch 'A' Part I before completion of work and the same shall be reflected in the completion certificate. The contractor shall furnish a guarantee in favour of GE for the efficiency of the water proofing treatment during the guarantee period. Security deposit towards the guarantee for all water proofing treatment work executed under this contract shall be retained from the payment due to contractor. Amount of Security deposit towards the guarantee , shall be calculated as per the scales laid down by MES for calculating individual Security Deposit on the amount of water proofing treatment at contract rates for the buildings described under Sch 'A' Part I. This amount shall be released after successful expiry of the guarantee period. The contractor may, however, furnish a fixed deposit receipt in lieu, from a Scheduled Bank pledged in favour of GE.

10.0 **GUARANTEE TO BE EXECUTED BY CONTRACTOR FOR REMOVAL OF DEFECTS AFTER COMPLETION IN RESPECT OF WATER PROOFING WORKS**.

This agreement made this ___day of _____ two thousand_____between _____of_____(son of _____ of_____ (hereinafter called guarantor on the one part) and the PRESIDENT OF INDIA (hereinafter called the Government on other part).

WHEREAS THIS agreement is supplementary to a contract hereinafter called the contract) dated _____ and made between the GUARANTOR of the one part and the

GOVERNMENT on the other part where by the contractor interalia, undertook render the buildings and structures in the said contract recited completely water and leak proof.

AND WHEREAS THE GUARANTOR agreed to give a guarantee to effect that the said structures will remain water leak proof for ten years from the certified date of completion.

NOW THE GUARANTOR hereby guarantees that water proofing treatment given by him will render the structures completely leak proof and the minimum life of such water proofing treatment shall be Ten years to be reckoned from the certified date of completion. Provided that the guarantor will not be responsible for leakages caused by earthquakes or structural defects or misuse of roof and sunken floor slabs or alterations and for such purpose.

a) Misuse of and sunken floor slabs shall mean by operation which will damage roofing treatment, like chopping of firewood and things of the same nature which might cause damage to the roof.
b) Alteration shall mean construction of any additional storey or any part of roof or construction adjoining to existing roof, where by roofing treatment is removed in parts. And removal of finishes of sunken portion of slabs.
c) The decision of the Garrison Engineer with regard to cause of leakage shall be final and binding. During this period of guarantee the guarantor shall make good all defects and in case of any defects being found, render the building water proof and finish the disturbedsurfaces as existed to the satisfaction of Garrison Engineer at his cost and shall commence the work for such rectification within seven days from the date of issue of the notice from the Garrison Engineer calling upon him to rectify the defects, failing which the work shall be got done by the department by some other contractor at the GUARANTOR'S risk and cost. The decision of the Garrison Engineer as to the cost, payable by the Guarantor shall be final and binding. That if the Guarantor fails to execute the water proofing or commit breach there under then the Guarantor will indemnify the Principal and his successors against all loss, damage, cost expense or otherwise which may be incurred by him by reason of any default on the part of GUARANTOR in performance and observing of this supplementary agreement. As to the amount of loss and/or damage and/or cost incurred by the Government, the decision of the Garrison Engineer will be final and binding on the parties.

PARTICULAR SPECIFICATIONS (Contd…)

IN WITNESS WHEREOF these presents have been executed by the obligator_____and by _____for and on behalf of the PRESIDENT OF INDIA on the day month and year first above written.

SIGNED sealed and delivered by OBLIGATOR in the presence of :-

1.

2.

SIGNED ON BEHALF OF THE PRESIDENT OF INDIA BY _____

11. **STEEL AND IRON WORK**

11.1 Quality of steel to be incorporated in the work shall be as follows:-

High strength deformed steel bars produced by Thermo Mechanical treatment process (In short called as TMT bars) for reinforcement.	Of grade Fe 500 / Fe 500D / Fe 550 /Fe 550D and meeting all other requirement of IS 1786-1985 (Reaffirmed) 1990. (Different grade of steel shall not be used in same building)
High strength deformed CRS bars produced by Thermo Mechanical treatment process (In short called as CRS TMT bars) for reinforcement.	Corrosion resistant steel TMT bars of grade Fe 500 / Fe 500D / Fe 550 /Fe 550D and meeting all other requirement of IS 1786-1985 (Reaffirmed) 1990. (Different grade of steel shall not be used in same building). **NOTE:- (i) Irrespective of whatever is shown elsewhere CRS TMT bars shall only be used in this work** **(ii) Reinforcement of whichever type and grade mentioned in these documents and drawings shall be CRS TMT bars only**
Mild steel for miscellaneous works	Shall conform to IS-432(Part-I). Grade wherever not shown/ indicated otherwise shall be of Grade I.
Structural steel	Steel for general Structural purpose shall be Grade Fe-410 WA ISI marked (IS-2062) for all type of steel Structures including those subject to dynamic loading. Structural steel ordinary quality shall confirm to IS-1977
Galvanised steel sheets (Plain & corrugated)	Conforming to IS-277.
Fabric reinforcement for concrete	Conforming to IS-1566

11.2 (a) For pricing deviations involving TMT bars the rates given in SSR part-II (subject to contractor's percentage) shall be applied irrespective of the grade of TMT bars.
(b) For pricing deviations involving steel for general structural purpose Gde Fe-410WA, the rate shall be applicable as per Gde Fe-410-W given in SSR Part II adjusted by applicable percentage for respective parts of Schedule 'A'.

11.3 **WELDING**

11.3.1 Welding of steel and ironwork shall be done in an approved manner with electric arc welding.

11.3.2 Electric welding shall be metal arc welding as specified in Para 10.15 of MES Schedule Part-I.

11.3.3 Welding electrodes shall be of quality suitable for welding of structural steel and shall comply with requirement of IS-814 for covered electrodes for metal arc welding of mild steel.

11.3.4 Unless otherwise specified/shown on drawings, the fillet welds shall be of 6mm size.

11.3.5 All butt welds shall be full penetration welds.

11.3.6 Testing of welds shall e as specified in drawing.

PARTICULAR SPECIFICATIONS (Contd…)

11.4 **REINFORCEMENT**

11.4.1 Reinforcement shall be bent to shape and provided as indicated on the drawings.

11.4.2 All laps and crossings shall be tied with mild steel wire (annealed) of size not less than 0.9 mm dia.

11.4.3 Unless otherwise shown on drawing, bends to a radius of 4 times the dia of bar shall be provided in respect of tor steel bars. In case of deviations involving reinforcement bars, the length of each bar for the purpose of calculating laps shall be taken as 10 (Ten) metres for omit purpose.

11.4.4 Unless otherwise shown on drawings, spacing of chairs shall be 75cm centre to centre below the cranked up bars to keep the bars in position.

11.5 **REINFORCEMENT STEEL AND STRUCTURAL/NON-STRUCTURAL STEEL**

11.5.1 **TYPE OF STEEL**

11.5.1.1**REINFORCEMENT STEEL**

(a) Wherever mild steel bars have been shown on drawings TMT bars of same size and grade Fe 500D /550D shall be used in lieu.

(b) Wherever deformed bar(s) have been shown in drawings, TMT bars of the same size and grade Fe-500D/550D shall be used in lieu.

(c) However 6mm bars (MS and CTD bars) wherever shown on drawings, the same shall be amended to 8mm TMT bars without changing spacing/numbers. The grade shall be as explained in (a) and (b) above,

The provisions in (a), (b) & (c) above shall be carried out by the contractor at no extra cost to the Govt.

11.5.1.2**STRUCTURAL STEEL**

(a) Unless otherwise shown in drawing / specified elsewhere, Structural Steel shall conform to IS 2062-1999, Grade Fe-410-WA for all types of steel structures including those subject to dynamic loading.

(b) Unless otherwise specified elsewhere, mild steel in grills/guard bars, railing, flat iron, hold fast and in various members of steel windows and steel doors shall be of ordinary quality conforming to IS 1977-1975 Grade Fe-310-O.

11.5.2 **SOURCE OF PROCUREMENT**

11.5.2.1 **REINFORCEMENTSTEEL** : All TMT steel bars shall be of grade Fe 500D / 550D meeting all other requirements of IS-1786 and shall be procured directly from the manufacturers as given in Appendix list of makes and manufacturers. However, the tenderer may procure the reinforcement of make if any approved by the HQ E-in-C branch during the currency of the work i.e. after bid submission end date, subject to compliance of tender provisions without any price adjustment.

11.5.2.2**STRUCTURAL STEEL**:- It shall be of grade Fe-410 WA ISI marked and shall be procured directlyfrom primary producers as approved by E-in-C's Branch i.e. **SAIL/ RINL/TISCO/JINDAL**. Further, in case of non availability of structural steel with primary producer manufactures, the same can be procured from approved producers / Secondary producers approved by E-in-C's branch New Delhi (till the date of receipt of tender) with minus price adjustment with prior approval of accepting officer. The minus price adjustment shall be done @ 5% of accepted rates i.e. SSR Rate enhanced by deviation percentage/ contractor's percentage as applicable as mentioned in the contract . In case any section is not being rolled / manufactured by primary producer, there shall not be any price adjustment on account of procurement of this section from approved secondary producers.

11.5.2.3 Galvanised Steel sheet & Fabric Reinforcement for concrete: These shall be ISI Marked and shall be procured directly from Main manufacturers as approved by E-in-C's Branch.

PARTICULAR SPECIFICATIONS (Contd…)

11.5.2.4 However, reinforcement steel, structural steel, Galvanised steel sheets and Fabric Reinforcement for concrete shall be permitted to be procured from authorised dealers of main producers in case the total smaller contracts of value not exceeding Rs 7.5 lakhs.

11.5.2.5 Steel sections for railing, gates, fencing, guard bars, grills, steel chowkhat, holdfasts etc., which do not constitute structural members, can be procured from main / primary /secondary producers/BIS marked manufacturers or their authorised dealers at the option of Contractor without any minus price adjustment. Test will not be insisted upon for such steel sections

11.5.2.6 The contractor should place their demand/requisition of steel with the producers with adequate lead time.

11.5.2.7 Steel, when procured from primary producers/approved secondary producers/authorized conversion agents of main/ primary producers (stipulated in above paras) shall have to be procured directly from the storage depots of the main / primary producers/approved secondary producers/authorized conversion agents and not from their authorized agents/ dealers.

11.5.2.8 The particulars of the manufacturer/supplier of steel shall be submitted by the contractor to the Garrison Engineer prior to procurement for every lot of steel separately. The original documents in support of purchase of steel shall be verified by the site staff and Garrison Engineer and certified true copy of the result shall be retained in GE's office. The form given here in after will be used for this purpose.

11.6 **TESTING OF STEEL**

11.6.1The manufacturer is to carry out inspection and testing of steel in accordance with the relevant BIS provisions. The contractor shall submit the manufacturer's test certificate in original along with the Test Sheet giving the result of each mechanical test as applicable and the chemical composition of the steel or authenticated copy thereof duly signed by the manufacturer with each consignment. The Engineer-in-Charge shall record these details in Steel Acceptance Register (as per proforma given hereinafter after due verification and send a certified true copy of test sheet to GE for his records. Any particular size of bar/steel section of any consignment, not meeting the requirement shall be rejected and the same shall be removed from the site by the contractor at his own cost and the contractor shall have no claim on this account.

11.6.2 Testing shall not be insisted upon for steel given in Para 11.5.2.7 above.

11.6.3 **INDEPENDENT TESTING**

11.6.3.1The GE shall also organize independent testing of random samples of steel drawn from various lots from National Test House, SEMT Wing CME Pune, Regional Research Laboratories, Govt approved laboratories, Zonal laboratories,NABL accredited labas per the recommended minimum frequency as in Para 11.6.4. In addition samples from each lot shall be tested for quality and elongation as per the parameters given in STEEL SUPPLY & ACCEPTANCE FORM. The elongation shall not be less than 18 %.

11.6.3.2. **REINFORCEMENT STEEL**

(i) Independent testing of steel by the GE shall be mandatory as per frequency laid here in after and testing charges shall be borne by the contractor irrespective of the outcome of test results.

(ii) The contractor at his cost shall provide all facilities required for the testing and cost of materials consumed in tests, samples, conveyance etc shall also be borne by the contractor.

11.6.3.3 **STRUCTURAL STEEL**

(i) Independent testing of structural steel by the GE shall be optional at the discretion of the GE in case of procurement from main producers and testing charges shall be borne in accordance with Condition 10 A of IAFW – 2249 i.e., testing charges shall be borne by the Deptt if the test results are found in order, otherwise these shall be borne by the contractor.

PARTICULAR SPECIFICATIONS (Contd…)

(ii) Independent testing of structural steel by the GE shall be mandatory in case of procurement from secondary producers/authorized conversion agents of main producers and testing charges shall be borne by the Contractor irrespective of the outcome of test results.

(iii) In both the cases at (i) & (ii) above, the contractor at his cost shall provide all facilities required for the testing and cost of materials consumed in tests, samples, conveyance etc shall also be borne by the Contractor.

11.6.4 **FREQUENCY FOR TESTS ON STEEL**

11.6.4.1 **REINFORCEMENT STEEL:** Nominal mass test, Tensile test, Bend and Rebound test shall be carried out as per minimum frequency given below: -

Nominal Size of Bars		Frequency
(a)	Bar size less than 10 mm	One sample (3 specimens) for each test for every 25 tonnes or part thereof.
(b)	Bar size 10 mm to 16 mm	One sample (3 specimens) for each test for every 35 tonnes or part thereof.
(c)	Bar size over 16 mm	One sample (3 specimens) for each test for every 45 tonnes or part thereof.

11.6.4.2 **STRUCTURAL STEEL**: The tests shall be carried out as per the frequency mentioned below:-

Type of Test		**Frequency**
(a)	Tensile Test	One test for every 25 tonnes of steel or part thereof.
(b)	Bend Test	One test for every 25 tonnes of steel or part thereof.

NOTE:

1. For various tests, Acceptance criteria, tolerance etc refer to Steel supply & Acceptance Form given here in before and relevant BIS Codes.
2. For steel items, for which, independent test is mandatory, as specified above, payment shall be released against these items only after satisfactory test report. Any item of steel, failing in test, shall be rejected and the particular consignment removed from the site by the contractor at his own cost.

11.6.5 **DOCUMENTATION**: The contractor shall submit original purchase vouchers from the manufacturer for the total quantity of steel supplied under each consignment to be incorporated in the work. All consignments received at the work site shall be inspected by the GE along with the relevant documents before acceptance. The original purchase vouchers and the test certificates shall be defaced by the Engineer-in-Charge and kept on record in the office of the Garrison Engineer duly authenticated and with cross reference to the consignment number recorded in the steel Acceptance Register. The steel Acceptance Register shall be signed by the JE, Engineer-in-Charge, GE and the Contractor. The entire quantity of all steel items shall be suitably recorded in the measurement book as not to be abstracted, before incorporation in the work and shall be signed by Engineer-in-charge and the contractor.

11.7 **STORAGE ACCOUNTING, PRESERVATION AND MAINTENANCE OF STEEL**

11.7.1 The storage, accounting, preservation and maintenance of steel supplied by the contractor shall be done as per standard engineering practice till the same is incorporated in the work and the cost of the same shall be deemed to be included in the unit rate/amount quoted by the tenderer. The GE shall inspect at regular interval to verify that steel lying at site is stored, accounted, preserved and maintained as per the norms. The steel shall be stored so as to differentiate each consignment separately. If the GE is not satisfied with the storage / preservation of any size of bar/section of steel, he may order for any test(s) of steel as applicable for that size of bar/section of steel and as specified in tender documents /relevant IS code to recheck the acceptability criteria for the same. The contractor shall bear the cost of necessary testing(s) in this regard and no claim whatsoever shall be entertained.

PARTICULAR SPECIFICATIONS (Contd…)

11.7.2 Movement of steel shall be recorded in In/Out steel register as per following proforma. Each entry in the register shall be signed by the contractor and Engineer-in-Charge.

IN/OUT STEEL REGISTER

Sl No	Date	Steel IN			Steel OUT			Qty Balance
		Qty (Tons)	Section	Control No	Qty (Tons)	Section	Reasons*	
1	2	3	4	5	6	7	8	9

* **Note**:

(i) The following reasons may be mentioned for taking out steel from storage: -

(a) For testing purpose

(b) For use in work

(c) Rejected steel taken out of site

(ii) All the transaction in the register shall be signed by Contractor / his representative and Engineer-in-Charge/JE.

11.7.3 **SCHEDULING AND SUPPLY** :-Schedule of supply of steel will be finalised by GE in consultation with contractor and the same will be incorporated in CPM chart so that supply of steel is monitored in a way to avoid any delay in completion of the work. The schedule of supply of steel will be vetted by Accepting Officer or Officer nominated by the Accepting Officer from time to time.

11.8 **MEASUREMENTS AND PAYMENT OF STEEL**

11.8.1 The entire quantity of all steel items shall also be suitably recorded in the Measurement Book for record purposes as 'Not to be abstracted' before incorporation in the work and shall be signed by the Engineer-in-Charge and the contractor.

11.8.2 The nominal mass conversion factors for various steel Sections/ size of finished baras given in relevant IS codes shall be considered standard for measurement.

11.8.3 The contractor shall plan to procure steel considering the time required for approval and testing of steel well in advance. No extension of time on grounds of delay in procurement of steel or testing/retesting of steel shall be admissible.

11.8.4 The payment of steel shall only be allowed after production of original purchase vouchers, test certificates by the contractor for each consignment of steel and results of testing carried out by the department are found satisfactory after testing as specified herein before.

11.8.5 **FABRICATION OF STEEL STRUCTURES**: - The structural steel work shall be carried out all as specified in MES Schedule Part-1, Section – 10 and in drawings. The quality of fabrication and erection at site shall be got certified to the satisfaction of the department by any of the certifying bodies as approved by Accepting Officer.

11.9 **HOLDFAST/LUGS**

(i) Flat iron holdfast/lug shall be provided by welding as and where shown on drawings except those to be provided to wooden chowkhat, which shall be fixed with screws as per details shown on drawings.

(ii) Flat iron holdfast shall be embedded in PCC (1:3:6) type C-1 block of size 15 cm (height of block) x30 cm (length of block) x width of masonry wall.

(iii) Holdfasts/lugs shall be hot tarred and sanded before fixing wherever doorframe and windows frame side is coming in contact with RCC column, dash fasteners shall be provided in lieu of holdfast.

PARTICULAR SPECIFICATIONS (Contd…)

STEEL SUPPLY & ACCEPTANCE REGISTER

1. CA No & Name of Work : _____
2. Contract No :_____
3. Name of Manufacture's TC No :_____
4. Manufacturer :_____
5. Random Test Details
   (a) Physical test report from _____ vide their letter No._____

   (Name of NABL approved lab / Govt .Engg College)
   (b) Chemical test report from _____ vide their letter No._____
(Name of NABL approved lab / Govt Engg College).
6. Type of Steel, Dia & Qty:
   (a) Type: TMT/CRS
   (b) Dia : ___mm
   (c) Actual Wt : ___MT
   (d) Conversion Wt : ___MT

	Chemical Test							Mechanical Test						
	Car-bon %	Sulphur%	Phosphorous%	Sulphur % + Phosphorus %	Manganese%	Silicon %	Corrosion Resistant element	Wt per Mtr	Yield Stress (N/mm^2)	Tensile Strength (N/mm^2)	Percentage Elongation (Min 18%)	Bend test	Re-bend Test	Remarks
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
As per IS-1786 – 2008														
As per Manufacture's test cert														
As per indep-endent Test														

Remarks with Signature Accepted/Rejected

Contractor Junior Engineer Engineer –in –Charge GE

Remarks of BOO/Inspecting officer/CWE----------

PARTICULAR SPECIFICATIONS (Contd…)

11.9.1**FAN HOOKS:** Fan hook shown in drawings shall be provided wherever ceiling fan/fan hook is indicated in drawings. The lump sum tendered by the contractor for the building under schedule 'A' Part-I shall be deemed to include for the cost of the same.

11.9.2 **PRESSED STEEL DOOR FRAMES**

(a) Pressed steel door frames wherever shown in drawings shall be fabricated with 1.25 mm thick MS plain sheet. Pressed steel door frames shall conform to IS-4351. Tolerance over profile shall be ± 2 mm.

(b) Space in pressed steel doorframes shall be filled with PCC (1:3:6) type C-0.

(c) Frames shall be supplied by manufacturers with a shop coat of zinc chrome primer.

(d) 12 mm square mild steel bar shall be welded horizontally at the bottom of frame. The 12 mm square bar shall be embedded in floors. In case of double shutter i.e. panelled door & gauzed door shutter, two nos. 12 mm square bar shall be provided at bottom of frame.

11.9.3 **STEEL FOR REINFORCEMENT**

Reinforcement shall be fabricated, placed in position all as shown on drgs and specified in clause 10.17 to 10.22 of MES Schedule (Part I) without application of heat.

12.0 FLOORING

12.1 General

12.1.1 Provisions contained in clause 13.25, 13.32, 13.38 & 13.39 of MES Schedule (Part I) are to be adopted for laying floors and pavements.

12.1.2 Floors shall be laid to levels or to falls as shown on drawings and as directed by the Engineer-in-Charge.

12.1.3 Floor finish shall be extended over dwarf walls, doors and other openings.

12.1.4 Floor topping of cast-in-situ cement concrete shall be finished even and smooth using extra cement as specified in clause 13.32.8 of MES Schedule Part I.

12.1.5 Pozzolana or pozzolana slag cement shall not be used for floor topping.

12.1.6 Floor finish over RCC slabs shall be laid all as specified in clause 13.32.5 of MES Schedule Part I.

12.1.7 Sub floor may not be laid in panels. The thickness of sub base concrete and hard core in the ground floor shall be as indicated in schedule of drawings. In case not indicated in drawings, it shall be 75mm thick PCC (1:5:10) over hard core of 100mm thick over rammed earth.

12.1.8 Under layer and topping layer of cast-in-situ cement shall be laid in panels conforming to pattern indicated in drawings. Length to width ratio not exceeding 1.5 times, not more than 3 dividing strips meeting at any joint. Dividing strips shall be of PVC width of strip shall be 2 mm less than thickness of PCC floor and 4mm nominal thickness. Top of dividing strips shall be finished smooth with the top surface of floor.

12.1.9 Floors shall be laid to level and/or to falls as shown on drawings and as directed by Engr-in-Charge

12.1.10 Floors shall be carried out through all opening in walls at floor levels, unless otherwise shown on drawings. In the case of different type of floor finish the dividing shall be the inner edges of the rebate of the doors unless otherwise shown on drawings.

12.1.11 The dividing line between the floors of different types wherever they so meet between adjoining rooms, shall be determined on the basis of the finish visible when the doors are closed and the applicable finish shall accordingly be provided.

12.2 **CEMENT CONCRETE FLOORS**:- PCC floors where indicated in drawings/Schedule of finishes shall be of the following composition :-

PARTICULAR SPECIFICATIONS (Contd…)

12.2.1 GROUND FLOOR:- Concrete floors shall be laid as specified in the schedule of finishes. The top layer/finishing layer shall be laid over a coat of cement slurry @ 2.5 kg per sqm over the sub base.

12.2.2 PVC DIVIDING STRIPS:- PCC floor topping shall be laid in bays/panels not exceeding 1.2mx1.2m or part thereof. Panels shall be formed with standard PVC dividing strips of 4mm thick and width of strips shall be 2mm less than the thickness of PCC floor topping (ie 38mm wide for 40mm PCC topping, 48 mm for 50mm PCC topping, etc). Bays shall be so formed that not more than 3 strips meet at a point i.e. only 'T' junction is formed.

12.2.3 Uses of all temporary fillets, side forms etc shall be dispensed with where glass-dividing strips occur. No price adjustment will be made on account of non-use of temporary fillets, side forms etc.

12.2.4 Surface of concrete floors, PCC skirting/ dado shall be treated with sodium silicate all as specified in para 13.35 on serial page 13-9 of SSR 1991 (Part-I).

12.2.5 RAMP: Ramps shall be provided as per details shown in drawing.

12.3 **CEMENT DADO/ SKIRTING**

12.3.1 Cement skirting/dado where shown on drgs shall be 6mm thick in cement and sand mortar (1:2) over 10mm thick cement and sand mortar (1:3) screed trawled even and smooth.

12.3.2 Height of skirting/ dado shall be as indicated on drgs. Height of skirting/ dado where not shown on drgs, shall be 150mm in case of skirting and upto window cill in case of dado. Dado/skirting shall be returned in Jambs, openings, exposed edges nitches etc.

12.4 **MACHINE CUT MIRROR POLISHED KOTA STONE FLOORING**

a) Machine cut mirror polished Kota stone flooring shall be provided in the locations indicated in drawings/Schedule of finishes.

b) The size and Thickness of Kota stone shall be as shown on drawings/ schedule of finishes. The tile/slab shall be laid over screed/ subbase as mentioned in Schedule of finishes. The stone slab flooring shall confirm and laid as specified in clause No. 13.47 to 13.47.4.5 os MES schedule Part-I.

c) Mirror Polished Kota stone slab shall be machine cut as mentioned in Schedule of finishes.The Kota stone shall be uniform in thickness & shall be uniform in colour.

d) The stone slabs laid on treads/ riser of steps shall be all as specified in schedule of finishes. Kota stone in treads shall be of one peice and edges shall be rounded/ bull nosed all as specified in schedule of finishes. Kota stone Risers/ Skirtings shall be laid over backing coat all as specified in schedule of finishes.

12.4.1 **MACHINE CUT MIRROR POLISHED KOTA STONE SKIRTING**

a) Kota stone skirting shall be provided at locations as shown on drawings. The height of skirting shall be as shown on drawing and if not shown shall be 125 mm height.

b) Kota stone tiles/slab shall be laid over 10mm thick in CM (1:3) irrespective of whatever specified elsewhere, jointed and pointed in grey cement. The exposed surfaces of Kota stone shall be mirror polished.

c) Kota stone tile/slab for skirting shall be of the selected quality hard, sound, dense and homogenous in texture, free from defects, machine cut and mirror polished all as specified in clause 13.12 of MES Schedule Part I. The size of the Kota stone tile skirting as shown in drawing and shall be 18mm to 20 mm thick. Where size does not fit on the same place, it shall be provided as per directions of Engineer-in-Charge. The Kota stone shall be uniform in thickness & shall be uniform in colour.

PARTICULAR SPECIFICATIONS (Contd…)

12.5 NON SKID CERAMIC TILES FLOORING

a) Non skid ceramic tiles in flooring shall be provided in the locations indicated in drawings/ Schedule of finishes.

b) Non skid ceramic tiles in flooring shall be 400 mm x 400 mm size and 7 to 8/ 9 to 10 mm thick unless otherwise specified in drawing, coloured of group B II (a)/B II (b) and of abrasion resistance PEI-III or above. Ceramic tiles shall be fixed with 15 mm thick screed bed 1:4 irrespective of what is mentioned in schedule of finishover PCC sub base / RCC slabs. Tiles shall be set and jointed in neat cement paste and shall be grouted with Epoxy grout "Ezy grout from Somany", "Ferro102 from Ferrous Crete", Bal Endura, Fevimate TG from Dr Fixit. Joints shall be thin & nearly indistinguishable. The workmanship in all respects shall be as specified in Clause 13.40 of SSR Part I.

c) Non Skid ceramic tiles shall be any one of the make as per make. Colour, pattern & design of tiles shall be as approved by GE.

12.6 GLAZED CERAMIC TILES IN DADO

d) Glazed ceramic tiles in dado shall be of 300x450 mm / 300x600 mm size and not less than 7 to 8 mm thick, coloured, group same as that of the floor and shall be provided in locations shown on drawing and the pattern shall be as decided by GE. Tiles shall be fixed with polymer modified tile adhesive "Ezy fix from Somany" , "Ferro 1111 from Ferrous Crete", " Favimate XL from Pidilite", Bal Endura. Joints grouted with Epoxy grout "Ezy grout from Somany", "Ferro102 from Ferrous Crete", Bal Endura, Fevimate TG from Dr Fixit.. Joints shall be thin & nearly indistinguishable. The laying, jointing and finishing shall be done in all respects as specified in Clause 13.40.1 to 13.40.2 of SSR Part I.

e) Dado shall be of full height I e, up to ceiling height unless otherwise shown on drawings/ schedule of finish.

12.7 VITRIFIED TILES

(a) Vitrified tiles double/ultra chargedshall beof size 800 x 800 x 9 to 10mm thick shall be used in flooring/ skirting in locations as indicated in schedule of finishes. The tile thickness can have a tolerance of ± 0.30mm. Tiles shall be of first quality product of the manufacturer.

(b) The tiles shall be procured from any of the manufacturers listed in annexure here-in-after.

(c) The colour/ printing/ texture/ shade and brand of tile shall be as approved by GE.

(d) The tiles shall be laid over screed/ sub base as mentioned in schedule of finishes. The tiles shall be fixed over backing coat as specified in schedule of finishes. The tiles shall be set and jointed in neat cement slurry and pointed with joint sealant to match the surface.

12.8 HEAVY DUTY OUT DOOR VITRIFIED TILES

Heavy duty vetrified tiles of size 400 x 400mm of 12mm thickness shall be used in parking areaand in locations as shown in drawings/ Schedule of finishes. The tiles shall be laid over screed/ sub base as mentioned in schedule of finishes.

12.9 COOKING PLATFORM: It shall be all as shown on drawing. Where indicated on drg, the cooking platform shall be provided with Black granite slab on top and sides as skirting. Granite slab shall not be less than 18mm thick, machine cut, pre-polished, set in cement mortar (1:3) including pointing in white cement mortar (1:2) using marble dust with admixture of pigment to match the shade of granite slab. The granite slab set on 15mm thick cement & sand mortar (1:4) screed. granite slab shall be in one piece. Cut-out shall be provided all as shown on drawing. The edges of Granite slab shall be rounded.

PARTICULAR SPECIFICATIONS (Contd…)

13.0PLASTERING AND POINTING

13.1 GENERAL

13.1.1 Cement: Refer specifications hereinbefore.

13.1.2 Fine aggregate for plastering and pointing work shall be river sand/manufactured sand/ manufactured fine aggregate conform to IS 1542-992 and as specified in Clause 14.5 to 14.5.2 of SSR Part I. Sand shall be obtained from source as approved by GE.

13.1.3 Particular attention of the contractor is invited to take note of local practices and the local availability of materials like bricks form work etc and cater for any extra quantities of mortar required for rendering smooth for extra dubbing required for touching up properly and for smooth and even surfaces. This shall be deemed to have been included in the lump sum.

13.1.4 Plaster and skirting/dado shall be returned in jambs, soffits of lintels and windows cills etc.

13.1.5 Where plaster on concrete surface shown to match the adjacent brick surfaces, the mix of plaster shall be as for the brick surfaces.

13.1.6 All plastered surface shall be trowel led to a smooth and even surfaces, without using extra cement.

13.1.7 Thickness of cement plaster mentioned hereinafter shall be finished thickness exclusive of dubbing. Dubbing may however be done in one operation with plaster.

13.1.8 Plaster on external surface shall be carried out upto 15cm below ground level except where Plinth protection/ramp etc are provided.

13.1.9 To avoid cracks at junction of RCC and Brick/ block masonry wall, 150mm wide GI chicken wire mesh of 0.9mm dia, 12mm mesh shall be nailed on the joints before plastering on external surfaces. Grooves between junction of RCC and masonry shall be formed on internal faces.

14.0 PLASTERING (INTERNAL)

14.1 Plastering shall be carried out all as specified in MES Sch Part I.

14.1.2 Cement plaster (internal) shall comprise of 10mm thick in cement and sand mortar(1:6) and finished even and smooth without using extra cement. Half brick walls shall be plastered with 10mm thick cement mortar (1:4).

14.1.3 PLASTERING (EXTERNAL)

14.1.4 Plastering (External) shall be done in two coats comprising cement and sand mortar(1:4) 5mm thick mixed with water proofing compound as specified here-in-before over a backing coat of cement and sand mortar(1:4) 10mm thick and finished with trowel to an even and smooth surface without using any extra cement.

14.1.5 Cement plaster on top of fascia/ chajjah etc shall be cement mortar (1:3) 10mm thick mixed with water proofing compound.

14.1.6 PLASTERING (CEILING)

14.1.7 Plastering shall be carried out all as specified in MES Sch Part I.

14.1.8 Cement plaster (ceiling) shall comprise of 5 mm thick in cement and sand mortar(1:3) and finished even and smooth without using extra cement.

14.1.9 TARRING: The backs of chowkhats in contact with PCC/brick work/plaster etc shall be given 2 coats of tar and sanded.

PARTICULAR SPECIFICATIONS (Contd…)

14.1.10 CEMENT SLURRY: Portions of MS bolts, lugs anchor bolts shall be applied with cement slurry.

15.0 WHITE WASHING

15.1 Three Coats of white(lime) wash shall be provided where indicated in drawing and all as specified as clause 15.12.5 of SSR Part I. For white washing on ceiling adequate quantity of Zinc oxide shall be added to lime for achieving white shade. Skirting and dado are not to be white washed. Surface preparation as per clause 15.12.1 of SSR part1 shall be done and after that a coat of wall care putty shall be applied and finished as per clause 15.13.1 of SSR part1 before application of white wash on internal surfaces

16. GLAZING

16.1 All glazing shall be with sheet glass of ordinary quality or pin head type or specified herein after. Glass shall be free from all defects viz bubbles, waviness and shall be homogeneous thickness.

16.1.1 Glazing to windows/doors etc as shall be as shown on drawing. Provide 3mm thick glass upto 0.5sqm and 4mm thick beyond 0.5 sqm in each pane unless otherwise shown on drgs. Glass used shall be plain sheet glass except in baths/WC/toilet where the same shall be figured glass pin head type.

16.1.2 Glass to wood frames shall be fixed with beads and golden putty as specified in clause 16.5 to 16.10.2 of SSR Part I.

17.1 PAINTING

17.1.1 All synthetic enamel paint shall be of Ist quality manufactured by any of the firms listed in Appendix attached here-in-after.

17.1.2 The contractor shall inform the GE, within three weeks of the acceptance of the tender, the brand names of the manufacturers of paint proposed to be used in the works and submit sample thereof and obtain prior written approval of the GE before their use in work.

17.1.3 Paints for priming coat under coat and finishing coat shall be of the same manufacturer.

17.1.4 Tint of paint, if not mentioned in drawings/schedules of finishes will be approved by the GE

17.1.5 Contractor shall execute painting under the guidance of the manufacturer's representative if so ordered by GE.

17.1.6 Members specified to be painted shall first be passed by the Engineer-in-Charge and marked as such before commencement of painting work.

17.1.7 If the undercoat of paint is not executed within six months after applying the priming coat of paint, the primer shall be redone by the contractor at no extra cost to the Government.

17.2 WORKMANSHIP

17.2.1All wood work required to be painted shall be smoothened sized and knotted and then applied with priming coat. Stopping and filling (filler coat) shall be done after priming coat and surface rubbed down to a level and smooth surface and thereafter under coat and finishing coat applied, all as specified in clause 17.6 of SSR Part I. Where painting to wood work is specified prepare surface and apply one priming coat of pink primer, one under coat and one finishing coat of synthetic enamel paint all as specified in MES Schedule.

17.2.2 All surface of timber exposed to view and for which no type of finished specifically catered for either on drgs or in these particular specifications shall be treated with 3 coats of paint as specified herein before.

17.3 PAINTING TO STEEL WORK

17.3.1 Steel and iron work shall be painted in the manner specified in clause 17.8 of SSR Part I after applying epoxy primer. Irrespective of what is indicated in Sch of finishes steel and iron work shall be applied with one coat of zinc chrome/Red oxide primer and two coats of synthetic enamel paint all as specified in the MES Schedule.

PARTICULAR SPECIFICATIONS (Contd…)

17.3.2 The steel surface which are required to be given three coats of paint, primer coat and under coat being given after fabrication but before assembly and erection.

17.3.3 Unless otherwise indicated on drgs and/or specified in this particular specification all surfaces of iron and steel work shall be treated with three coats(one priming coat, one under coat and one finishing coat) of paint as specified hereinbefore.

17.3.4 Following shall not be painted: -"All galvanized sheet/galvanized iron articles/aluminium anodized article reinforcement bars"

17.3.5 Other steel surfaces exposed to view but not covered in schedule of finishes or their finishing is not specifically indicated in any other drawings or specified elsewhere shall be given two coats of synthetic enamel paint over a coat of zinc chromes primer of tint as directed by the GE.

17.3.6 Bottom of door shutter shall be given one coat of primer only.

17.3.7 Where oil painting is indicated/specified it shall be amended to read as synthetic enamel paint only.

17.3.8 Prepare surface and apply two coats of tar mixture to the following: -

(a) Timber surface in contact with or burred in concrete/brick work, stone masonry and plaster.
(b) Mild steel (except reinforcement bars) surface in contact with or buried in brick work.
(c) Hold fast shall be given 2 coats of tar and sanded.

17.3.9 Tarring: The back of wooden chowkhats in contact with brick/concrete work/plaster etc shall be given 2 coats of tar.

17.3.10Cement Slurry: Portion of MS bolts, lugs, anchor bolts etc embedded in concrete/brick work shall be applied with cement slurry.

17.3.11Synthetic enamel paint:- Synthetic enamel paint and primer shall be any one of the makes as listed here in after.

17.4 ACRYLIC EXTERIOR EMULSION (ANTI FUNGAL PAINT): Two coats of Acrylic exterior emulsion paint over a coat of anti fungal primer(Exterior) shall be provided on external faces of walls in locations as indicated in drawing all as per manufacturer's instructions. Preparation of surfaces and application

shall be done all as per manufacturers or his authorised agent instructions using latest equipment/tools. This shall be of Make Asian Paints (Apex Ultima Protek)/ Nerolac (Excel rain guard) /Berger (Weather coat allguard). Both the Contractor and the applicator Shall give five years guarantee for acrylic emulsion paint. Anti fungal paint shall be got applied through authorized applicator of manufacturer only.

(a) The contractor shall ensure that all the buildings painted with acrylic paint free from fungal infections, light patches, discolor, etc. during the guarantee period of 05 years and he shall be responsible to upkeep the entire surface for a period of 05 years from the certified date of completion of work. The security deposit towards the guarantee for efficiency of treatment (acrylic painting) shall be retained from the RAR/Final bill amount. The **security deposit** amount shall be calculated as per the scales laid down by MES for calculating Individual Security deposit on the total amount of exterior painting carried out at contract rates. This amount shall be released to the contractor after satisfactory expiry of five years guarantee period. The GE may accept BGB from schedule bank or a fixed deposit receipt (pledged in GE's favour) from any Nationalized Bank for the said guarantee amount for a period of **five years**, in which event no further amount will be recovered from the final bill/RAR on this account. The defects liability period under condition 46 of General conditions of contract (IAFW-2249) shall be deemed to be amended to the extent mentioned above for acrylic painting works.

PARTICULAR SPECIFICATIONS (Contd…)

(b) Necessary guarantee shall be obtained from the manufacturer applicator on non judicial stamp paper giving five years guarantee from the certified date of completion of contract to keep the building free from fading, peeling off paint during guarantee period. The guarantee does not however, absolve the contractor from his responsibility in respect of this specialized work as per contract conditions. The contractor shall be responsible to ensure that the buildings are kept free from fading or peeling off the acrylic paint during guarantee period.

(c) STAGEWISE APPLICATION FOR ACRYLIC PAINT
Work for acrylic emulsion paint shall be carried out as per stage given below. Before starting the work for next stage prior approval of Engineer-in-charge shall be obtained.

Stage I	:	Preparation of surface.
Stage II	:	Applying priming coat
Stage III	:	Applying first coat of acrylic emulsion paint
Stage IV	:	Applying second coat of acrylic emulsion paint

17.5 Distempering:- Where OBD/Acrylic distemper is indicated on Schedule of finishes/drawings apply two coats of Acrylic washable distemper over one coat of approved interior wall primer as per manufacturer's instruction over 3 coats white cement based putty. The entire work shall be executed form manufacturers or his authorized agents/applicator using latest equipments/tools.

17.5.1 Acrylic distemper shall be first quality

17.5.2 Alkali resistant primer shall be provided all as indicated in drawings SSR Part I/ as directed by the Engineer–in-Charge.

17.6**External finishes** :- All external surface shall be finished with two coats of wheather coat including preparation of newly plastered /concrete surface over one coat of primer complete all as specified in the Schedule of finishes

17.7CEMENT BASED PUTTY

17.7.1 White cement-based putty of 3 coats shall be provided over a coat of alkali resistant primer in locations as shown on sch of finishes. White cement-based putty shall be obtained ready mixed form JK wall putty/ Birla wall care putty as approved by GE. White cement-based putty shall be in the form of dry homogeneous powder free from odours of putrefaction and when mixed water.

17.7.2 Preparation of surfaces, preparation of putty, wetting surfaces, application of putty and precaution to be taken while application shall be as per 15.19.2 to 19.9.6.4 of SSR Part –I2009

18.0 **SUNDRIES**

18.1 **MS RUNGS:-** Whether shown on drg or not, provide MS rungs with 20 mm dia TMT bars bent to shape size of rungs 300mm wide 150 mm projected and 100 mm inserted in PCC 1:2: 4 blocks. Overall length of rungs shall be 800 mm, spacing of rungs shall be 300 mm C/C in staggered , painted with 2 coats of anti-corrosive paint over a coat of metal primer. Location of rungs shall be in shaft and to RCC water tank all as directed by Engr-in-charge.

18.2 **NUMBER PLAQUE:-** All buildings shall be provided with a diamond square 30 cm x 30 cm x 20mm thick plaster plaque in cement mortar (1:2) finished smooth using extra cement duly painted in black (three coats) of synthetic enamel paint including priming coat and number letter (each of 12 cm height) as directed by Engineer-in-Charge. The number of each building on main entrance door shall also be painted as directed by Engineer-in-Charge.

PARTICULAR SPECIFICATIONS (Contd…)

18.3 **INFORMATION PLAQUE (TABLET)**

(a) All building/structure shall be provided with plaque (tablet) furnishing the following information in plaster engraving in and painted as directed by GE

(i) Job No, CA No, Year and name of work
(ii) Name of contractor, GE and Engineer-in-Charge
(iii) Date of commencement, completion phase wise and expiry of maintenance period
(iv) Date of expiry of warranty period against ATT and water proofing treatment

(b) Plaster shall be 20 mm thick in cement mortar (1:4)

(c) Location of plaque shall be as decided by the GE

18.4 **OPENING FOR EXHAUST FAN** :-Opening for exhaust fan shall be provided all as specified and shown on drawings. In case location is not shown on drawings, the same shall be decided by Engineer-in-Charge.

18.5 **RAIN WATER PIPE (RWP)**

(a) The location, size & type of rain water pipe & fittings shall be as shown on drawing, if not indicated on drawing the same shall be of UPVC (Unplastic PVC) type `A`, 100 mm dia conforming to IS: 13592-1992 all as specified in clause 18.27A of MES SSR Part-I (2009). Jointing of pipe shall be carried out as per manufacturer's instructions. Pipe & fittings shall be secured to walls at all joints with mild steel holder bat clamps all as specified in clause 11.37.5 of MES SSR Part-I.

(b) Grating and shoe shall be provided at inlet and outlet of RWP and splash stone shall be of 20mm thick cuddappa stone 600x600mm size as shown on drawings.

19.**PLUMBING DRAINS & SANITARY FITTINGS**

19.1.1 MATERIALS/ FITTINGS/ FIXING ACCESSORIES:- All Sanitary fittings (except flushing cistern) shall be of WHITE vitreous china best quality ISI marked. The samples of sanitary fittings, fixing accessories shall be approved by the GE before bulk procurement for incorporation in the work. Sanitary fittings shall be of one of the manufacturers as specified in List of Manufacturer, conforming to the catalogue numbers where referred to in these tender documents. ISI marked fittings of these brands shall be provided instead of those without ISI marking.

19.1.2 The unit rates of buildings included in Sch 'A' Part I include for the entire and complete sanitary installation, internal plumbing and drainage which comprise interalia the provision of sanitary and toilet appliances, gully traps fittings and fixing accessories, soil, waste, vent anti-syphonage pipes and specials, ducks foot bends etc. All the specials viz single & double branch pieces, bends, ducks foot bends etc shall be provided with access door fitted with 3mm rubber washer screwed with set screws. The outlet of 'P' trap of WC shall be jointed with connecting pipe/branch piece.

19.1.3The contractor shall employ qualified & licensed plumber for supervision of installation & testing of the sanitary fitting & plumbing. The connection shall be provided as indicated in drawings to conform to sound engineering practice. The layout of soil, waste vent & anti syphonage pipes shall be marked on the walls in coloured chalk & approved by the GE before execution of work. The unit rates of buildings given in Sch'A' Part I also include interalia the provision of the followings: -

19.1.3.1 All sanitary fittings etc shall be of Ist quality conforming to relevant IS and shall be ISI marked.

19.1.3.2 All soil pipes within the building & upto 3 metres beyond the outer face of wall including ducks foot bend upto and including first manhole shall be PVC SWR (ISI marked) pipes.

19.1.3.3 All waste pipes upto & including provision of gully traps duly embedded in concrete below ground level. Waste pipe from gully traps to Ist manhole shall PVC pipe 110 mm bore duly embedded in cement concrete below and confirming to IS,.

19.1.3.4 Vent pipes & anti syphonage pipes.

PARTICULAR SPECIFICATIONS (Contd…)

19.2 STAINLESS STEEL SINK WITH DRAINING BOARD: Stainless steel sink and draining board shall be made out of 1mm thick sheet. The size of sink shall be as shown on drawings. Unless otherwise shown on drawing overall size shall be 1155mm X 520mm with bowl size 510X405X230 mm and of make as listed in list of manufacturers.

19.3 STAINLESS STEEL SINK: Stainless steel kitchen sink shall be made out of 1mm thick sheet. The size of sink shall be as shown on drawings. Unless otherwise shown on drawing overall size shall be 495 X 420mm with bowl size 410X330X160 mm and of make as listed in list of manufacturers. Wherever double sinks are shown on drawing, provide double sink of overall size not less than 1067x 510x230mm.

19.3.1 All sinks shall be provided with 40 mm dia CP waste coupling with grating and connected with 32 mm dia PVC waste pipe. The stainless steel sink shall be supported on RCC platform having suitable cut for the sink. All stainless steel shall be Salem steel grade 304.

19.4 Wash Hand Basins: All rectangular wash hand basins shall be of size 550x400 mm white vitreous china Ist quality and shall be any one of make as listed list of manufacturers. These shall be complete with following:

(i) CP brass waste 32mm dia coupling
(ii) CP brass pillar taps 15mm
(iii) CP brass angle valves with LDPE PVC connection pipes with nuts and washers on both ends.
(iv) GI Waste pipe 32mm dia

19.4.1 Wash hand basin shall be supported on suitable size of a pair of painted cast iron bracket including wooden plug embedded in wall.

19.4.2 Where counter sunk wash basins are shown on drawing the same shall be as per Cat No. CNS-WHT-WB03 of Jaguar or its equivalent as approved in other makes as per list of manufacturers. The size of basin shall be 600mmx425mm x195mm Ist quality. However where stainless-steel trough type WHB are shown on drawing the same shall be of size as shown on drawing, the steel used shall be of salem stainless steel 304 gde 1mm thick.

19.5 EUROPEAN TYPE WATER CLOSET AND CISTERN: European type water closets shall be with P or S trap as required at site of any one of the makes mentioned in List of Manufacturer. These shall be complete with the followings: -

(i) Irrespective of what is shown on drg, provide Soft touch dual flow concealed type flushing valve set for EWC with auto shut off all as per Cat No. SFT-STL-085 of Jaquar or equivalent of other makes mentioned in List of Manufacturer here-in-below.

(ii) Plastic solid type seat and cover ISI marked type IA (IS-2548-1983) with CP brass hinges of make mentioned in List of Manufacturer here-in-below of white colour.

Note: Where flushing cistern cannot be fixed on wall due to window/lintel as per sanitary plan, the same shall be fixed in the side wall with additional bend, the cost of which shall be deemed to be included in the lump cost of buildings given in Sch ‘A’ Part I.

19.5.1 INLET CONNECTION: Connection to inlet of flushing cisterns/wash hand basin shall be done with readymade low-density polyethylene pipe for cold water services (conforming to ISS 3076) comprising 16mm dia (outer dia) and 450mm long polyethylene pipe having brass union at both ends with one end to be screwed to inlet of flushing cistern/pillar cock, bib cock and other end to GI pipe and dimension given in ISS 3076 for low density polyethylene pipes for cold water services. The weight of pair of brass union shall be not less than 40 gms.

19.6 **WATER TANK OVER ROOF**

19.6.1 Provide water tank all as shown on drawing. The lump sum quoted by the tenderer for Schedule ‘A’ Part I Item No. 1 shall be deemed to include for water tanks complete as shown on drawings.

PARTICULAR SPECIFICATIONS (Contd…)

19.7SOIL/WASTE/VENT PIPES/FITTINGS/ACESSORIES:- Irrespective of what is indicated in drawings all soil/waste/vent pipes/fittings/accessories shall be PVC[SWR] ISI marked approved quality. All the soil pipes shall be 110mm bore and waste/vent pipe shall be 75 mm bore, PVC vent cowl shall be provided at the end of vent pipes.

19.7.1 JOINTING**:** - All PVC pipes and fittings below / concealed under floor/wall shall be jointed as specified in MES Schedule

19.7.2 Fixing of pipes and fittings shall be fixed to walls all as specified in MES Schedule Part I.

19.7.3 The vent pipe shall be taken from soil pipes to a height of 120 cms, height measured from top of roof (last storey) to the bottom of the cowl.PVC cowl shall be provided on top of vent pipes and shall be fixed as per IS.

19.7.4 Soil pipes shall be taken up to 45 cms below ground level and provided with duck's foot bend.And waste pipe shall be taken up to 15 cms below plinth protection level and provided with duck's foot bend.

19.7.5 Soil /drain pipes from duck's foot bend shall be of PVC up to first manhole and waste pipe from gully traps to first manhole.

19.7.6 All exposed PVC bends/branches and other junctions shall be provided with access doors as per IS make with air and water tight.

19.8 NAHANI/FLOOR TRAPS**:-** Nahani /floor traps shall be provided in situations as shown in drawings these shall be PVC as per IS. Floor trap shall be fixed with SS gratings.

19.9 GULLY TRAP: - Gully trap shall be of PVC rebated top and 100mm dia (bore) outlet and stainless-steel grating of size 15cm x 15 cm. It shall be set as per manufacturer's instructions. Gully trap shall be provided in the situations where shown on the drawings.

19.10 SOAKAGE PIT **-** Soakage Pit shall be provided all as shown on drawing.

20.0 INTERNAL WATER SUPPLY

20.1 Scope of Work. The work includes all pipes/fittings and all as described in Sch `A' and as indicated in drawings.

20.2 CPVC Pipe and fittings

20.2.1 CPVC pipes and fittings shall comply with the requirement of relevant IS. Laying of CPVC pipe shall be in accordance with Clause 18.51 of MES Schedule. The pipes (for supply of water), with all fittings, shall run concealed in the walls except where otherwise specified. Where pipes crosses the wall, sleeve piece of suitable dia/length shall be provided to accommodate the pipe and cost of same shall be deemed to be included in the lumpsum cost of building. The contractor shall use proper bends, elbows, tees etc at turning corners. Contractor shall provide screwed plugs to all open ends of pipe on completion of days work. Make of CPVC pipe shall be SFMC, RELIANCE, SUPREME.

20.3 Bib Taps and Stop valves. These shall be as per IS-781/1984. All fancy fittings where specified shall be as per IS 8931- 1978. These shall be as specified in clause 18.14 & 18.15 of MES Schedule (Part I) polished bright.

20.4 Laying and fixing of pipes

(a) The pipes for supply of water to all fittings in the buildings shall be run on the walls except otherwise specified in these tender documents or shown on drawings, connected to various fittings and shall be brought in the rooms at point/position approved by the Engineer-in-Charge at site. Where pipe crossed the wall a GI sleeve piece of suitable length/dia (bigger) shall be provided to accommodate the PPR pipe. The cost of the sleeves shall be deemed to be included in the lumpsum cost of the building.

(b) Where pipes are laid underground, the trenches shall be excavated as directed by the Engineer-in-Charge. The pipes running along face of the wall shall be clamped to the walls as specified in SSR.

PARTICULAR SPECIFICATIONS (Contd…)

(c) All water supply works shall be carried out through licensed/skilled plumber/ fitter as approved Engineer-in-Charge.

20.5 Workmanship.
(a) Laying of pipes (internal works) shall be in accordance with clause 18.51 of MES Schedule (Part I). All-important provision of IS-2065-1983 as applicable shall be complied with in consultation with EIC.

(b)The contractor shall use proper bends, elbows tees etc at turning corners, bending of pipes shall not be allowed in any case.

(c) Pipes shall not run diagonally.

(d) Contractor shall provide screwed plugs to all open ends of pipes on completion of days' work.

20.6 Testing. Testing of pipes shall be carried out all as per clause 18.50.4 of MES Schedule (Part I). After completion of work, entire water supply installation shall be tested as per clause 18.55 of MES Schedule Part I without any extra cost to the Government Any defects noticed shall be rectified by the contractor to the entire satisfaction of Engineer-in-Charge. The pre-priced lumpsum against Schedule `A' Part II include for the above provision.

20.7 Record Drawings. Three copies of line plan of complete work indicating the line of pipes, size, position of fittings etc shall be submitted by the contractor to the Engineer-in-Charge on completion of the work.

20.8 Excavation and earthwork involved for this work shall be carried out as specified in the Sch `A'

20.9 Trench bed shall be got approved in writing by the Engineer-in-Charge before pipes are laid.

20.10 Excavation in trenches for pipes shall be the actual width of excavation or authorized width specified in the MES Sch whichever is less.

22.11 No extra payment shall be made for excavation through existing roads, paths etc and making good to match the existing work.

20.12 Timbering to uphold the sides of excavation shall beprovided by the contractor, if considered necessary and ordered by the GE, without extra cost to the Government.

20.13 Returning, filling shall be as specified in clause 3.20 of MES Schedule Part I.

20.14 Wherever trenches are dug across road etc, the contractor shall provide all necessary precautions against danger with danger lights, flags etc as directed by the Engineer-in-Charge

20.15 GATE VALVES: This shall be of approved make high pressure type with ISI mark.

21.0 INTERNAL ELECTRIFICATION

21.1 General

23.1.1 The scope of work included under Sch'A' consists of providing internal electric wiring to the buildings and external electrification as described in Sch'A' and as specified/Shown on drawings.

21.1.2 The general layout of wiring points and fittings are as shown on drawings. The exact location of fittings may be altered by the GE/Engineer-in-Charge to suit local requirements and the contractor shall not claim any extra on his account for point wiring and connected accessories.

21.1.3 All electrical fittings and wiring must run clear of door, windows and openings No diagonal run shall be allowed. These must always be parallel or perpendicular to the ground.

21.1.4 The specification and general rules/conditions laid down in the MES Schedule Part I and II including errata and amendments as applicable will be generally applicable to the whole work unless otherwise specified hereinafter.

PARTICULAR SPECIFICATIONS (Contd…)

21.1.5 The work shall be carried out in strict compliance with the latest edition of Indian Electricity Act the Indian Electricity Rules and IEE Regulations and the latest edition of IS732. It shall be of high standard and approved construction used in a modern electrical work both as regards design and workmanship. Complete work shall be suitable in every respect of type of voltage specified and shall be to the satisfaction of Engineer-in-Charge.

21.2 WORKMANSHIP

21.2.1 The work under internal electrification work shall be carried out only by licensed wireman under the supervision of a qualified Electrical Engineer. The contractor shall produce (for inspection) license of electrician/wireman if ordered by the Engineer-in-Charge.

21.2.2 Protection of wiring from mechanical damage shall be as per clause 19.119 of MES Schedule Part I.

21.2.3 Passing of cables through walls, floors etc shall be protected as specified in clause 19.120 and 19.121 of MES Sch Part I.

21.3 SAMPLES

21.3.1 Before starting the work, the contractor shall produce samples of all materials including accessories and approved by the GE. Samples that are approved will be retained by GE until the work is completed and accepted. The contractor will not be allowed to commence the work before the samples are produced and approved. The contractor shall ensure that the materials used in the work are identical with approved samples and are uniform throughout. All materials shall be accordance with latest edition of the IS/BS specification and shall be of the best indigenous make approved by the GE

21.4 MATERIALS: All materials shall be accordance with latest edition of the relevant IS/BS specification and shall be of the be indigenous make approved by the GE.

21.5 CABLES; Cables shall be single core PVC insulated and PVC sheathed/unsheathed (all as described in Sch'A' with multi stranded copper conductor conforming to specifications laid down in clause 19.25 (a) (ii) and 19.25(b) (ii) of MES Schedule Part I. All cables shall be 1100 volts grade for point wiring and shall be of such size as to be capable to carrying the maximum current which will normally flow through them without the respective rating being exceeded as laid down in IEE Regulations. Cables shall be delivered at site in makers wrappers etc, with the seal intact and shall not be installed unless approved by the Engineer-in-charge. If cables of size as specified in Sch 'A' as per relevant IS are not available, cables of equivalent size with identical current carrying capacity and with multi stands shall be got approved from accepting office before incorporation works without extra cost to Govt.

21.6 MODULAR FLUSH BOXES

21.6.1 Modular GI flush box of suitable size with inner and outer white plastic plate (same make of approved switch and socket) shall be provided.

21.6.2 MODULAR SWITCHES/SOCKETS. Modular switches/ switches and socket shall be as per schedule and shall be of any one make as listed in manufacturer list.

21.7 LAMINATED SHEET

21.7.1 Laminated sheet shall be 3mm thick fixed in MS pressed steel terminal boxes by cadmium plated iron screws. Make of laminated sheet shall be 'HYLAM' or equivalent

21.8 CIRCUITS

21.8.1 Separate circuits shall be provided for lighting and power wiring.

21.8.2 Lighting sub circuits shall not be loaded with more than 8 light points.

21.9 BOXES: These shall be flush as specified in clause 19.30 of MES Schedule Part I.

21.10 CONDUIT AND CONDUIT ACCESSORIES: All conduits, conduit fittings and accessories shall be rigid PVC conduit and fittings conforming to IS-9537 Part III of 1983 of (Medium grade).

PARTICULAR SPECIFICATIONS (Contd…)

21.11 CEILING ROSES: Ceiling roses shall be flush type all as specified in Clause 19.32 of MES Schedule Part I.

21.12 SOCKET OUTLETS: These shall be modular type all as specified in of SSR Part I.

21.13 MINIATURE CIRCUIT BREAKERS AND DISTRIBUTION BOARDS

21.13.1 MCB shall be of approved make and shall be obtained form any one of the manufacturers listed in manufacturer list.

21.13.1The unused portion of DBs are to be blocked by the blank piece.

21.14 FAN REGULATOR: Fixing of fan regulator and hanging ceiling fans shall be all as specified in clause 19.115.4 of MES Sch Part I. All ceiling fans and regulators shall be earthed effectively by means of 2mm bare GI iron conductors. Cost of this earthing to fans and regulators shall be deemed to be included in the rate for point wiring for fans.

21.15 CEILING/BRACKET LIGHT FITTING: These shall be all as specified in Sch'A' and as shown on drgs. Light fittings as per E-in-C's TI No 3 of 1986 shall be conform to IS and obtained from any of the manufacturer listed in Appx'' and shall be fixed to walls/roofs etc with proper size of clamps/screws and connecting cables etc complete.

21.16 LED LIGHT FITINGS: These shall be of make as indicated in Sch'A' and shall be provided all as specified therein.

21.17 PAINTING: Unless otherwise specified all exposed surface of steel works like terminal boxes LT panel board MS clamps etc shall be painted with 2 coats of synthetic enamel paint over a coat of red oxide primer. Metallic suspenders of tube light fittings shall be black painted.

21.18 SYSTEM OF WIRING: - Wiring shall be carried out with PVC insulated cable and shall run as far as possible on walls, ceilings so as to be easily accessible and capable of being inspected. Power wiring shall be kept apart at minimum distance of 2mm unless they are enclosed in earthed metal conduit suitably marked to indicate the risk of dangerous shock due to voltage in the conductor.

21.19Concealed wiring: Provide concealed wiring all as described in Sch 'A'. PVC conduit shall be rigid PVC conforming to relevant IS. The make of PVC conduit shall be as specified in Appendix 'B'. PVC conduit shall be adequate size to draw required No for cables.

21.20 TESTING

21.20.1 On completion of the work, the entire electrical installation shall be tested by the contractor for the following tests which shall be carried out in accordance with clause 19.146 of SSR Part I in the presence of and to the satisfaction of the Engineer-in-Charge:-

(a) Continuity
(b) Insulation resistance
(c) Any other test prescribed by the GE.

21.20.2 All testing equipment/apparatus, material, labour etc required for above tests shall be provided by the contractor at his own expenses through his sources, works for which test results do not conform to standard will be redone by the contractor at his own expenses.

21.21 RECORD DRAWINGS: On completion of work, wiring diagram showing the internal electrification (including high frequency supply) and layout of the buildings shall be prepared by the contractor in conformity with clause 19.146 of MES Schedule Part I and submitted to Engineer-in-Charge in triplicate.

21.22 EARTHING

21.22.1 Earthing shall be carried out as specified in Sch 'A" and electrical plate No 3 of MES Sch Part I. Earthing shall be with earth plate electrode as mentioned in Sch 'A' buried directly in ground (earth pit) not less than 2.25m deep below ground level connected to GI strip of size 25 x 4 mm/32x6mm/4mm dia GI wires with top edge of the plate not less than 1.5m below ground level connected to earth wire as

PARTICULAR SPECIFICATIONS (Contd…)

mentioned in Sch 'A' by means of bolt nut and washers of galvanized iron or steel, and protected by GI pipe light grade of suitable size. Excavation and earth work shall be in soft/loose soil. All surplus soil to be removed to a distance not exceeding 50m. Cement concrete for PCC chamber shall be in PCC 1:3:6 type C1 and chamber cover shall be RCC 1:2:4 type B1 50mm thick, reinforcement shall be with 8mm dia CTD Bar @ 100mm centre to centre both ways including necessary form work. etc. The unit rate of relevant item of Sch 'A' include the cost of earth work, PCC Chamber, RCC cover, testing etc complete.

22.0 ROAD/PATH/HARDSTANDING

22.1 EXCAVATION AND EARTHWORK

(a) Excavation and earth work shall be carried out all as described in Clause 20-A.15 of MES Schedule.

(b) Before commencement of excavation of earth filling, the representative of the GE and the contractor will be required to take the levels jointly, of the existing ground surfaces at intervals as decided by the GE (the decision of the GE being final and binding in this respect) and plot the same on longitudinal and cross sections to be prepared by the Engineer-in-Charge. These crosssections shall also show the proposed formation levels after consolidation and shall be signed by the GE and Contractor in token of their acceptance.

(c) For the purpose of filling/cutting, the entire area where filling/cutting activity is to be done shall be divided into grids of suitable sizes and quantities shall be computed by using Simpson's rule for areas and Prismoidal formula for computing volumes.

22.3 **HARD CORE**

The material for hardcore (broken stone,Granite, trap or Basalt) shall be as per sample kept in GE's office and from the sources as described in Appendix 'C' to particular specifications. Hardcore shall be stones of gauge not exceeding 63mm grade. Hardcore shall be deposited, spread and leveled in layers n exc 15cm thick and watered, well rammed to a true surface and compacted. The thickness of the hardcore after consolidation shall be as per schedule A.

22.4 **SOLING**

Where soiling (hard core) shown on drawings for roads provide soiling (or bottoming) of stoned as specified hereinbefore. Soiling shall be provided on the prepared sub grade in conformity with lines, grade thickness and cross section as shown on drawings or specified. The edge of soling shall be marked out by straights and shall be carried out as specified in SSR Part I vide clause 20.4.20.1.

22.5 INTER LOCKING PAVER BLOCKS

24.5.1 Provide Interlocking paver block all as specified in schedule. Pre cast interlocking paver block shall be of grade M35 grade if nothing is indicated elsewhere. The paver block shall confirm to IS 15658-2006. Paver block shall be sound and free from cracks and other visual defects. The tolerance in thickness shall be +3mm. Water absorption shall not be more than 6% by mass. Size and shape and colour shall be as approved by GE.

23.0 SEWAGE DISPOSAL

23.1 Excavation and earth work: Irrespective of the width of trenches for the pipes excavated, the width for the purpose of payment shall be the authorized width as defined in clause 3.2 of SSR (Part II). Other requirements specified hereinbefore in the MES Sch as applicable shall be complied with.

23.2 Sewer pipes

23.2.1 Pipes complete all as specified in schedule 'A'/BOQ. If nothing is indicated elsewhere then the pipes shall be 150mm class NP2 RCC.

23.2.2 Concrete Foundations, Beds, etc. Concrete Foundations to the drain and sewer pipes and the haunching or encasing to the pipes shall be provided as indicated. No work shall be covered over or surrounded with concrete until it has been inspected and approved by the EIC after testing as specified.

PARTICULAR SPECIFICATIONS (Contd…)

23.2.3 Bedding: The thickness of concrete bed below the barrel of the pipe where indicated shall be 10 cm for pipes 100 mm and 150/160 mm dia and 15 cm for pipes 200 mm dia and over. Width of concrete bedding shall extend laterally 15 cm beyond either side of the pipe i.e. outer diameter of drain pipes (not of sockets/collars). Where bedding only is provided, the concrete shall be brought up at least to the invert level of the pipe to form a cradle.

23.2.4 Haunching shall consist of concrete bed as described for bedding with full width of bed carried upto the level of horizontal dia of the pipe and splays from this level carried up on both sides of the pipe from the sides of the bed to meet the pipe barrel tangentially.

23.2.5 Lowering and laying of Pipes

23.2.5.1 The laying and jointing of pipes shall conform to IS 783-1985, pipes shall be joined by collar joints.

23.2.5.2 The trench shall be checked for proper level, gradient and alignment before lowering the pipes.

23.2.6 Lowering
The pipes shall be lowered cautiously to prevent disturbance of the bed and side of the trench. The heavy pipes shall be lowered by means of proper legs chain pulley blocks as directed by Engineer-in-charge. Great care should be taken to prevent sand etc. Fromenteringthe pipes.

23.2.7 Laying
23.2.7.1 Laying of pipes shall proceed up-grade of slopes. The error of grade shall not be rectified by packing up earth underneath the pipes. If required, concrete shall be used for packing.

23.2.7.2 The ends of the pipes shall be kept closed to keep dirt, mud and foreign materials out. Adequate provision shall be made to prevent floating of pipe in the event of flooding of trenches.

23.2.7.3 The body of the pipe for its entire length shall rest on an even bed in the trench and places shall be excavated to receive the collar for the purpose of jointing.

23.2.8 Covering pipes
No work shall be covered over or surrounded with concrete until it has been inspected and approved by the EIC after testing as specified.

23.8Testing: Drains and sewers shall be tested as per clause 18.79.2 and 18.79.3 to 18.79.5 of MES Schedule Part I.

23.9 Manhole: Manhole shall be built as per the relevant items given in the respective section of Sch'A' and as shown on drawings and as per Clause 18.78 of MES Schedule Part I.

24.0 **EXTERNAL ELECTRIFICATION**

24.1.1 General: The work under this contract shall be carried out in accordance with Schedule 'A' this particular specification and the drawings forming part of this contract and the General Specification and other provisions in the MES Schedule in conjunction with each other.

24.1.2 Scope of Work: The scope of this contract consists of laying of cables, providing LT Panel board etc all as described in Schedule 'A'.

24.2 STANDARD

24.2.1 All materials incorporated in this work shall be of the make specified in Sch 'A' or this particular specification.

24.2.2 All materials to be incorporated shall conform to relevant Indian Standards or if they are not existing, other relevant International Standards. Copies of test certificates shall have to be produced and got accepted by the Engineer-in-Charge before the materials are brought to site.

PARTICULAR SPECIFICATIONS (Contd…)

The workmanship shall be of the highest order and standard and shall conform to all the provisions of relevant IE rules and other statutory requirements. All electrical works shall be executed by properly skilled and trained electricians under the supervision of qualified Supervisors. Contractor is required to produce evidence of such qualifications on demand by Engineer-in-Charge.

24.3 UNDERGROUND CABLES

(a) Underground XLPE cables shall be all as specified in Sch 'A' & shall conform to IS 7098 Part II including amendments. The cables shall be laid, jointed and tested all as specified in clauses 19.74 to 19.94 of MES Schedule Part I

(b) The quantity of cable given in the schedule 'A' is only approximate. The contractor is instructed to visit the site before taking action for procurement of the cable.

(c) The rate quoted for cable also includes connecting the cable terminals to the LT panels and control switches etc with suitable size of cable lugs, bolts nuts and washers.

(d) No straight through joints permitted on cables of length upto 400 mtrs and maximum straight through joints permitted for length of cable more than 400 mtrs is strictly repatriated to two.

(e) All length of cable shall be tested before laying and after laying and the satisfactory test results are to be signed by the contractor and the Engineer-in-Charge

(f) Before accepting the lot of cable for incorporation in the work, acceptance tests of XLPE cable as stipulated in clause 18.2.1 of IS –7098 (Part-II) shall be made by the contractor to ensure the quality of cable. testing charges including transportation of material shall be borne by the contractor without extra cost to the Govt.

24.4 APPLICABILITY OF RULES, REGULATIONS AND CODE OF PRACTICE :-

24.4.1 The entire electrical installation under this contract shall comply with the requirements of Indian Electricity Rules, Acts and other regulations such as those made under factories act. Fire Insurance Act may be applicable from time to time.

24.4.2 PAINTING AND PROTECTION OF EQUIPMENTS: FITTINGS ETC

Each item of equipment, fittings etc shall be painted or protected as detailed hereinafter. It is essential that before any coat of paint is applied either in the shops or at the site, the surface shall be clean and free from all dirts and shall be completely dry. The manufacturers guarantee that, each coat of paint is compatible with the previous and subsequent coats so that oiling, flaking or other faults do not occur.

24.4.3 Steel fabrication shall be thoroughly scrapped to remove mill scales, rust, dirt etc and wire brushed, One coat of red oxide primer shall then be applied. After erection at the site, the steel work shall be painted with one under coat followed by one finishing coat of oil paint. The colour of all paints above floor level shall be light grey or any other shade as approved by the GE.

24.4.4 Where it is usual practice of the manufacturer or special items such as switch gear, control cabinets etc to apply a high standard of protective enamel paint in the shop before despatch, this will be acceptable provided any damage to the said paint which occur during transit or erection is made good by applying same tint and type of finishing paint by the contractor at his cost before the equipment.

24.4.5 After erection at site and before being connected to the power supply all items of electrical equipment and all circuits shall be fully tested to prove correct connections, insulation resistance, continuity effective earthing etc. The installation shall be tested by an authorized Electrical inspector in addition to the test as specified herein, any defects pointed out shall be rectified immediately by repairing or replacing the defective parts of equipments at no extra cost. Installations shall be taken over by the department after successful commissioning.

PARTICULAR SPECIFICATIONS (Contd…)

24.4.6 If owing to storage or other causes, the electrical insulation resistance has deteriorated, the equipment, cable etc shall be thoroughly dried out replaced/ or other steps taken to restore proper insulation resistance before connecting the same to the power supply. After connecting the power supply, the whole of the equipment shall be tested to demonstrate its stability to operate satisfactorily.

24.4.7 All instruments and appliances, oil or other materials necessary for the complete testing shall be provided by the contractor at no extra cost.

24.4.8 Approval by the department for materials, workmanship, etc during manufacture or approval at the site will not relieve the contractor from his obligation to comply with all the requirements of the contract.

24.5 **GUARANTEE**

24.5.1 The tenderer shall submit the tender to suit the over all specification given in the tender and shall guarantee that materials and workmanship of the equipments and connected items offered and installed by him under the contract are of high class in every respect and that he will make good the defects damage which are not attributed to normal wear and tear and will be responsible for repairing/replacing at his own cost any parts which area rendered defective and unserviceable during the period of maintenance. The decision of the GE in this regard shall be final and binding.

24.5.2 **EQUIPMENTS**

24.5.2.1 The entire equipments shall be from standard manufacturers with high class workmanship and finished. The tenderer shall state while submitting his tender, the name of the manufacturer whose equipments be would offer for incorporation in the work. As far as possible, one type of equipment shall be from the same manufacturer to ensure interchange ability.

24.6 **EARTHING**

24.6.1 Earthing shall be carried out as specified in Sch ‘A” and electrical plate No 03 of MES Sch Part I. Earthing shall be with earth plate electrode as mentioned in Sch ‘A’ burried directly in ground (earth pit) not less than 2.25m deep below ground level connected to galvanised earth top edge of the plate not less than 1.5m below ground level connected to earth strip as mentioned in Sch ‘A’ by means of bolt nut and washers of galvanised iron or steel, and protected by GI pipe light grade of suitable size. Excavation and earth work shall be in soft/loose soil. All surplus soil to be removed to a distance not exceeding 50m. Cement concrete for PCC chamber shall be in PCC 1:3:6 type C1 and chamber cover shall be RCC 1:2:4 type B1 50mm thick, reinforcement shall be with 8mm dia CTD Bar @ 100mm centre to centre both ways including necessary form work. etc .The unit rate of relevant item of Sch ‘A” including the cost of earth work, PCC Chamber, RCC cover, testing etc complete.

24.7 Record of Materials

27.7.1 Original voucher/invoices of all materials required for execution of work shall be produced by the contractorbefore claiming RAR payment. The vouchers/invoices shall be from the manufacturers and/or from their authorised agents for full quantity of respectiveitems required for the work under this contract. These vouchers shall be enclosed, dated and initialled by the Engineer-in-Charge giving the contract number and certified true copy of each such vouchers signed both by Engineer-in-Charge and contractor and shall also be kept on MES record.

24.7.2 General requirement of concrete work covered under this schedule shall be all as specified in this shall comply with IS-1554(Part I) 1976 and shall be of make specified in Appendix ‘B/Schedule ‘A’.

24.8 LT PANEL &PDB’s :

24.8.1 Main LT Panel & PDB will be free standing, front operated, cubical in construction having multi-tier arrangement of the incomers and feeders as per details given in the schedule of Quantities. The panel will have cable entry and cable exit (as per site condition).

PARTICULAR SPECIFICATIONS (Contd…)

(a) Selection of the components : All the electrical components selected for constructing the various modules of boards must have sufficient ratings to perform the duty they will be required to do consistently without any deterioration in their normal life and safety of the other equipment including the safety of the operators. However these ratings will not be less than the ones specified in Schedule 'A'.

(b) General arrangement and layout : Each vertical section of the panel will contain 5 to 6 modules only for 125 Amps and above and will be limited to 7 modules for all modules current rating lower than 125 Amps . Minimum size of bigger module should not be less than 400 mm (height) * 450 mm (width). The height of the panel should be limited to 2400 mm. Operating levers handle etc. of the highest unit shall be at a height less than 1800mm and should not be less than 250mm of the lowest unit. All the cables incoming to the main L.T panels will be entering from the top. For this purpose a removable gland plate will be provided at the bottom. A minimum distance of 250 mm will be provided between the gland plate and the nearest terminal for proper dressing and termination of the cable. All the components of a module will be mounted on a component plate using machine screws and taped holes (excepting the components mounted on the door). These component plates should be fixed with bolts for easy replacement. Standardisation will be adopted while making these plates so that the component plates of same size modules can be changed from one module to other. No air circuit breaker will be placed in top compartment in one vertical section having only one air circuit breaker. Not more than two circuit breakers will be placed in any vertical cubical and in no circumstances three air circuit breakers will be placed in one vertical cubical.

(c) Fabrication :The Main LT Panel & PDB's will be metal clad, cubical in construction, free standing, floor mounted, indoor duty sheet metal enclosure. It shall be totally dust and vermin proof. All the cubicles will be equipped with front located, outward opening, lockable doors having hidden hinge and a bolted back cover both using non-deteriorating neoprene rubber gaskets to provide IP 52 protection and necessary louvers shall be provided. The board will have structural frame as well as all sides, partitions and enclosures made out of shaped CRCA sheet of minimum thickness 2 mm. In case of panels of lengths more than 2.4 meters the fabrication of any single section will be limited to a maximum length of 2.4 meters for the purpose of shipping and shifting at the site. These sections will be assembled at the location of installation with the help of nuts and bolts on a supplied common frame. While making general arrangement consideration will be given to the place of sectionalisation to select the location where the minimum electrical connections are transferred from one section to the other section. These connections will be required to be terminated on a connector terminal block (stud type) and the connections will be numbered and marked for joining at the site. All nuts, bolts, handles, meters, knobs etc. appearing on outside of the panels should be locatedin symmetry so as to give a neat appearance. All knibling work shall be carried out by automaticmachines. For rigidity the panel will be provided with a base frame made out of MS channel ofsize not less than 100 mm and the same will be bolted to the panel.

All the hardware used in theassembly will be electroplated for protection and appearance. Each part of the fabricated panelwill be subjected to seven tank treatment in the order as below:

(i) De-greasing.
(ii) De-rusting
(iii) Rinsing.
(iv) Phosphating.
(v) Drying.
(vi) Primer Spraying.
(vii) Oven drying and baking.

Powder coating paint of minimum 65 microns shall be applied after primer and shade of the colour will be decided by Engineer-in-charge. Paint thickness & paint adhesion test will be carried out & witnessed.

The various components, assemblies and sub equipments of the panel will be as per the details given below.

PARTICULAR SPECIFICATIONS (Contd…)

(d) Bus bar and Connections: The main bus bars of the panel will be shaped out of high conductivity, electrolytic conductor Al91E grade, and shall be liberal sized. The conductivity of aluminium shall be minimum 58% and shall be ascertained with the help of conductivity meter. To maintain the extension capability at both ends of the panel the horizontal main bus bar will run from left end of the panel to the right end of the panel without any taper or reduction in the size. The bus bars will be covered with the coloured, heat shrinking, PVC sleeve throughout (Except at joints). The three phases will be identified with red, yellow and blue colour bands provide on both side of a joint. The size of bus bar calculations must be approved by GE for rated fault level keeping in view the overall temp rise to 85°C. The distance between two SMC support shall be placed to suit the rated fault level. In any case the distance between SMC supports shall not exceed 450 mm. These SMC distance calculations also must be approved by GE prior to any fabrication work and factor of safety shall be considered as two for thermal stress calculation

(i) A minimum de rating factor of safety of .66 on account of temperature rise and enclosure shall be taken and exact calculations shall be done as per site condition subjected to a minimum of 0.66 de rating. While doing calculations exact rating chart of manufacturer shall be attached along with the calculations.

(ii) The bus bars will run in a separate bus bar chamber using bus insulators made of non-deteriorating, vermin proof, non-hygroscopic materials such as SMC. The interval between the two insulators will be designed after considering the following points:

(iii) Strength and safe load rating of the insulator.

(iv) The vibrating force generated during a fault having fault level of 50 KA or as specified in Schedule 'A'.

(v) A set of SMC support.

(vi) To limit the temperature, rise in the bus bar chamber a set of louvers can be provided at stratergical places considering the air circulation. The louvers provided will have a brass wire mesh covering from inside with minimum 100 openings per sq inch. The overall temperature of bus bars shall not exceed 85°C. Each vertical section of modules will be given power supply by using a set of completely enclosed vertical bus bars taping off from the main bus bars. These vertical bus bars will be of adequate size and rating to carry the full load current of all the modules and will not be less than220 square mm in cross section. The electro-galvanised high tensile steel nuts, bolts, plain washers and spring washers of suitable size will be used in connecting the various section of the bus bars. A minimum of 1.6 times the width of bus bar will be the lapping length of each joint.

(e) The Control wiring: The control wiring of all the panels will be done with PVC single core flexible copper wires of cross section 1.5 sq mm. and 2.5 sq mm. All the wiring involving current transformers and DC circuits will be wired with minimum 2.5 mm cross section wire and the others with 1.5 sq mm. The control wiring will be done by properly dressing all the wires in a laminar manner either in a PVC duct of liberal size or bunched together by PVC strapping tapes at a distance not exceeding 150 mm. Each wire will terminate with a copper ferule crimped to the wire.

The PVC ferules will be used to identify each wire of the circuit and the same numbers will be marked on the drawing for the corresponding wire. Only Elmex type clip on style connectors with suitable rails will be used for ending the control wiring. The decision will ensure that only one out going wire will be connected to one connector. When the control wiring is crossing from fixed parts to moving parts such as doors etc. the wire will be run in PVC sleeve of suitable size and the same will be mechanically clamped at both the ends ie. one end of the fixed part and the other on the moving part.28.6.1.

(f) Under no circumstances the wiring should be under any kind of stress for which sufficient length of control wiring in the PVC sleeve should be provided. The colour code for the wiring will be as follows

(i) For +ve of DC circuits RED
(ii) For -ve of DC circuits BLUE
(iii) For CT and PT circuits RED/YELLOW/BLUE following the phase codes
(iv) For remaining BLACK
(v) For earthing GREEN

PARTICULAR SPECIFICATIONS (Contd…)

24.9CONTROL DEVICES:

a) Push buttons: The push buttons used in this panel will be rated for more than 415 Volts and 2 Amps. All the push buttons will be mounted on the front door. The assembly will be in two part comprising of contact blocks. The design should permit addition of extra contact blocks where ever required functions is detailed below

Function	Colour	Type
Starting/switching ON	Green	Shrouded
Stopping/switching OFF	Red	Shrouded
Resetting	Black	Shrouded
Emergency OFF (locking in pressed pressed position & shall be key resettable)	Red	Mushroom

All the push buttons will be mounted on the front door of the cubicle in regular symmetrical fashion as per the general norms being practiced. Only one make of push buttons will be used in the assembly of all the panels.

b.) Indicating lights : The indicating lights used in the panel will be pleasant looking, indicating lights in round shape having following features :

(i) A separate front lens for it's easy replacement.
(ii) Facility to replace the bulb from the front.
(iii) Shall comprise of LED lights.
(iv) The shape of the lens to allow viewing from sides.
(v)Series resistance with use of low voltage bulb for longer life.
(vi) Clear and distinct indication for light ON and OFF with difference of brightness of the lens in ON and OFF state.
(vii) Loss of not more than 1.5W per lamp.
(viii) The selection of the colours of the indicating lamps will be as follows :
(ix) Red for system in operation
(x)Amber for system ready for operation
(xi) Green for system being put off
(xii) Red, Yellow and blue for incoming supply indication.

c.) Control Switches : - All control switches shall be rotary , back connection. All selector switches shall have make before break feature. The selector switches shall have four positions, three for reading phase and fourth is OFF position. All switches will be rated for 16 amps and poles and various positions (ways) will be as per SCHEDULE 'A'.

(d) Control & selector switches shall be of the rotary type with enclosed contacts adequately rated for the purpose intended (min. acceptable rating is 10A continuous at 240V AC and 1A (inductive break) 220V DC).

(i) Provided with escutcheon plates clearly marked to show the positions of the spring return to normal type provided with pistol grip type handles
(ii) Selector switch shall be of the maintained contact stayput type.

24.10 MINIATURE CIRCUIT BREAKER : - It shall comply IS: 8828-1996 / IEC 898-1995. The MCB shall be suitable for DC and 400 Hz application. The MCB shall be made of self-extinguishing tropicalised (95% humidity 55°C material. The MCB shall have a trip free toggle mechanism. The contact closing shall be independent of operator speed. The terminal shall be protected against any finger contact to IP20 degree of protection with no restriction for line and load. The breaking capacity shall be 10KA in accordance with IS 8828-1996, watt loss per pole shall be as per IS8828-1996. The rated impulse voltage (Vimp) of MCB shall be 6kV. The MCB shall be capable of being used as incomer and isolator application and shall have provision of fitting in future accessories like auxiliary switch, alarm switch, shunt trip and under voltage trip.

PARTICULAR SPECIFICATIONS (Contd…)

The manufacturer shall define the tripping characteristics of their MCB'S and furnish the respective tripping curves. High voltage feature, wherever applicable shall be preferred. The terminal shall have box type terminal for sound electrical connections.

24.11MOULDED CASE CIRCUIT BREAKERS: - Moulded case circuit breakers shall be of adjustable current setting type with trip free manual closing mechanisms. MCCB's shall conform to IS: 13947 -1, 2 (1993). They shall be of single break type and shall be trip free from all positions and temperature compensated for thermal type only. Suitable discrimination shall be provided between upstream and downstream breakers. All moulded case circuit breakers will have front operated extended rotary handle offering IP54 protection. They shall have minimum watt loss per pole. The closed, minimum trip position and open positions must be clearly indicated as per clause in IS part-I of the IS standard. MCCB cover and case shall be made of high strength heat resistant and flame resistant thermosetting insulating material, operating handle shall be quick make, quick break type. Front operating handle shall have a common operating handle for simultaneous operations and tripping of all three phases. All MCCB's shall have indication light for ON.

24.11.1 MCCB shall be extra current limiting type and manufacturer shall furnish let through energy curves. All capacities taken are service breaking short circuit capacity. The MCCB shall be suitable for aluminium terminations without any link or component heated up. All the MCCB's shall have variable trip setting facility which could be adjusted easily from front at site. It shall have clear TRIP position marked in front. Electrical life of the MCCB shall be as follows :

upto 100 amps - 6000 operations
upto 250 amps - 3500 operations
upto 400 amps - 3000 operations

24.11.2 MCCB shall have provision of adding in future accessories like under voltage trip, shunt trip, alarm switch, auxiliary switches, earth fault and earth leakage for fitting if required. Whenever interlocking is required, suitable selection shall be made. The MCCB shall have extended links with phase barriers and temp. of links shall not reach beyond the limits as specified in IS.

24.12PRODUCTION OF TEST CERTIFICATES**: -**The contractor shall submit manufacturer's test certificates for the following material/equipment along with original purchase vouchers to Garrison Engineer before incorporating the same in the work and before claiming the payments.

(a)LT Cables
(b) Steel tubular poles
(c) XLPE Cable
(d)MCCBs
(e) ACBs
(f) Relays
(g)Feeder Pillars

24.13CATALOGES & BOOKS/ RECORD OF INSTALLATION

24.13.1 On completion of work, the contractor shall submit to the Engineer-in-charge 6 copies of books duly bounded at no extra cost, containing the following: -

(i) Diagram of equipments & connection thereto (complete wiring diagram of VCBs/ ACB's/ MCCB's/ panels etc) installed in substation switch room/rooms.
(ii)Details/layout of bus bar ducts / trunking, tap off boxes and underground cables.
(iii) Details of switch gear i.e. its capacity etc alongwith cables/cable boxes/feeder pillar-boxes etc.
(iv) Manufacturers catalogue of transformers, switch gears HT/ LT/ OT /PT, ACB/ MCCB / DG set, various instruments & protection devices and streetlights etc.

24.13.2 Six copies of maintenance/operation manual of all equipments installed shall be supplied by the contractor free of cost.

PARTICULAR SPECIFICATIONS (Contd…)

24.14GENERAL ARRANGEMENT & MARKING DRAWINGS

24.14.1 The contractor shall submit detailed marking drawings for HT switchgear panels & LT switch gear panels, BBTs etc incorporating all the specified equipments/accessories, controls & instruments for the equipments for the approval of GE before commencement of work.

24.14.2 The duct sizes & layouts in the sub-station for cables and equipments etc and the foundation pits for panels required will be submitted by the contractor alongwith these equipment drawings (not later than 3 months after acceptance for approval).

24.15TAKING OVER OF COMPLETED INSTALLATION

24.15.1 The installation shall be taken over by MES after initial inspection approval and commissioning only of equipments viz HT/LT switch gears, transformers and Gen set etc is carried out by electrical inspector of the department. Testing of DG set shall be carried out by the rep of Acceptance Officer.

24.15.2 The complete work from initial sources and to the equipment tail end shall be carried out keeping correct phase sequence in all cables BBT's and switchgear connections etc.

24.16TESTING OF INSTALLATION

24.16.1 The contractor shall carryout the tests of completed installation in the presence of Engineer-in-Charge as per relevant IS and test results shall be recorded before taking over installation.

24.16..2 All expenses incurred on such tests including all labour and provision of testing apparatus required to carry out the above mentioned tests shall be borne/arranged by the contractor without any extra cost. However, electric energy for testing will be given free of charge. If the results of tests are not found satisfactory, the contractor shall at his own expense rectify/replace the defective equipment/material or any part there of as directed by the Engineer-in-Charge. The decision of Accepting Officer shall be final and binding in this respect.

24.16.3 Six copies of report of site tests carried out on all items and signed by representative of contractor and Engineer-in-Charge shall be submitted to GE for approval.

24.17 PRECAST CONCRETE CABLE COVERS

Precast concrete cable covers shall comply with IS:5820-1970, Specification for precast concrete cable covers and shall be of class and type as indicated. It shall be as specified in clause 19.20 of MES Schedule Part I.

25.0 **EXTERNAL WATER SUPPLY**

25.1 GENERAL

Works under this contract shall be carried out in accordance with the drawing and these particular specifications read in conjunction with the specifications, general rules, special conditions and all preamble contained in MES Standard Schedule of Rates 2009 Part I and SSR -2010 Part II. In case of variance the provisions of particular specifications shall take precedence over the aforesaid provisions in the MES Schedule Part I

25.2 SCOPE OF WORK

Laying of DI/ GI (medium grade) water tube with necessary fittings and connections from existing pipe line to building/ structure

25.3 MATERIALS

All materials incorporated in this work shall conform to the relevant Indian Standard Specifications and shall be best indigenous make of reputed firm approved by GE.

25.4. **GI WATER TUBING**

25.4.1 The requirement pertaining to materials conform to with National Building Code, workmanship, testing, recording of installations shall be all as specified in MES Schedule clause 18.40 and 18.41.Laying and jointing of GI pipe shall be in accordance with Clause 18.50 and 18.51 of MES Schedule. All pipe work shall be laid or fixed to be completely airtight and water tight as specified.

PARTICULAR SPECIFICATIONS (Contd…)

25.4.2 Testing of pipe shall be carried out as specified in clause 18.50.4 of MES Schedule. Record of testing shall be maintained separately for each block.

25.4.3 Pig lead and wool shall comply with IS-702, 1978 specified for caulking lead.

25.4.4 Pig lead shall be uniform quality, clean and free from foreign materials and shall be of uniform softness and capable of being easily caulked and driven.

25.4.5 Lead wool shall not contain sulphur and shall not be manufactured from discarded accumulator battery plates. The lead wool shall consist of fine strands or plate ribbons of lead. The cross section of individual strand shall be flat. The dimensions in sectional plane shall not be less than 0.13mm and not more than 0.9mm and of length same as the length of rope.

25.4.6 Run lead joint shall be provided as per clause 18.48.2 of SSR Part- I

25.5 TESTING OF PIPE LINES

Testing of pipelines shall be carried out as stated in Clause 18.48.7 and 18.50.4 of MES Schedule 2009, Part I and result of test shall be recorded.

25.6 CONTROL VALVE: Control valve provided in the ground shall be housed in a valve chamber.

25.7 SLUICE VALVE

Sluice valve shall conform to IS 14846-2000 long body type class I and tested to 300 psi pressure. The spindle of the valve shall be made of high tensile bronze and the valve scaling and facing shall be of gunmetal. All other parts shall be closed grained cast iron and shall be of make mentioned in Appendix 'B'

25.8 VALVE CHAMBER

Valve chamber shall be constructed to the size as indicated in Schedule 'A' and as per details shown on drawings. The specifications for various items as described in Schedule 'A' and shown on drawings shall be as described here in before.

25.9 TESTING OF PIPE LINE AFTER LAYING

Testing shall be carried out by the contractor all as specified in clause No 18.54 and 18.55 of MES Schedule Part-I (2009) in the presence of Engineer-in-Charge. If any fittings, specials, joints leak during testing, the same shall be replaced / rectified by the contractor without any extra cost to the Government. The lumpsum amount quoted by the tenderers against the Schedule 'A' is deemed to include for the above provision and no extra claim will be entertained on this account.

25.10 MAKING GOOD

Roads, footpath including side drains if any cut through for trenches shall be made good by the contractor to match the original specification as directed by the Engineer-in-Charge. When roads shave to be crossed, half the width of the road shall be dug at a time and proper warning notices, signs and lights shall be displayed and watchman posted by the contractor at his own cost.

25.11 Gun metal gate/globe valve shall have IS certificate marks and of make mentioned in Appendix 'B' and as approved by GE.

26.0 **ARBORICULTURE**

26.1 The Contractor shall provide all horticultural operations and services specified in the schedule 'A'/ BOQ of quantities including:

(a) **Sweet Earth**

(i) The soil shall be fertile, friable, natural topsoil typical of locality, and shall be obtained from a well-drained site that is free of flooding. It shall be without admixture of subsoil or slag and shall be free of stones, lumps, plants or their roots, sticks and other extraneous matter and shall not be delivered or used while in a frozen or muddy condition.

(ii) The soil as delivered to the site shall have an acidity range of PH 6.5 to 7.5 and shall not contain less than 5% organic matter. Sufficient Limestone shall be added to topsoil to bring it to a range of PH 6.5 to 7.5.

PARTICULAR SPECIFICATIONS (Contd…)

(b) Grading

(i) Grade lawn areas to finish grades, filling as needed or removing surplus dirt and floating areas to a smooth, uniform grade as indicated. All *lawn* areas shall slope to drain. Where no grade are shown, areas shall have a Smooth and continual grade between existing contours (such as walks, curbs) catch basin, elevation at steps or building) and elevation shown on plans. RoH, scarify, rake and level as necessary to obtain true, even lawn surfaces, all finish grades shall meet approval of Engr-in-Charge, before lawn is laid.

(ii) Chemical fertilizer shall be spread on areas to be seeded at the rate of 20 lbs. per 100 square feet to fertilizer shall be 50% organic and shall be applied from a suitable spreader.

(iii)Natural fertilizer — Manure shall be well-unleached, cattle manure free from sawdust shavings, refuse and harmful chemicals, manure shall contain no lumps which will not pass a 2 inch sieve. Spreading of 1 inch thick layer of manure on the ground and mixing with the good earth.

26.2 After incorporating of fertilizer in the soil, the lawn bed shall be fine graded to remove all ridges and depressions, and surface cleared of all stones 1 inch or more diameter and of other debris. Atleast 1 inch layer of manure to be spread on the ground and mix with earth for grassing. Ref given specification for lawn bed preparation.

26.3 PLANT MATERIAL TREES, SHRUBS, GROUND COVERS, CREEPERS, VINES, ETC.

a. Plant List Plants are listed in Sch 'A'/ BOQ.

b. Nomenclature - The names of the plants confirm to standardize botanical names.

c. **Quality and General Requirements of Plants**.

(j) Plants shall be typical of their species and variety, have normal growth habits, well developed branches, densely foliated with vigorous and fibrous root systems.

(ii) Plants shall be free form defects and injuries & Barks shall be free from abrasion.

(iii) Plants shall be freshly dug and nursery grown. Plants shall have been grown under climatic conditions similar to those in locality of project, or have been acclimated for at least 2 years to conditions of project location. Nursery grown plant shall have been at least once transplanted.

(iv) Each bundle of plants and all separate plants shall be properly identified by weather proof labels securely attached there to before delivery to project site.

d. **Size of Plants.**

(i) All plants shall be equal to or exceed the sizes given in the Sch 'A'/ BOQ which are minimum acceptable sizes. Plant shall be measured before planting, with branches in normal position.

26.4 **SOIL:** Topsoil shall be friable, fertile, typical or the cultivated topsoil of the locality, containing at least **2%** of decay organic matter. Top soil shall be taken from well drained arable site, shall be reasonably free of subsoil, stones. weeds, earth, clods, sticks, roots or other objectionable extraneous matter or debris and shall contain no toxic materials.

26.5 AFTER PLANTING CARE

Watering - Watering for trees and other plants by flooding twice within first 24 hours of the time of planting. Tree requires 6 gallons per plant per day for fist 45 days from the date of planting and subsequently every alternate day for the next 6 months. Thereafter watering twice a week at the rate of 10 gallons per plant.Water required for Plants/ tree during this period shall be own arrangement of contractor either from borewell installed at site or from anywhere else.

26.6 GUARANTEE

(a) Any plant during currency of work plantation found missing/ dead not true to name or size as specified or not in satisfactory growth shall be replaced immediately by contractor with a plant of approved size and shape. All cost shall be borne by the contractor on this account and nothing extra shall be admissible on this account.

(b) Period - All plants shall be guaranteed by the contractor for 36 (Thirty six) months after the certified date of completion for replacement of damaged/ defective plants.

PARTICULAR SPECIFICATIONS (Contd…)

(c) Replacement - All replacement shall be plants of the same species, variety and size as specified in the Sch 'A'/ BOQ.

27.0 CLEANING OF SITE

27.1 The Contractor shall properly clean the work site/s as work progresses and shall remove all rubbish and debris from the site time to time when necessary and when directed. On completion, the Contractor shall ensure that the premises and/or site are cleaned off surplus material, debris, etc. to the satisfaction of the Garrison Engineer. The quoted rates shall inclusive of above and nothing extra shall be paid on this account.

28.0 DEMOLITION/ DISMANTLING:

28.1 Demolition/ dismantling shall be carried out as directed by Engineer-in-charge. All unserviceable/ serviceable material shall be kept separately and they shall be responsible of the contractor for safe custody. All precautions shall be carried out while demolition/ dismantling being carried out that any damage occurred shall be made dood by contractor without any extra cost to the government. All the unserviceable materials not used, obtained from dismantling/ taking down shall be returned to MES stores/ removed from site of work as directed by Engineer-in-charge.

29.0 List of Drawing: Drawings listed in next pages shall form part of these tender documents

Signature of Contractor **AAD (Contracts)**
Date: **For Accepting Officer**

PARTICULAR SPECIFICATIONS (Contd…)

LIST OF DRAWINGS

SN	DESCRIPTION	DRG. NO	SH NO	DATE OF DRAWING	DATE OF REV
1	2	3	4	5	6
1	LIST OF DRAWING	CE(CG)GOA/LD-1061/2023	01/01	16.10.2024	21.01.2025
	SITE PLAN				
2	SITE PLAN SHOWING EXTERNAL B/R SERVICES	CE(CG)GOA/SP-1061/2023	01/02	15.10.2024	
3	SITE PLAN SHOWING EXTERNAL E/M SERVICES	CE(CG)GOA/SP-1061/2023	02/02	16.10.2024	
	ARCHITECTURAL DRAWINGS				
4	GROUND FLOOR PLAN AND FIRST FLOOR PLAN	CE(CG)GOA/AR-1061/2023	01/08	14.10.2024	09.01.2025
5	SECOND FLOOR PLAN, ROOF PLAN AND MUMPTY FLOOR PLAN	CE(CG)GOA/AR-1061/2023	02/08	14.10.2024	09.01.2025
6	SECTION A-A, B-B & C-C	CE(CG)GOA/AR-1061/2023	03/08	14.10.2024	09.01.2025
7	FRONT, REAR ELEVATION, RIGHT SIDE AND LEFT SIDE ELEVATION	CE(CG)GOA/AR-1061/2023	04/08	14.10.2024	
8	KITCHEN DETAILS	CE(CG)GOA/AR-1061/2023	05/08	14.10.2024	
9	TOILET DETAILS	CE(CG)GOA/AR-1061/2023	06/08	14.10.2024	
9	TYPICAL DETAILS OF COOKING PLATFORM, OVERHEAD CABINET AND CUPBOARD	CE(CG)GOA/AR-1061/2023	07/08	14.10.2024	
10	MISCELLENEOUS DETAILS	CE(CG)GOA/AR-1061/2023	08/08	14.10.2024	09.01.2025
	STRUCTURAL DRAWINGS				
11	STRUCTURAL GENERAL NOTES	CE(CG)GOA/STR–GEN-1061/2023	01R/01	21.01.2025	
12	STRUCTURAL STANDARD TYPICAL DETAILS	CE(CG)GOA/STR-STD-1061/2023	01/02	11.10.2024	
13	STRUCTURAL STANDARD TYPICAL DETAILS	CE(CG)GOA/STR-STD-1061/2023	02/02	11.10.2024	
14	LAYOUT PLAN AND SCHEDULE OF FOOTINGS AND DETAILS OF FOOTING	CE(CG)GOA/STR-1061/2023	01/05	11.10.2024	
15	LAYOUT PLAN AND SCHEDULE OF COLUMNS	CE(CG)GOA/STR-1061/2023	02/05	11.10.2024	
16	LAYOUT PLAN AND SCHEDULE BEAMS AND SLABS AT 1ST& 2NDFLOOR LEVELS	CE(CG)GOA/STR-1061/2023	03/05	11.10.2024	21.01.2025
17	LAYOUT PLAN OF ROOF BEAMS , MUMPTY BEAMS, MIDLANDING BEAMS AND SLAB AT ALL FLOOR LEVEL	CE(CG)GOA/STR-1061/2023	04/05	11.10.2024	21.01.2025
18	LAYOUT PLAN OF LINTEL BEAMS AND TYPICAL DETAILS OF LINTEL BEAMS	CE(CG)GOA/STR-1061/2023	05/05	11.10.2024	21.01.2025

PARTICULAR SPECIFICATIONS (Contd…)

LIST OF DRAWINGS (Contd)

1	2	3	4	5	6
	B/R DRAWINGS				
19	SCHEDULE OF FINISHES	CE(CG)GOA/SCH-1061/2023	01/02	15.10.2024	
20	SCHEDULE OF FINISHES(SPECIFICATIONS)	CE(CG)GOA/SCH-1061/2023	02/02	15.10.2024	
21	SCHEMATIC DIAGRAM OF INTERNAL WATER SUPPLY	CE(CG)GOA/SCH-1061/2023	01/01	15.10.2024	
	E/M DRAWINGS				
22	GROUND FLOOR PLAN SHOWING INTERNAL ELECTRIFICATION	CE(CG)GOA/EM-1061/2023	01/02	16.10.2024	
23	SECOND FLOOR PLAN, ROOF PLAN SHOWING INTERNAL ELECTRIFICATION & SCHEMATIC DIAGRAM OF INTERNAL ELCETRIFICATION	CE(CG)GOA/EM-1061/2023	02/02	16.10.2024	
	TYPICAL DRAWINGS OF CE (CG) GOA				
24	ARCHITECTURAL NOTES	CE(CG)GOA/TD-101	1/1	31-01-2020	
25	TYPICAL DETAILS OF FLUSH DOOR WITH WOODEN FRAME	CE(CG)GOA/TD-102	1R/1	24-03-2022	09.01.2025
26	DETAILS OF BRACKET & OTHER MISC DETAILS FOR VERTICAL PIPE & HORIZANTAL PIPE SUPPORTING SYSTEM	CE(CG)GOA/TD-A-109-2023	1/3 to 3/3	22-08-2023	
27	TYPICAL DETAILS OF UPVC WINDOWS & VENTILATOR	CE(CG)GOA/TD-A-2024-110	01/01	17.09.2024	
28	TYPICAL DETAILS OF UPVC WINDOWS WITH FLY & MOSQUITO PROOFING	CE(CG)GOA/TD-A-2024-112	01/01	06.10.2024	
	TYPICAL DRAWINGS OF CE (NAVY) KOCHI				
29	DETAILS OF PELMET BOXES AND <1 BRACKETS FOR CURTAIN ROD	TD/A/118	1/3 to 3/3		
30	DETAIL OF WALL PEGS AND HOOKS	TD/A/136	1/1		
31	FUNCTIONAL HEIGHTS OF TOILET/LAVATORY FITTINGS	TD/A/146	1/1		
32	ARCHITECTURAL NORMS HEIGHT OF VARIOUS FITTINGS	TD/A/147	1/1		
33	DETAILS OF ROAD, HARDSTAND AND FOOT PATH	TD/A/170	1/1		
34	TYPICAL DETAILS OF STEPS AND PLINTH PROTECTION AND RAMP	TD/A/190	1/1		
35	TYPICAL DETAILS OF ALUMINIUM WINDOWS/VENTS	TD/2004/6	1/3 to 3/3		
36	DETAILS OF DRAINS AND COVER SLABS	TD/A/167	1/1		
37	TYPICAL DETAILS OF HUME PIPE CULVERT	TD/A/168	1/1		
38	LEGEND FOR INTERNAL ELECTRICAL INSTALLATION & TYPICAL DETAILS OF FIXING FAN HOOK, TYPICAL SECTIONAL ELEVATION SHOWING HEIGHTS OF DIFFERENT ELECTRICAL FITTINGS.	TD/E/401	1/1		

PARTICULAR SPECIFICATIONS (Contd…)

1	2	3	4	5	6
39	TYPICAL DETAILS OF SWITCH BOX, FAN HOOK, SWITCH CUPBOARD AND WATER METER BOX	TD/99/55	1/5 to 5/5		
40	DETAIL OF FIXING EXHAUST FAN	TD/E/411	1/1		
41	TYPICAL FIXING DETAIL OF HDPE WATER STORAGE TANKS OVER RCC ROOF SLAB	TD/A/156	1/1		
42	DETAILS OF VALVE CHAMBER	TD/A/169	1/1		
43	TYPICAL DETAILS OF WC & WASTE WATER PIPE CONNECTIONS MANHOLE ETC	TD/77/4	1/1		
44	TYPICAL DETAIL OF CRUMPLE / EXPANSION JOINTS	TD/S/85/8	1/1		
45	DETAILS OF RWP	TD/A/150	1/1		
46	ALUMINIUM SLIDING WINDOWS	TD/A/197	2/3		
	TYPICAL DRAWINGS CE (NAVY) MUMBAI				
47	SEPTIC TANK 150 TO 500 USERS	TD/2007/10	1/4 to 4/4		

Signature of Contractor
Date

AAD(Contracts)
For Accepting Officer

PARTICULAR SPECIFICATIONS (Contd…)
YARD STICK
SCHEDULE 'A' PART-I, ITEM NO.1
(Name of Building: JCOs/ SOS Single Living Accommodation)

Sr. No	Stage of Works	% as per Estimate
1	**Stage - 1 : Upto Plinth Level**	
	Surface dressing, excavation, PCC (1:4:8) in Foundation,& PCC (1:3:6) in Combined Footing, PCC (1:2:4) For Plinth beams, Plinth beams & dwarf wall, RCC M-30 in Foundation, plinth beam, Column including steel reinforcement, form work, Return filling in foundation & back filling under floor complete upto Plinth level, PCC Block Masonry upto plinth level including ATT complete	15.00
2	**Stage - 2 : Plinth Top To G. Floor Lintel Top Level**	
	RCC M-30 in Column including reinforcement, form work from Plinth Top To G. Floor Lintel Level, PCC Block Masonry, RCC Lintel Band & Chajja including reinforcement, form work, complete.	4.00
3	**Stage - 3 : G. Floor Lintel Top To G. Floor Slab Level**	
	RCC M-30 in Column including reinforcement, Form work from G.Floor Lintel Top To G.Floor Slab Level, PCC Block Masonry, RCC M-30 in Beams, RCC slab, Parapet Wall, Staircase, Form work, complete.	10.50
4	**Stage - 4 : G. Floor Slab Top Level To 1st Floor Lintel Top Level.**	
	RCC M-30 in Column including reinforcement, form work from G. Floor Slab To F. Floor Lintel Level, PCC Block Masonry, RCC Lintel Band & Chajja including reinforcement, form work, complete.	5.30
5	**Stage - 5 : 1st Floor Lintel Top To 1st Floor Slab Level**	
	RCC M-30 in Column including reinforcement, Form work from F. Floor Lintel Top To F. Floor Slab Level & Mumty slab level, PCC Block Masonry, RCC M-30 in Beams, RCC slab, Staircase including reinforcement, form work, complete.	10.90
6	**Stage - 6 : F. Floor Slab To S. Floor Lintel Level**	
	RCC M-30 in Column including reinforcement, form work from G. Floor Slab To F. Floor Lintel Level, PCC Block Masonry, RCC Lintel Band & Chajja including reinforcement, form work, complete.	5.24
7	**Stage - 7 : S. Floor Lintel Top To Roof Slab Level**	
	RCC M-30 in Column including reinforcement, Form work from F. Floor Lintel Top To F. Floor Slab Level & Mumty slab level, PCC Block Masonry, RCC M-30 in Beams, RCC slab, Staircase including reinforcement, form work, complete.	14.50
8	**Stage - 8 : Plaster Work (G. Floor)**	
	10mm thick in CM (1:6) to internal surface of walls , 5mm thick in CM (1:3) on Ceiling & 15mm thick in two layers 10mm thick in 1:6 and 5mm thick finishing layer in cm 1:4 to external surface of wall Cement etc.	1.50
9	**Stage - 9 : Plaster Work (1st Floor)**	
	10mm thick in CM (1:6) to internal surface of walls , 5mm thick in CM (1:3) on Ceiling & 15mm thick in two layers 10mm thick in 1:6 and 5mm thick finishing layer in cm 1:4 to external surface of wall Cement etc.	2.30
10	**Stage - 10 : Plaster Work (2nd Floor)**	
	10mm thick in CM (1:6) to internal surface of walls , 5mm thick in CM (1:3) on Ceiling & 15mm thick in two layers 10mm thick in 1:6 and 5mm thick finishing layer in cm 1:4 to external surface of wall Cement etc.	2.56

PARTICULAR SPECIFICATIONS (Contd…)

YARD STICK (Contd..)

11	**Stage - 11 : Joinery (G. Floot To 2nd Floor)**	
	Wooden door frame with Flush door shutter ,UPVC Window & Steel ventilator frame's, Shutters, FRP Doors etc.	7.96
12	**Stage - 12 : Surface Finishes (G. Floor)**	
	Painting Work For Internal, External, Ceiling Newly Plastered Surfaces	0.90
13	**Stage - 13 : Surface Finishes (1st Floor)**	
	Painting Work For Internal, External, Ceiling Newly Plastered Surfaces	1.25
14	**Stage - 14 : Surface Finishes (2nd Floor)**	
	Painting Work For Internal, External, Ceiling Newly Plastered Surfaces	1.40
15	**Stage - 15 : Flooring (G. Floor)**	
	PCC Flooring sub base to floors, Screed bed, Coloured Non-skid Ceramic Tiles, Mirror Polished Kota Stone Flooring, Backing Coat, Coloured Ceramic Tiles for skirting, Glazed Ceramic Tiles dado, etc. complete	3.80
16	**Stage - 16 : Flooring (1st Floor)**	
	PCC Flooring sub base to floors, Screed bed, Coloured Non-skid Ceramic Tiles, Mirror Polished Kota Stone Flooring, Backing Coat, Coloured Ceramic Tiles for skirting, Glazed Ceramic Tiles dado, etc. complete	3.56
17	**Stage - 17 : Flooring (2nd Floor)**	
	PCC Flooring sub base to floors, Screed bed, Coloured Ceramic Tiles, Coloured Non-skid Ceramic Tiles, Mirror Polished Kota Stone Flooring, Backing Coat, Coloured Ceramic Tiles for skirting, Glazed Ceramic Tiles dado, etc. complete	3.60
18	**Stage - 18 : Water Proofing**	
	Roof Treatment Complete	1.60
19	**Stage - 19 : Sanitary Fixtures**	
	Sanitary Items & Fixtures complete	1.30
20	**Stage - 20 : Misc Items**	
	Site Clearance, Plinth protection & other unforeseen items	2.83
	Total	**100%**

Note:

(a) The yardstick is purely for payment purpose and description of various stages is in brief and shall neither be used for intention of contract nor to decide scope of work.

(b) The assessment of work of each stage shall be done by Engineer-in-charge and in case of any dispute, decision of Accepting officer shall be final and binding.

(c) In case of dispute on scope of each stage for payment purpose the decision of Accepting officer shall be final and binding.

Signature of Contractor **AAD (Contracts)**
Date: **For Accepting Office**

PARTICULAR SPECIFICATIONS (Contd…)

MAKES / NAMES OF FIRMS FOR PRODUCTS TO BE INCORPORATED IN WORK

Appendix 'A'

SL No.	ITEMS	MAKES
01	CEMENT	M/s Associated CementCompaniesLtd.**Brand: "ACC"**
		M/sUltraTechCementLtd**Brand: "ULTRATECH"**
		The India Cement
		Dalmai Cement (Bharat) Ltd. **Brand**: "**DALMIA INFRA PRO"**
		Century Cements **Brand: "CENTURY**"
		Saurashtra Cement **Brand: "SAURASHTRA**"
		The Ramco Cements Ltd. (Formerly Madras Cement) **Brand:** "**RAMCO**"
		Mangalam Cement Ltd **Brand : "MANGALAM"**
		Birla Corporation Ltd**Brand: "BIRLA"**
		Orient Cement**Brand: "BIRLA-A1"**
		Nuvoco Vistas Corporation Ltd (Formerly Lafarge Cement) **Brand: "NUVOCO"**
		Shree Cement Ltd**Brand: "SHREE"**
		J K Cement**Brand: "J K"**
		J K Lakshmi Cement Ltd**Brand: "J K LAKSHMI"**
		Jaypee Rewa Cement **Brand: "JAYPEE"**
		Ambuja Cement Ltd **Brand: "AMBUJA"**
		M/s JSW Cement Ltd, A. P **Brand "JSW PSC"**
		M/s Kesoram Industries Ltd. **Barnd: "BIRLA SHAKTI"**
		M/s Sagar Cements Limited, Hyderabad **BRAND: "SAGAR"**
02	Reinforcement steel (CRS-TMT)	Rashtriya Ispat NigamLimited (RINL) **Brand: "RINL"**
		Tata Iron & Steel Company (TISCO, orTata Steel)**Brand: "TATA"**
		Steel Authority of India Limited (SAIL)**Brand: "SAIL"**
		M/s Kamachi Industries Ltd. **Brand: "KAMACHI"**
		M/s Shyam Metalics & Energy Ltd.**Brand:"SEL"**
		M/s Electrotherm (India) Ltd. **Brand: "ET TMT"**
		M/s Tulsyan NEC Limited **Brand: "TULSYAN TMT"**
		M/s Incredible Industries Limited (Formerly M/s Adhunik Industries Limited.) **Brand: "ADHUNIK Fe 500 SD"**
		M/s Gallantt Ispat Ltd, Gujarat **Brand: "GALLANT TMX"**
		M/s Rashmi Metaliks Limited, W.B. **Brand: "RASHMI TMT"**
		M/s Real Ispat & Power Limited, Chhatishgarh **Brand: "G K TMT"**
		M/s Steel Exchange India Ltd.(AP)**Brand: "SIMHADRI TMT"**
		M/s Super Smelters Ltd, Kolkata. **Brand: "SUPER SHAKTI"**
		M/s Shyam Steel Industries Ltd.(WB)**Brand: "SHYAM"**
		M/s Jai Balaji Industries Ltd.(WB)**Brand: "BALAJI SHAKTI"**
		M/s Bajrang Power & Ispat Ltd Raipur
		M/s MSP Steel & Power Limited, Chhatishgarh **Brand: "MSP TMT 500D"**
		M/s. Sugna Metals Ltd, Hyderabad
03	**Structural Steel:**	Rashtriya Ispat NigamLimited (RINL) **Brand: "RINL"**
		Steel Authority of India Limited (SAIL)**Brand: "SAIL"**
		Tata Iron & Steel Company (TISCO, orTata Steel)**Brand: "TATA"**
		M/s Jindal Steels and Power Ltd. **Brand: "JINDAL"**

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A (Contd..)

SL No.	ITEMS	MAKES
JOINERY AND HARDWARE		
1.	Factory made wooden shutter/Flush doors/Frames	Century Ply Industries, Greenlam (Brand : Mikasa Door Frames) Jain Doors, A-1 Teak Products Pvt Ltd
2.	Panelled doors/Skelton door shutters	Pioneer Timbers A-1 Teak Products Pvt Ltd Goyal Industries Doorking Industries M/s MP Wood Products Jain Wood Industries "Jain Brand" Indian Timber Product
3.	Flush Doors	Archid Ply, Duro Doord, Bhutton tuf, Swastik, Durian
4.	Factory made PVC / FRP door frames and shutter	Qute Extrusions Pvt Ltd (Brand Qute), OM Industries Sintex Industries Ltd, Rajshree Plastiwood Ltd, Duroplast Extrusion, Fenesta, Durian Sarvesh Multi Plast India Ltd
5.	UPVC Doors, Windows, Kitchen cabinets and Ventilators, Ceiling, Wall Panelling and duct cover	Fenesta, Qute Extrusions Pvt Ltd (Brand Qute) Madhu Industries, H2O Solutions, Prominance, NCL Veka Limited (KPT), ALUPLAST, Rehau India Pvt Ltd, Astral Limited, Sarvesh Multi Plast India Ltd, Yashashri Polyextrusion LTD, Duroplast Extrusion, Astrapia
6.	Wood Plastic Composite (WPC) Doors & Frames	Kalinga, Ecoste, Hardyplast, Qute Extrusions Pvt Ltd (Brand Qute) Duroplast, Sarvesh Multi Plast India Ltd
7.	Steel windows, Ventilators, Door frames, Shutters (Z/Box type)	Godrej & Boyce Steel Mfg Mumbai, Madhu Industries, Ashwani & Sons Gaziabad, Multiwyn Industrial Corporation Kolkata, Agew steel Ahmedabad, Shiv Mullar Ahmedabad, Sen Harvic Mumbai, Steel man Industries Jammu
8.	Wood based partitions	Novapan, Bhutan board, Green Ply, Mangalore, Mysore &Chipboard, Century Ply board India Pvt. Ltd, Kit Ply, Yamuna Interiors Pvt Ltd, Alexia Panels (ALEX)
9.	Dash tru-hold Expansion fasteners	Dash fasteners Pvt Ltd., C-16, South Extension Part II, New Delhi M/s Mon Traders, C-16, Tardeo, Air conditioned Market, Bombay - 34
10.	Steel Rolling shutters & collapsible gates	Welroll, Gandhi Automation Pvt. Ltd, Swastik rolling shutters Mumbai Metal King, Indian Entrance Automation, Indian Rolling Shutters Bangalore, New Karanataka Rolling Shutter Bangalore, New Dynamic Rolling Shutter Chennai, Akash Rolling Shutter Hydrabad, Access Automation Hydrabad, Shri Krishna Associates Panchakula, Eshwar Industries Meerut, Steel man Industries Jalandhar
11.	Mortice locks	Harrison, Godrej & Boyce Co Ltd. Door set India, Spider Metal product Ltd, ENOX, Jainson
12.	Hydraulic door Closers	Godrej Hafele, Ozone Door Set, Spider, IPSA door closer
13.	Sheet Glass- Plain/Frosted	Modiguard, Saint Gobain, Atul Glass, Gold plus Glass industries Ltd,
14.	Heat absorbing glass & Reflective solar control film	Modiguard, Saint Gobain, Atul Glass Industries, Garware
15.	Rough cast/Pin headed wired glass	Modiguard, Saint Gobain, Atul Glass Industries

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A (Contd…)

SL No.	ITEMS	MAKES
16	Oil Putty	Gold Mohar, Shalimar Hardware, Atul Dyes and Chemicals, UK Paint Industries, Jenson & Nicholson, MYK Arment, Berger
17.	Aluminium Sections of shutters/ frames for door/ Windows/ Ventilators	Jindal , Hindalco, Associated Group (Elesar Focci Nivrav Commercial Ltd), Maan Aluminium, Century Extrusions, Banco Alluminum Ltd, Sterlite, Ajit India
18.	Drapery rod	Vista Levolor, MAC-DÉCOR, Deco- Window, Jayesh Metal Corporation (JMC)
19.	Aluminium section of shutters/ frames for door/ window / ventilators	Associated Group (Elesar Focci- Nivrav Commercials Ltd HINDALCo, JINDAL, Maan Aluminium, Century Extrusions, Bance Alluminum Ltd, Sterlite, Ajit India
20	Aluminium Tower Bolt/ Aldrops/ Door Handle/ Butt Hinges / Towel Rail	Jindal, ESSESS, Aluminium Udyog (Brand GLOBAL)
21	Fencing (Chain link, Concertina coil, Weldmesh, Electric, Gabions, Honeycomb, Razor wire, Razor blade tape, UNICO Weld mesh system, Tree guard, Barbed wire, Anti climb and security gates	A-1 Fence Products Company Pvt Ltd, Shankar weld mesh Pvt Ltd, Tata steel Global wires, Secure fencing Delhi, Eco netting Nagpur, OM wires and wire products Kolkatta, Just fence Hydrabad, Rajesh Indudtrial Corporation Chennai, Zeetmaull Janichandlaal Madras Pvt Ltd, Oswalwiremesh Co Bangalore, Mysore wires netting CO Pvt Ltd, Asian Fence Wire Products Hydrabad. M/s. Shiva Engineering works
22	Aluminium & brass mongery	Argent Industries, Crown Industries, Mech India, Nu-lite Industries, Perfect Fabricators, Classic, Aluminium udyog
23	Gypsum Based ceiling	Gypsum board, Arm strong, Ramco, Bison
24	Aluminium section for false ceiling	Jindal, Indal, Hindalco
25	Venetian Blinds	Aerolux, MAC, VISTA Levolor
26	Laminated sheets	Formica, Sun Gloss, Sunmica, Backlyte Hylum.
27	Particle board and Pre laminated particle board	Green Ply (Greenlam MFC Board), Kitlam, Anchorlam, Century Plywood/Sharon
28	Perforated Particle Board/tiles for insulation and acoustic	Anchor ceiling Tiles, Armsrong, Anutone, Aerolite ceiling system, Kitply ceiling ties, Knauf,
29	Plywood/Block Boards	Anchor, Century Ply wood, Mysore Chip Board, Mysore, Green Ply, Archid Ply, BhutanTuff India
30	False ceiling	Anutone, Armstrong, Aerolite ceiling system, Saint Gobain Gyproc, Diamond ceilings, Ramco Hilux
31	False ceiling members (Perimeter, Ceiling Section, Intermediates, Angles etc	Armstrong, Aerolite ceiling system, Saint Gobain Gyproc, Gridsquare, Ramco Hilux
32	Particle Board/ fibre board cement based	NCL Industries Ltd (Bison Panel) Everest Industries Ltd (Everest) Hyderabad Industries Ltd (HIL)
33	Thermal /Acoustic Insulation (Nitrile Rubber) & Insulation covering system/Acoustics panels	Armacell India Pvt Ltd (ARMAFLEX), Lloyd Insulation, Rock wool India Pvt Ltd, Insulation India Engineering, Himalayan Accoustics
34	Adhesives	Pidilite, Fevicol, Vamicol
35	Veneered Particle Board	Greenply, Kitply, Associate Particle Board, Century, Sharon
36	Aluminium Composite Panels	Aluco Bond, Alstone, Alstrong, Viva, Aludecor

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A (Contd…)

SL No.	ITEMS	MAKES
37	Pre moulded non bituminous joint filler board	Elcon, Duran Board HD-100
38	Pre moulded bituminous joint filler board	STP Ltd., Tikitar Industries Ltd, Capital Steel and Chemicals Ltd., Garlex Joint Filler by M/s Garlick & Co Pvt Ltd, Roof Rites Delhi , SN Industries, Delhi
39	Mangalore Tiles	Charminar, Raja, RECHO, Prajapat, Kerala tile Works Trisur
40	Cool roof tiles	Johnson tiles, Somany, Nitco, Asian, Kajaria
41	Pre painted Galvalume/ Galvanised corrugated steel sheets	JSW Colouron+, Tata Bluescope steel, Lloyd insulation, Tata steel BSL, Kirby Building Solutions (I) Ltd
42	Colled rolled close annealed (CRCA) steel sheets Non Asbestos Fibre reinforced (Polypropylene) 6mm cement corrugated sheets	JSW MI, JSW Steel, Tata steel, Nezone Charminar, Ramco, Everest Industries Ltd (EVEREST)
43	AC rain water pipes	Everest, Rajastan asbestos Cement, Hyderabad Asbestos Cement, India Asbestos Products, Rohtas, Sabarmangala Manufacture Co., Hyderabad Charminar
44	Ready mix plaster	Accoplast, Birla Aerocon, Ascolite
45	Ready mix joint mortar	Accoplast, Birla Aerocon, Ascolite
	KITCHEN / BATH / SANITARY FITTINGS	
01	Kitchen chimneys	Bajaj, Sunflower, Butterfly, Glen, Sunflame
02	Stainless steel sink ISI marked	Nirali, Prayag Polymers, JAYNA, Parryware, Shakti, Pheonix, Jaquar (Bathing), Jainko
03	EWC, IWC, Wash Hand Basin, Urinal and other vitreous china sanitaryware	Johnson, Kohler, Parryware, Cera, Jaquar, Neycer, HINDWARE
04	Towel Rails/Brackets and toilet paper holder	Johnson, Kohler, Parryware, Cera, Jaquar, Neycer, HINDWARE
05	PVC Low level flushing cistern	Johnson, Kohler, Parryware, Cera , Jaquar, Commander, Prayag Polymers, Neycer, Jainko
06	Seat and cover for WC	Commander, Parryware, Cera, Neycer, PRAYAG Polymers, Jaquar, Kohler
07	CP/Copper/Brass alloy, Pillar tap, Bib tap, Wall mixer, Shower, Diverters, Waste coupling, Bottle trap, Health faucet, Angle cock, Stop cock, and other related accessoriees	Jaquar (Bathing), Somany, Kohler, Grohe, Hindware
08	PVC toilet cabinet	Commander, Jasmine, Cipla Plast, Rotamax, Polytuff
09	Hot &cold water supply fittings	M/s Jaquar, M/s L&K, Goldline, M/s Crabtree
10	Mirror	Atul, Kohinoor, Swastik, Saint Gobain, Modiguard, Asahi (AIS) Glass
11	PTMT Bathroom and Hardware fittings	Prayag Polymers, Vectus, Jainko, Jaquar (Bathing)

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A (Contd…

SL No.	ITEMS	MAKES
12	Shower rose	Jaquar (Bathing), Kohler, ESS-ESS, Soma, Crabtree
13	CI/Brass/SS Sall cocks	Leader, Zoloto, Prayag Polymers, Jainko,
14	Wall mouted cloth drying stand	Bathla Aluminium, Homewell
CHEMICALS / WATER PROOFING COMPOUNDS		
01	Chemical for anti-termite treatment for structures (Chloropyrifos)(20EC)	Montari Industries Ltd., 78, Nehru Place, New Delhi - 110019 4-H, Clubs (india), 487 A, Gandhi Nagar, Jammu-04,TRISHUL bearing ISI Mark conforming to IS specifications, M/s Dara Chemicals Industries Ltd. D-3, Lajat Nagar-01, New delhi-24' DURSBAN' manufactured by DENOCIL marketed at Mumbai by DE-NOCIL Corporation Pretection Ltd, 1st& 2nd Floor, Priojshah Nagar, Eastern Express Highway, Vikroli (East) Mumbai – 79 ,M/s Kanonia chemicals & Industries, PO Renukoot-231217, Dist-Sonebhada (UP) M/s India Pesticides Ltd., E-17 to 23, UOSIDS Industrial Area, Deva Road, Chinal Lucknow (UP)-227105, M/s Pest Control (Rajastan) Guruhari Niwas D-232 Marg Bani Park Jaipur, Firm having membership of IPCA and holding Valid licence
02	Water proofing compound	Pidilite Industries Ltd, taloja, raigad District, super Aquacem (India) Ltd., New delhi M/s Krishna Conchem Products Pvt Ltd , STP LTD, SICO NO 1 ,IMERMO ,EXCOT,SCOT NO 1 FOSROC,M/s Bauchemimic (India)
	FLOORING/TILING	
01	Mosaic/Cement flooring tiles/Terrazzo tiles	Nitco, Hindustan Tiles, Ultra Tiles, Duracrete
02	Glazed wall tiles / Non skid ceramic tiles	Johnson Tiles, Somany Tiles, Euro Tiles, Kajaria Tiles, AGL Tiles
03	Vitrified Tiles	Johnson Tiles, Somani Tiles,Nitco Tiles, Kajaria , Orient, Bell, URO Tiles
04	Acid resistant Tiles	Johnson Tiles, Somani Tiles, Kajaria Tiles Burns Standard Co. Jabalpur, M/s Parshuram Pottery Works, Rajkot
05	Inter locking tiles/ Paver blocks	Ultra Tiles, NITCO, Topaz Tiles, Godrej Boyce Mfg Co Ltd
06	PVC sheet and tile flooring	Krishna vinyl Tiles, Armstrong, Marblex Tiles, Polyfin Tiles, Square Foot
07	Laminated Wooden Flooring	Action Tesa, Armstrong & Bruce, Pergo, Greenply Industries Ltd (Green Floormax)
08	AAC Blocks	Ultra Tech, Birla Aerocon, Ascolite, Siporex, Magicrete, Godrej & Boyce Mfg Co

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A (Contd…)

SL No.	ITEMS	MAKES
	PAINTS/FINISHES	
1	Primer	Asian paints, Nerolac, Shalimar, J&N, Berger Synthetic Enamel
2	Synthetic enamel paint	Berger Paints (Luxol), Nerolac (Synthetic Enamel Hi Gloss), Asian Paints (Apcolite Premium Gloss), Johnson & Nicholson, Akzo Nobel Dulux (Super Gloss), MYK Arment
3	Anti corrosive paint on structural steel	Asina Paints, Nerolac, Berger,Dulux, Shalimar, Johnson Nicholson
4	Cement Base Paint	SUPER SNOWCEM, DUROCEM, AQUACEM, SHALIMAR, Berger
5	Interior Emulsion paints	Berger Paints (Rangoli Acrylic emulsion), Asian Paints (Premium emulsion), Nerolac Paints (Beauty Gold acrylic emulsion), Shalimar Paints
6	Exterior Emulsion paints	Asian Paint (Apex/Ultima), Berger Paint (Weather coat all guard) ICI Dulux (Whether shield max), Kansai Nerolac (Excel Top Guard),
6a	Interior Plastic emulsion paints	Asian Paints (Apcolite premium Emulsion Nerolac (Beauty Gold) Berger (Rangoli Total Care)
7	Acrylic Washable distemper	Asian Paints (Tractor Uno), Kansai Nerolac Paints (Beauty), Berger Paints (Jadoo), Shalimar Paints, Jonson & Nicholson. MYK Arment
8	Colour anodizing for Aluminium sections	Conpaptex, Vadodara, KrishnaMetal industries, Mumbai. Mansi chemical, ahemdabad, Anand tech Process, Bengaluru
9	Cement Putty	Birla Cement, J K White, Golden Mohar, Asian paints, JK Cement, Snowcem Paints, MYK Arment
10	Hot applied thermo plastic road marking paint (As per Clause 803.4 of MORTH)	M/s Shivalik Primo India Pvt Ltd. New delhi, Asian Paints Ltd, SHALIMAR, STP Ltd, SAI Thermoplastic Paints Ltd..
PLUMBING		
1	CI Pipes (centrifugally cast & vertical cast) and fittings	NECO, Kesoram, Electrosteel, Kejriwal,
2	GI and MS pipes	Tata, Swastik, Jindal, Prakash, Zenith,
3	PVC SWR Pipes	Kissan, Prince, Supreme, Finolex, Infra, Astral
4	Gun metal Globe/Gate valve/Angle valve	Leader, Zoloto, Bir, Kirloskar
5	PVC water tanks/Polythylene (ISI marked)	Sintex, Polycon, Rolex, Infra, Astral Limited, Supreme Industries Ltd
6	PVC Float Valves and accessories	Prayag polymers, GMP, Neta, Zolota, Symet, Jaquar (Bathing)
7	Loft Tanks	Sintex, Polycon, Plasto, Ganga
	MISCELLANEOUS	
1	APP membrane	Asian Paints Ltd, STP, IWL, Torchtar, Tiki Tar, Shivam Tar Products, Bengal Bitumen
2	Protective Coating to Reinforcement Bars	PSL Ltd, Zary Causeway, KAchigam, Daman (UT) or equivalent, Pidilite, Zinga (Calcutta),
3	Bituminous Sealing Compound	STP Ltd, SN Industries, Tikitar

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A (Contd..)

S.No	ITEMS	MAKES
	ELECTRICAL ITEMS	
01	Transformers 66/11KV, 333/11, 33/0.433 KV, 22KV/ 11KV, 22/0.433KV, 11/0.433KV copper wound oil/ dry type cast resin of all capacity incl and above 500 KVA	ABB, SIEMENS, BHARAT BIJLEE , CROMPTON GREAVES, ALSTOM (GEC), BHEL,
02	Transformers 66/11KV, 33/0.433 KV, 22KV/11KV, 22/0.433KV, 11/0.433KV copper wound oil/dry type cast resin of all capacity incl and below 500 KVA	VOLTAMP, INDIAN TRANSFORMER & ELECTRICAL (ITEL), Esennar Transformer (P) Ltd, PACTIL, Andrew Yule
03	Isolation Transformer	VINITEK ELECTRONIC, INDIAN TRANSFORMERS & ELECTRICAL
04	Air Break Switch (Gang Operated Device),33/11 KV	PACTIL, JAIPURIYA BROTHERS, Mysore Electrical Industries (MEI), YAMUNA POWER INFRASTRUCTURE, BHEL
05	Air Break Switch (Isolators)	Mysore Electrical Industries (MEI), YAMUNA POWER INFRASTRUCTURE, Southern Switchgear, Andrew Yule,CROMPTON GREAVES
06	HT 11 KV, 3Ph Automatic Switch fuse unit	ABB, CROMPTON GREAVES, SCHNEIDER, YAMUNA POWER INFRASTRUCTURE, INDIAN TRANSFORMERS & ELECTRICAL, SIEMENS,
07	VCB (Indoor/Outdoor)	BHEL, ABB, SIEMENS, CROMPTON GREAVES, SCHNEIDER, ALSTOM. INDIAN TRANSFORMERS & ELECTRICAL,
08	HT Switch gear protection relay electronics	ABB, SCHNEIDER, C&S, SIEMENS, L&T
09	HT Trivector Meter	SECURE, L&T, ABB (ELSTER), SCHLUMBERGER, SIEMENS
10	HT Ring main unit, SF6, 11 KV	ABB, SIEMENS, CROMPTON GREAVES, SCHNEIDER
11	Package/unified Sub-Station	ABB, SIEMENS, CROMPTON GREAVES, SCHNEIDER
12	Cable jointing kit	RAYCHEM, M-SEAL, DENSON, BIRLA 3M, YAMUNA CABLE ACCESSORIES
13	CT & PT, 11 KV/33 KV	PRAGATI, AE, KAPPA, SCHNEIDER, L&T (AS PER HT PANEL MANUFACTURERS DESIGN), SIEMENS, ABB,
14	HT Armoured XLPE, PVC insulated cable Aluminium conductor for system upto and including 11KV grade 33	Havells, RPG, Gloster, Finolex, SBEE Cables (india) Ltd, KEI Industries
15	HT Armoured XLPE, PVC insulated cable Aluminium conductor for system upto and including 11KV grade and upto 66KV grade	Havells, RPG, Gloster, Finolex, SBEE Cables Ltd (india) (only approved for 11/33KV Cables), KEI Industries (only approved for 11/33KV Cables)
16	Insulator HT/LT Disc/ Pin/ Shackel/ Loop/ any type	BHEL, WS INSULATORS, JAYSHREE, MODERN INSULATOR,YAMUNA POWER INFRASTRUCTURE, SOURTHEN INSULATORS, MEI
17	HT Armoured XLPE cable 132 KV grade	Havells, Gloster, Finolex
18	HT/LT Steel tubular swaged pole	THE NATIONAL TUBING CO KANPUR, SINGH PROFILE Pune,
19	Pre stressed concrete Poles	Cement Fabric India Jodhpur, Concrete Udyog Jhansi, Sankla Udyog Jhansi, HINDUSTAN Pre stressed Concrete Faridabad, INDIAN PCC POLES

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A (Contd..)

S No	ITEMS	MAKES
20	Bus Bar Trunking	L&T, SIEMENS, SCHNEIDER, ABB, GE, LEGRAND, C&S
21	Lightning Arrestor LT/HT	OBLUM, ELPRO, ALSTOM, CROMPTON GREAVES, BHEL, YAMUNA POWER INFRASTRUCTURE
22	LT Panels	SIEMENS, CROMPTON GREAVES, L&T, ABB, HPL INDIA LTD, DOLAR, EPLEC, LEGRAND, STANDARD
23	HT PANEL	L&T,SIEMENS,SCHNEIDER,ABB,GE,EPLEC LEGRAND,C&S
24	Microprocessor based MCCB/RCCB LT 415 Volts	LEGRAND, SIEMENS, L&T, SCHNEIDER,ABB, HAVELLS, C&S LMITED, INDO ASIAN, LUKER ELECTRIC
25	ACB LT 1100 VOLTS	GEC, SCHNEIDER, SIEMENS, C&S limited, L&T, ABB, CROMPTON GREAVES
26	Contactor	SIEMENS, ABB, L&T, GE, SCHNEIDER, BCH, C&S,LEGRAND
27	MCB/ RCBO/ RCCB/ DB/ Isolators	L&T (HEGAR), LEGRAND (LEXIC), SIEMENS (BETAGUARD), SCHNEIDER (MG-MULTI-9), ABB, GE, HAVELLS, HPL, CGS, C&S, MDS
28	DBs/ MCB/ MCCB/RCCB	L&T, Legrand, ABB, Siemens, Schneider, Havells, Polycab, C&S Limited, Indo Asian, Luker Electric, Orient Electric, Bentec
29	Voltmeter/Ammeter/ Frequency meter/ Meters/ Power factor Meter	Automatic Electric, IMP, L&T, HAVELLS, MECO
30	Digital meters with built in Selector switches for Voltmeter/ Ammeter/ Frequency meter/ Energy, KW, Power Analyser	L&T, Automatic Electric, SECURE Meter, Havells, C&S Limited
31	SCADA including PLC (Programmable Logic Controller)	L&T, Schneider, Siemens, Rockwell Automation, Honeywell, ABB
32	Selector Switch	SIEMENS, KAYCEE, C&S. L&T, LEGRAND, AUTOMATIC ELECTRIC, SECURE METER, HAVELLS
33	Street light timers	L&T, LEGRAND, GE, BAJAJ, SIEMENS, ABB, BCH,
34	LT Cables, 1100 Volts, XLPE	RPG, UNIVERSAL CABLES, KEI, FINOLEX CABLES, POLYCAB, HAVELLS INDIA LTD, HAVELLS INDIA LTD, RR KEBLES, RPG CABLES (KEC) ,UNIFLEX, PLAZA, ANCHOR,NICCO, POLYCAB
35	Change over Switch/ Starter/ Contactor DOL / Star Delta/ Synchronous/Single phase preventor	L&T, SIEMENS, CROMPTON GREAVES, GE, HAVELLS, BCH, C&S, ABB
36	Main switch Iron clad	L&T, SIEMENS, CROMPTON GREAVES, GE, HAVELLS, Indo Asian
37	Tamper proof electronic meter	SECURE, L&T, ACCURATE METERS, HAVELLS, HPL
38	Automatic Voltage Stabilizers/ Servo controlled Voltage Stabilizer	Automatic electric, VINITEC, Indian Transformer and Electricals, Emerson, Andrew Yule, APLAB, V GUARD, VOLINA
39	(b) Voltage Stabilizers up to 4 KVA	VINTEC, SINETRAC, MICROTECH, POWER WARE, APLAB, V GUARD, AE, VOLINA
40	(c) Voltage Stabilizers above 25 KVA (Servo Controlled)	ANDREW YULE (BRENTFORD), AE, APLAB
41	UPS	TATA LIEBERT, APLAB, LUMINOUS, MICROTECH, SUKAM, APC
42	DG SETS (Engine)	CUMMINS, KIRLOSKAR, ASHOK LEYLAND, GREAVES COTTON, CATERPILLAR
43	DG SET (Alternator)	KIRLOSKAR, CROMPTON GREAVES, JYOTI, STAMFORD, ALSTOM, BHARAT BIJLEE

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A(Contd..)

Sl. No	ITEMS	MAKES
44	DG Sets assembled with sound proof canopy	KIRLOSKAR, JACKSON, SUDHIR, GREAVES COTTON, CUMMINS INDIA.
45	Induction Motors	CROMPTON Greaves, KIRLOSKAR ELECTRIC, BHARAT BIJLEE, SEIMENS, NGEF, ABB.
46	Precast concrete cable covers	Mehtab, Supreme tiles, Sukhi Enterprises
47	Compact Sub-station	Sudhir, Kirloskar, ABB, C&S, Indian Transformer and Electricals (ITEL)
48	Power factor improvement Capacitor Banks	SIEMENS, L&T, ABB, EPCOS, GE,
49	APFC Panels	L&T, GEC, SIEMENS, ABB, EPCOS, INDIAN TRANSFORMERS & ELECTRICAL
50	LT Cable, XLPE, PVC Insulated Copper/ Aluminium Wires & Cable 650/1100V of all types	FINOLEX, Nicco, RPG, HAVELLS, PLAZA, ANCHOR, SBEE Cables (India) Ltd, KEI Industries
51	MS Conduit	AKG, BEC Industries, BHARAT, POLYCAB, KALINGA, JINDAL
52	PVC casing capping & acessories	PRESTO FIT, PRECISION, SUPREME, MODI. POLYCAB, PLAZA
53	PVC Conduits (Rigid or flexible) and fittings	PRECISION, ANCHOR, MODI, PRESFIT, ASTRAL, AVON, POLYCAB, ASTRAL Limited
54	Electric Acessories Piano Switches, Ceiling rose, Call bells, Buzzers, Lamp holders/ Socket oulet etc	LEGRAND, LEADER, ANCHOR, CRABTREE
55	Modular Switches/ sockets	ANCHOR (ROMA), LEGRAND, CRABTREE, Indo Asian, C&S Limited
56	PVC Insulating tape	ANCHOR, 3M, AKG, Steelgrip
57	Street light luminaries HPSV/Metal Halide	PHILIPS, CROMPTON GREAVES, BAJAJ, WIPRO, C&S, HALONIX, HAVELLS, SHAKTI
58	LED Lamps/Tube lights/High Bay / Flood /Street /Façade / Bollard/ Post top/ Step/ Aviation/ Down/ Search/ Commercial/ Landscape/ Industrial/ Area/Panel lightings/ High bay Fittings	Bajaj, Philips, Wipro, Crompton, Havells, Haloxix Technologies Pvt Ltd, Crompton Greaves (Consumer Electricals Ltd), Jaguar (Lightening Solutions), Luker electric, Orient Electric, FortuneArrt, Syska LED lights, Bentec
59	Ceiling fan/Wall fan	CROMPTON, HAVELLS, KHAITAN, BAJAJ, ORIENT ELECTRIC, LUKER ELECTRIC, Haloxix Technologies Pvt Ltd, Crompton Greaves (Consumer Electrical Ltd),
60	Exhaust fan/Air circulators	CROMPTON, HAVELLS, USHA, BAJAJ, ALMONARD, ORIENT ELECTRIC, LUKER ELECTRIC, Haloxix Technologies Pvt Ltd, Crompton Greaves (Consumer Electrical Ltd),
61	Smart/BLDC Energy Efficient ceiling, wall, Pedestal and Exhaust fans	CROMPTON, POLAR, KHAITAN, BAJAJ, ORIENT ELECTRIC, LUKER ELECTRIC, ATOMBERG
62	Fan Regulator	CROMPTON, ANCHOR, KHAITAN, BAJAJ,HAVELLS, USHA, ORIENT ELECTRIC,
63	Geyser/Water Heater	BAJAJ, AO SMITH, USHA, VENUS, ORIENT ELECTRIC, CROMPTON, HAVELLS, LUKER ELECTRIC
64	PVC Pole Boxes	SINTEX, FINOLEX, UNIVERSAL
65	Flame Proof light Fittings LED/CFL) /Fan/Well glass/Bulk head incl accessories	BALIGA, SUDHIR, Flexpro Electricals, Nasik, Bajaj, Crompton, Shyam Switchgears
66	Electric/ Photoelectric switch for auto Op of street lights	L&T, GE, Siemens, Bajaj, Legrand, Indo Asian
67	ACSR Conductor	ALL-IND, ICC UJALA, NICCO, BHARAT CONDUCTORS, INDIANALUMINIUM CO

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A (Contd..)

Sl. No	ITEMS	MAKES
68	Laminated Sheet Cover	HYLAM, FORMICA, GREEN LAMINATED BOARDS, ANCHOR, BAKELITE
69	Aviation Obstruction Light	BAJAJ, CROMPTON GREAVES, WIPRO, PHILIPS
70	Thimbles/Studs/Lugs	DOWELLS, JAIPURIA BROTHERS, AXIS, INDIRA
71	Lifts	OTIS ELEVATORS, MITSUBISHI, SCHINDLER, HITACHI, TOSHIBA, FUJITEC INDIA LTD
72	LT Changeover Switch	L&T, SCHNEIDER, HAVELLS, STANDARDS, C&S, LEGRAND, ABB, GE
73	Battery Charger	VOLTSTART, AE, BCH
74	Heavy duty EOT/HOT cranes any type	MUKUND, JESSOP, CRANEX
	WATER SUPPLY	
01	GI Pipes	JINDAL, TATA, SWASTIK, ZENITH. PRAKASH
02	HDPE Pipe & FITTINGS	FINOLEX, SUPREME Industries Ltd, Jain Irrigation Sys, PRINCE, Astral Limited
03	CI Pipes & fittings	ELECTRO-STEEL, KESORAM, KEJRIWAL, NECO
04	DI Pipes & fittings	ELECTRO-STEEL CASTING, JINDAL, TATA METALIKS, SAW PIPES
05	CPVC Pipe	DUTRON, FINOLEX, ASTRAL, SFMC, SUPREME Industries Ltd, Prayag Polymers
06	PVC Soil waste, rainwater (SWR) & Drainage pipes	SUPREME Industries Ltd, Prince, Kisan, Finolex, Infra, Astral Limited
07	PPR Pipe	SUPREME Industries Ltd, PRINCE, FINOLEX.
08	PVC pipes & fittings/PVC –O & Fittings	Prince, Finolex, SUPREME Industries Ltd, Astral, Jain irrigation Sys, Ashirwad
09	MS/ERW Pipes	TATA, JINDAL, ZENITH
10	UPVC Pipes & Fittings	SUPREME Industries Ltd, Prince, Finolex, Astral Limited, Jain irrigation Sys, Astral Limited, Prayag Polymers, Supreme
11	Polyethylene/Aluminium/ Polyethylene composite pressure pipes	Prince, Finolex, Supreme
12	Plastic Pipe (for non pressure Drainage & Sewarage)	Prince, Finolex, Supreme
13	CI-Soil waste rainwater (SWR) & Drainage pipes	NECO Nagpur, Kesoram, Electro Steel
14	AC-Soil waste rainwater (SWR) & Drainage pipes	M/S Everest Asbestos Hyderabad, Vishakha India Ltd, M/s Hyderabad Asbestos Cement (Charminar), M/s Ramco
15	RCC Pipes, Drain pipes	Indian Hume Pipes, Everest, Himalaya, Thuluvananikal pipes
16	Sluice Valves	KIRLOSKAR, L&T, AUDCO, LEADER, UPADHAYAY, BIR
17	Reflex Valves/Non Return Valve/Check valves	KIRLOSKAR, LEADER, Upadhyay, AUDCO, L&T, Jainko
18	Pump Sets-Mono Block	KIRLOSKAR, CROMPTON GREAVES, BEST & CROMPTON (BEACON), MATHER & PLATT, GROUND FOSS, BEACON, KSB, JYOTHI
19	Centrifugal Pumps	KIRLOSKAR, CROMPTON GREAVES, BEACON, MATHER & PLATT, GROUND FOSS, KSB, JYOTI, CRI Pumps

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A (Contd..)

Sl. No	ITEMS	MAKES
20	Submersible Pumps/ Open well pumps	KSB, KIRLOSKAR, GROUND FOSS, MATHER &PLATT, JYOTI, CRI Pumps, Suguna Pumps
21	Vertical turbine pumps/Horizantal split case pump set	KSB, KIRLOSKAR, MATHER &PLATT, JYOTI, CRI Pumps,
22	Non clog sewage submersible and VT pumps/Dewatering submersible pump sets	KSB, KIRLOSKAR, GROUND FOSS, MATHER &PLATT, CRI Pumps,
23	Pumps for fire fighting	KIRLOSKAR, CROMPTON GREAVES, MATHER & PLATT, Bharat Bijlee
24	Submersible Cable	FINOLEX, NICCO, HAVELLS, PLAZA
25	Motor Starter	L&T, BCH, SCHNEIDER, SIEMENS, GE
26	Electric Motors	BHARAT BIJLEE, KIRLOSKAR ELECTRIC, CROMPTON GREAVES, ABB, SIEMENS
27	Single Phase Preventer/ Phase Sequence corrector	L&T, SIEMENS, MINILEC, SINETRAC
28	Silver Ionization Plant	SIEMENS, BHEL, BHARTI WATERS, JYOTI WATERS, PIOUS WATERS
29	Dual Ionization Plant	SIEMENS, BHEL, BHARTI WATERS, JYOTI WATERS, PIOUS WATERS
30	Air Release Valve	BIR, KIRLOSKAR, LEADER, AUDCO, L&T, UPADHYAY, SANT
31	Foot valve	KIRLOSKAR, LEADER, AUDCO, UPADHYAY, SANT, Prayag Polymers, Jainko
32	Butterfly Valves/Disc Valves	L&T , AUDCO, KIRLOSKAR, UPADHYAY, Jainko
33	Battery	EXIDE, AMCO, AMRON, TATA GREEN
34	CI Fitting	UPADHAYAY, KEJERIWAL, BIR, SANT, VENUS
35	Water Meter	CAPSTON, DESHMESH, KEYCEE, CAPITAL, KIRLOSKAR
36	Gate Valves	ZOLOTO, LEADER, GEM, SEICO, SANT, Jainko
	AIR CONDITIONING	
01	Central Air Conditioning system any type	BLUE STAR, VOLTAS, DAIKIN, LEEL ELECTRICALS Ltd (Lloyd)
02	Air Handling Unit	BLUE STAR, VOLTAS, DAIKIN, NATIONAL, ZECO
03	Air curtains	Almonard, Air Con, Crompton Greaves
04	Cooling Tower	MIHIR, PAHARPUR, DELTA
05	Chillers/Condenser	VOLTAS/ BLUE STAR/ CARRIER
06	Compressors	DANFOSS/ KIRLOSKAR, BLUE STAR, VOLTAS, CARRIER.
07	Pumps (Chillers & Condensers)	KIRLOSKAR/GROUNDFOSS/MATHER & PLATT/BEST & CROMPTON (BEACON)
08	Motors	SIEMENS/ ABB/ CROMPTON GREAVES/ KIRLOSKAR ELECTRIC/ BHARAT BIJLEE/ NGEF
09	Pot/Y-Strainer	EMERALD/RAPID COOL/DANFOSS/SPARLAN
10	MS Piping	TATA/BST/JINDAL
11	(a) Butterfly Valve	AUDCO/SANT/ NORMAX/LEADER
12	(b) Check Valve/ Non Return Valve	AUDCO/SANT/ BIR/ KIRLOSKAR/LEADER

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A (Contd..)

Sl. No	ITEMS	MAKES
13	(c) Gate Valve	LEADER/ AUDCO/ITT BELL & GOSSETT/ HM/ZOLOTO
14	Balancing Valve	AUDCO/DANFOSS/FLOWCON
15	Pressure Gauge	FIEBIG/ BAKER MERCER/DANFOSS/L&T
16	Dial Type Thermometers	FIEBIG/ TEDDINGTON/DANFOSS
17	Insulation	
	(i) Glass wool	OWENS CORNING/UP TWIGA/LLOYD
	(ii) Insulation (Expand Polystyrene)	CAPRICORN/BEARDSELL/PENGUIN/LLOYD
	(iii) Insulation (Cross Polyethylene Foam)	SUPREME/TROCELLANE/PARAMOUNT/LLOYD
18	(a) Controls (3 Way Valves Actuator & Motor)	SIEMENS/HONEYWELL/JOHNSON/ANERGY
	(b) Thermostats	ENERGY/ SIEMENS/ HONEYWELL/ JOHNSON/ ANERGY
19	Extruded Aluminum Grills Diffusers	DYNA CRAFT/ RAVISTAR/ MAPRO/ CARRYAIRE/ COSMOS
20	(a) Control Cable	NICCO/ASIAN CABLES/UNIVERSAL/FINOLEX/POLYCAB
21	(b) Power Cable	GLOSTER, ASIAN, UNIVERSAL, ELEKTRON
22	Contactors/Starters	L&T/SIEMENS/ABB/BCH
23	LT Relay numerical static	SIEMENS, ABB, SCHNEIDER, L&T, EPCOS
24	Time Delay Device	L&T/SIEMENS/SCHNEIDER/BCH
25	Single Phasing Preventer	L&T, SIEMEN, MINILEC, SCHNEIDER
26	GSS Sheets	SAIL/TATA/JINDAL
27	AIR CURTAIN	SAN PRODUCTS,EURONICS,CRISIL
28	DEHUMIDIFIER	BRYAIR,OSTER BIONAIRE, NOVITA
29	Fire Dampers/ Grill/Diffuser	DYNA/RAVISTAR/MAPRO/CARYAIRE/ATE
30	Expansion Valve	DANFOSS/SPORLAN/RANCO
31	Switchgears MCCBs/ ACBs/ MCBs	LEGRAND/L&T/SCHNEIDER (MERLIN GERIN)/ GE/ SIEMENS/ BCH/ C&S
32	LT Control Panel	FACTORY FABRICATED CONTROL PANEL WITH CPRI TEST CERTIFICATE
33	LT & Potential transformer (CT & PT) 11KV and above range	SIEMENS, BHEL, Schneider, Crompton Greaves, L&T
34	LT & Potential transformer (CT & PT) below 11KV range	Automatic Electric Ltd (AE), Pragati Electricals Pvt Ltd, HPL, C&S
35	HP/LP/Op Cut Out	PENN/INDOFOSS/RANCO/DANFOSS/JOHNSON
36	Digital/Analog AC Static Watt hour Meters (Post paid & Pre paid)	L&T, Jaipuria Meters, Havells, Secure Meters, Zenmeter Solutions Pvt Ltd
37	Centrifugal Fans (for AHU's)	NICOTRA/KRUGER/COMBIFREE/OEM'S MAKE
38	Strip Heaters	RACOLD/DASSPASS/UE
39	Humidistat	PENN, DANFOSS, HONEY WELL, JOHNSON, INDOFOSS
40	Filter	JOHN FOWLER/THERMODYNE/TENACITY/PUROLOTER

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX-A (Contd..)

Sl. No	ITEMS	MAKES
41	Factory Built Duct	ROLASTAR/TECHNO AIR/CAM DUCT
42	Selector Switch	L & T, LEGRAND, C&S, OMEGA, KAYCEE
43	Indicating light	SIEMEN, CONCORD, C&S, LEGRAND
44	Digital Voltmeter/ Amp meter	AE, EMERCON, SECURE, L&T,C&S
45	Current Transformer	AE, KAPPA, SCHENIDER
46	Flexible Connector	CORI ENGINEER, RESISTO FLEX
47	Hot Water Generator	RAPID COOL/ROSS THERMAL SYSTEM PVT LTD/ REYNOLD/ RAPID CONTROL
48	Air Washer	BREEZE AIR/THERMODYNE/CHEMPAC/KLENZOID
49	Non Chemical Type Electrostatic Scale Preventer	SCALEOFF (MFD BY M/S WELDON), SCALEX (MFD BY M/S TBI SYSTEM), HYDROCON
50	Window AC	CARRIER, DAIKIN, BLUESTAR, VOLTAS, O-General, Hitachi
51	Split AC	CARRIER, DAIKIN, BLUESTAR, VOLTAS,
52	Smoke Detector	APPOLO, EDWARD, MARLAY SYSTEMS, SAFEX, SENSOR
53	Manual Call Point	MINIMAX, APPOLO, VIJAY, SAFTEX, HARDIK ENGGS HONEYWELL
54	Fire Hydrant Valves	MINIMAX, VIJOY, SAFEX, CEASE FIRE
55	Fire Hose Pipe/ Reel	MINIMAX, SAFEX, VIJAY, DUNLOP
56	Fire Extinguishers	MINIMAX, SAFEX, VIJAY, SAFEZONE, CEASE FIRE, MONARC ENGINEERINGS
57	Branch Pipe Nozzle and Couplings	MINIMAX, SAFEX, VIJAY, SAIFEZONE, CEASE FIRE, MONARC ENGINEERINGS
58	Control Panels	CEASE FIRE, VIJAY FIRE PROTECTION SYSTEMS, EX-FLAME. HONEYWELL
59	AIR WASHER	BREEZE AIR, THERMODYNE,CHEMPAC,KLENZOID
60	SOLAR WATER HEATER SYSTEM	RACOLD SOLAR WATER HEATER, V-GUARD SOLAR WATER HEATER, TATA POWER SOLAR SYSTEM, WAAREE ENERGIES LTD
61	SOLAR POWER PANELS (PV)	TATA SOLAR, BHEL, HHV SOLAR, WAAREE ENERGIES LTD, VIKRAM
62	SOLAR STREET LIGHT	TATA POWER, HAVELLS, BAJAJ, Sol Mitra Power, Jai Sunlight System
63	Fire alarm and fire fighting system	Nitin, Agni
64	Fiberglass (FRP/GRP) lighting pole, cable tray, enclosure, Safety ladder	SUMIP, ERCON
	MISC	
01	SODIUM HYDROCHLORIDE	AQAFID, ALTOCH DECLIBAC, BIROJING
02	BLEACHING POWDER	KANORIA CHEMICALS, SREERAM BENGAL CHEMICALS, KASHYAP INDUSTRIES
03	ALUM	SREERAM BENGAL CHEMICALS, JYOTI, INDIAN ABRASIVE IMPEX INDIA, DANEC INTERNATIONAL
04	GLAND PACKING	CHAMPION, COMMANDER, NAVEEN ENGINEERING
05	WATER GASKET	CHAMPOIN, CHARMINAR, COMMANDER
06	FLOAT VALVE	PRAYAG, SATYAM, TECHNO, GOLDLINE

PARTICULAR SPECIFICATIONS (Contd…)

Sl. No	ITEMS	MAKES
07	GI PIPE FITTINGS	ZOLOTO, LEADER, UNIK
08	Fire alarm Control Panel	GE Edwards, Notifier, Siemens Simplex, HONEYWELL
09	Manual Call Point	GE Edwards, Notifier, Siemens, HONEYWELL

NOTES:

1. Items shall be considered whichever is applicable only and if not specifically mentioned in Sch 'A'.

2. Sources indicated are only for guidance and approval of the Garrison Engineer shall be taken in proper time before procurement of materials and its incorporation.

3. The above list is not exhaustive but indicative of all items required for work under the contract.

4. Sources of materials shall be as above or in the vicinity thereof. The tenderer shall ascertain the actual position/exact location of source before submitting his tender and no additional payment shall be made on account of misunderstanding of its distance from site of works. Contractor may bring material conforming to contract specifications from other sources without any price adjustment after obtaining written approval of the Garrison Engineer.

5. The tender shall amongst other things also ascertain all information such as royalties, taxes duties and other charges etc. on the materials and no additional payment shall be made on account of the foregoing.

6. The name of approved manufacturer is not shown in the list attached, but same has been added/ deleted by E-in-C, ADG or any Chief Engineer command or by CE (CG) goa during execution are also deemed to be added/ deleted in the approved list.

Signature of Contractor

AAD (Contracts)
For Accepting Officer

PARTICULAR SPECIFICATIONS (Contd…)

APPENDIX – B'

RECOVERY RATES OF TESTING CHARGES

Sl no	Materials	Test	Method of testing	Frequency of Tests			Level of test	Rate per Test Rs. P	Remarks
1	2	3	4	5			6	7	8
1.	Bricks	(i) Compressive strength	IS-3595	AS per IS-5454 a given under			`A'	330/-	Checks for visual and dimensional characteristics shall also be carried out
			(Part-II)						
		(ii) Water absorption	- do -	Lot size	Sample Size	Permissible No of defective bricks	`A'	330/-	As per IS-5454
		(iii) Efflorescence	- do -	1001 to	5	0	`A'	330/-	
			(Part I)	10000					
				10001 to	10	0			
				35000					
				35001 to	15	1			
				50000					
2.	Coarse aggregates	(i) Sieve analysis	IS-2386	One test for every 15cum of aggregates			`A'	660/-	
			(Part-I)	or part thereof brought to site					
		(ii) Flakiness Index	- do -	- do -			`A'	250/-	
		(iii) Estimation of deleterious materials	IS-2386	One test for every 100cum of aggregates			`A'	600/-	
			(Part-I)	or part thereof					
		(iv) Organic impurities	- do -	One test per source of supply			`B'	275/-	
		(v) Moisture content	- do -	Regularly as required			`A'	330/-	
			(Part - II)						

PARTICULAR SPECIFICATIONS (Contd…)

RECOVERY RATES OF TESTING CHARGES (CONTD…)

1	2	3	4	5	6	7	8
		(vi) Specific gravity	- do -	One test for each source of supply	`B'	330/-	-
3.	Fine aggregate	(i) Sieve analysis	IS-2386 (Part-I)	One test for every 15CUM of FA or part when brought to site.	`A'	660/-	-
		(ii) Test for clay, silt and impurities	- do - (Part-I)	- do -	`A'	500/-	-
		(iii) Specific gravity	- do - (Part-I)	One for each source of supply	`B'	330/-	-
		(iv) Moisture content	- do - (Part – II)	Regularly as required subject to 2tests/day when being used	`A'	330/-	-
		(v) Test for organic impurities	- do - (Part - II)	One test for each source of supply	`B'	275/-	-
4.	Structural Concrete (M-25 grade and above)	(i) Slump-test or compacting factor test or vee-bee time	IS-119	The minimum frequency of sampling of concrete of each grade shall be as under:	`A'	300/-	Random sampling shall be carried to cover all mix units
		(ii) Compressive strength	IS-516	Qty of concrete In the work (Cubic Meter) — No of samples 1----5 — 1 6----15 — 2 16---30 — 3 31---50 — 4 51 and above — 4+1 for each Addl 50m or part thereof	`A'	900/-	-

PARTICULAR SPECIFICATIONS (Contd…)

RECOVERY RATES OF TESTING CHARGES (CONTD…)

1	2	3	4	5	6	7	8
5.	(a) PCC Block for walling (Hollow Block)	(i) Compressive strength	IS-2156 – 1984 (Appx-`B')	08 Blocks out of 20	`A'	900/-	Sample: 20blocks from consignment of every 5000blocks or part thereof
		(ii) Water absorption	- do - (Appx-`B')	03 Blocks out of 20	`B'	330/-	-
		(iii) Density	- do - (Appx-`A')	03 Blocks out of 20	`B'	330/-	-
	(b) PCC solid Block for walling	(i) Compressive strength	IS-2185	08 Blocks out of 20	`A'	900/-	Sample: 20blocks from consignment of every 5000 blocks or part thereof
		(ii) Water absorption	- do -	03 Blocks out of 20	`B'	330/-	These blocks to be checked for dimension and weight
		(iii) Density	- do -	03 Blocks out of 20	`B'	330/-	-
6.	Cement Flooring Tiles / Terrazzo Tiles	(i) Water absorption	IS-1237 (Appx-`D')	06 tiles out of 18	`B'	330/-	Sample of 18tiles from each source of supply selected at random
		(ii) Wet transverse strength	- do - (Appx-`E')	- do -	`B'	660/-	-

PARTICULAR SPECIFICATIONS (Contd…)

RECOVERY RATES OF TESTING CHARGES (contd…)

1	2	3	4	5	6	7	8
7.	Cement	(i). Setting time			'B'	500/-	
		(ii). Soundness			'B'	550/-	
		(iii). Compressive Strength			'B'	550/-	
		(iv). Fineness			'B'	275/-	
8.	Reinforcement Steel	(i). Physical tests up to 16mm dia (normal mass, tensile elongation, bend and rebend)			'B'	2500/-	
		(ii). More than 16mm dia			'B'	2750/-	
9.	Timber	(i) Specific gravity and weight	IS-1708-1960	Minimum `3' samples from a lot of 4 Cum or 250pieces of seasoned timber.	`B'	120/-	-
		(ii) Moisture content	- do -	-	`A'	120/-	-
10.	Water for construction purpose	(i) Test for Acidity	IS-456 & 3015	Once at the stage of approval of source of water	`B'	240/-	Also refer clause 4.3 of IS-456 and its subsequent sub clauses regarding suitability of water

PARTICULAR SPECIFICATIONS (Contd…)

RECOVERY RATES OF TESTING CHARGES (contd…)

1	2	3	4	5	6	7	8
		(ii) Test for Alkalinity	- do -	-	`B'	240/-	-
		Test for solid content	- do -	-	`C'	300/-	-
11.	Welding of steel work	Visual inspection test	IS-822-1970 Clause 7.1	100% by visual inspection	Work site	360/-	Specialized tests their method and frequency to be decided on consideration of their importance by the Accepting Officer
12.	Timber paneled and glazed Door/ Window & shutters (Including factory made shutters)	(a) Dimensions, sizes, workmanship and finish	IS-1003-1977 (Part I)	Frequency of sampling from each lot shall be as under: Lot size — Sample size 26 to 50 — 5 51 to 100 — 8 101 to 150 — 13 151 to 300 — 20 301 to 500 — 32 501 to 1000 — 50 1001 and above — 80	`A'	180/-	-
		(b) Strength test (i) Slamming	IS-1303-1990	Form the each lot 5% of the factory made shutters shall be manufacturer tested for strength tests	-	-	-
		(ii) Impact indentation	- do -	-	-	-	-
		(iii) Shock resistance	- do -	-	-	-	-
		(iv) Edge loading	- do -	-	-	-	-

PARTICULAR SPECIFICATIONS (Contd…)

RECOVERY RATES OF TESTING CHARGES (contd…)

1	2	3	4	5	6	7	8
13.	Plywood (IS-303-1989)	(a) Moisture content	IS-1734-1983 (Part-I)	Six tests pieces cut from each of the boards selected as per table shall be subjected to tests	`C'	240/-	Sampling shall be as per IS-7835-1975 Tables
14.	Wood particle board (medium density) IS-3097-1985	(a) Density	IS-2360-(Part-III)	Three test specimens from each sample (size 150mm x 75mm)	`A'	60/-	Sampling shall be as per IS-3487-83 with moisture meter
		(b) Moisture Content	- do -	- do -	`A' & `B'	60/-	-
		(c) Water Absorption	- do - (Part 16)	- do – (Size 300mm x 300mm)	`A'	60.-	-
		(d) Swelling due to surface absorption	- do - (Part-17)	- do – (Size 125mm x 100mm)	`A'	60/-	-
		(e) Swelling in water	- do -	- do – (Size 200mm x 100mm)	`A'	60/-	-
		(f) Modulus of rupture	- do - (Part-4)	Three test specimens as per IS-2380-77	`B'	90/-	-
		(g) Screw withdrawal strength	- do - (Part-4)	- do – as per IS-2385	`C'	120/-	-

NOTE:

The rate given in Appendix- 'D' above shall be enhanced by 70%

Legend: A-Site lab, B-Zonal lab, C-National test house / SEMT wing Pune / Govy. Engineering College

Signature of Contractor

AAD (Contract
For Accepting Officer